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About this document

Purpose

The *NSP Data Collection and Analysis Guide* introduces NSP Data Collection and Analysis functions to operators and administrators by describing usage and features.

Scope

This document covers the NSP Data Collection and Analysis functions. Some feature sets described in the document require the purchase and configuration of additional feature packages.

See the *NSP System Architecture Guide* for more information about feature packages and installation options.

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Part I: Data collection, analysis, and visualization

Overview

Purpose

The chapters in this part of the document describe telemetry, baseline, and indicator management in NSP.

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1 Overview

1.1 How does NSP support telemetry management?

1.1.1 Definition

NSP Data Collection and Analysis functions provide collection, processing, and analysis for MDM and Cloud Native (CN) telemetry. NSP supports configuring telemetry subscriptions and using baseline analytics, NSP indicators, and OAM tests for anomaly detection, metric customization, and network tests.

The type and scope of statistics support are dependent on the NE source. NSP supports the following telemetry providers:

- **MDM-based telemetry**
- **cloud native telemetry**
- **statistics collection for classically managed NEs**

The following table provides definitions of telemetry terms in use in NSP.

i **Note:** If you are using adaptor artifacts from previous releases, telemetry functions are supported at the level of the release your adaptor artifacts are for; see the artifact guide for the adaptor artifacts you are using.

Term	Definition	References	Notes
Telemetry providers			
MDM telemetry	Collection using SNMP for multivendor NEs and for model-driven NEs using artifacts from releases prior to 23.8.	See the artifact guide for the NE type and NSP release.	Contact Nokia for information about adding adaptation for third-party devices.
Cloud Native telemetry (CN telemetry)	Collection using gNMI for Nokia and multivendor NEs. CN telemetry can be used with classic or MD managed devices that support gNMI. gNMI uses gRPC as the underlying protocol, with NSP acting as the gRPC client, and the NE acting as the gRPC server.	For information about using NSP for cloud native telemetry, see the chapters in this document. For information about using APIs for cloud native telemetry, see the Telemetry tutorial on the Network Developer Portal . For information about the artifacts that support CN telemetry, see the artifact guide for the NE type and NSP release.	Cloud native telemetry requires a gRPC mediation policy in the discovery rules used to discover and manage the NEs. The client/server gRPC communication must be authenticated and encrypted.

Term	Definition	References	Notes
Telemetry providers			
Telemetry for classically managed NEs	Collection using SNMP for Nokia classically managed NEs	For more information about statistics management using the NFM-P, see the <i>NSP NFM-P Statistics Management Guide</i> .	—
Related terms			
Vendor agnostic (va)	Telemetry definitions included with the NSP that allow for collection of statistics on multiple NE types.	For information about the telemetry definition bundles for a specific model-driven NE type, see the artifact guide for the NE type. See the API Documentation on the Network Developer Portal to find the list of applicable telemetry definitions. For information about telemetry artifacts, see 2.2 "How do I prepare for telemetry collection?" (p. 29).	The adaptation artifact bundle for an NE includes telemetry mappings for the NE that align with the telemetry definitions packaged with the NSP.
Device telemetry	Device-specific telemetry, including both YANG telemetry definition files and device adaptation custom resources for CN telemetry.		The device telemetry bundles for an NE contain telemetry definitions and adaptation for the supported NE-specific counters. Nokia recommends that you only install the telemetry YANG files required for your use cases.
Accounting telemetry	Collection of accounting files using cloud native telemetry		The adaptation artifact bundle for an NE contains telemetry mappings for accounting collection on the NE. Accounting telemetry is supported for model-driven NEs using SFTP only. SFTP and FTP are supported for classic NEs. It requires a file transfer mediation policy in the discovery rules used to discover and manage the NEs.

1.1.2 Functionality

NSP provides the following using collected telemetry data:

- visualization— showing information in the form of telemetry charts
- aggregation—combining measurements for analysis by Analytics
For aggregation information to be available, the auxiliary database must be installed.
- baseline analytics and anomaly detection—defining trends using telemetry data and identifying outliers

- NSP indicators—defining and tracking KPIs using customized metrics
- OAM tests—creating tests using templates

The telemetry functions described in this document are distinct from:

- Cflowd AA statistics and other flow statistics collected via Flow Collectors
- Analytics report catalogs, although statistics collected using DCA aggregation functions are available in Analytics. See the *NSP Analytics Report Catalog* for information about report generation and visualizations, and reference lists of the available report catalogs. Baseline analytics and analytics reporting are different functions, each with their own installation options.

Telemetry artifacts

The telemetry types for NEs are defined using YANG files in the NE artifact bundles. NSP uses custom resource (CR) definitions to translate the YANG data, map incoming telemetry to managed objects in the network, and output data to Postgres, Vertica, and Kafka as required. These resource definition files can be viewed as part of the artifact archive. For information about the CRs that accompany your NE adaptors, see the relevant artifact guide.

See [2.2 “How do I prepare for telemetry collection?” \(p. 29\)](#) for more information.

 **Note:** You must install the relevant adaptation artifact bundles before managing the associated NE types to use the telemetry definitions. NSP also includes a limited, generic set of default telemetry. Without installing adaptor artifacts, you will only see the default telemetry. To use device telemetry counters, you must also install the device telemetry bundle.

Telemetry subscriptions

A subscription defines the parameters of telemetry collection and uses a filter definition to select the NEs/objects. NSP deploys the specified data to the NE and registers the Kafka subscription, which is used to transfer telemetry data from the NE to NSP.

See [2.3 “What is a subscription?” \(p. 31\)](#) for more information.

1.1.3 Accounting telemetry

Accounting telemetry requires retrieval of accounting file information from NEs. The NSP polls the NE directories specified in the device helper custom resource for accounting file updates. The file on the NE is transformed to JSON format, compressed, and saved in the NSP file server.

If communication with the NE is lost, files that were stored on the NE during the loss of communication are retrieved when communication is reestablished, provided the files have not been purged by the NE.

By default, accounting files are retained on the NE after processing by NSP.

1.2 What are the best practices for telemetry data collection?

1.2.1 Data collection prerequisites

Many functions, including baselines and indicators, depend on telemetry data. Telemetry data collection must be enabled before any downstream function can be used.

NSP has the following prerequisites for telemetry data collection.

- The NEs from which you want to collect data must be discovered, managed, and reachable by the NSP.
 - Mediation policies must be configured in the discovery rules used to discover and manage the NEs:
 - SNMP telemetry requires an SNMP mediation policy
 - gNMI telemetry requires a GRPC mediation policyFor unsecure telemetry collection, disable the Secure toggle in the mediation policy creation form.
 - Accounting telemetry requires a file transfer mediation policy.
- See “Device discovery” in the *NSP Device Management Guide* for more information about NE discovery and mediation.
- For unsecure telemetry collection, unsecure collection must be enabled on the NE using CLI; see the NE documentation.
 - If you want to use a telemetry subscription to manage MIB statistics from classic NEs, statistics collection must be configured in the NFM-P.
 - Cloud native telemetry requires the gnmiTelemetry installation option.
 - Cloud native accounting telemetry requires the accountingTelemetry installation option.
 - All required NE adaptors and telemetry custom resources must be installed on the NSP server. For the list of required telemetry artifacts for an NE family, see the artifact guide for the NE adaptor artifact bundle.
 - If you want to use statistics for Analytics reports with granularity other than raw data, you need aggregation rules for the telemetry types you want to collect; see the Telemetry tutorial on the [Network Developer Portal](#).
 - You need to [configure the aggregation time zone](#).

1.2.2 Telemetry subscription best practices

A telemetry subscription is configured to collect telemetry data from the NE, process it via NSP, and, if needed, output data to Kafka and/or the database.

When configuring a subscription, it is recommended to align the subscription collection interval with the intended use case and output option.

- Monitoring or troubleshooting use cases normally require shorter collection intervals such as 10-60 s with notification (Kafka output) for plotting purposes.
Creating a custom notification topic may be useful for these use cases.
- For historical analysis such as Analytics reports a longer collection interval such as 5 or 15 min with database output is recommended.

Combining use cases in a single subscription, such as the monitoring and historical analysis use cases, is not recommended. Combining them with the longer collection interval would impede the monitoring use case. Combining with the shorter interval may overload the NE CPM and/or quickly exceed database storage capacity and impact reporting performance.

1.2.3 Accounting telemetry

Accounting telemetry requires retrieval of accounting file information from NEs. The NSP polls the NE directories specified in the device helper custom resource for accounting file updates. The file on the NE is transformed to JSON format, compressed, and saved in the NSP file server.

If communication with the NE is lost, files that were stored on the NE during the loss of communication are retrieved when communication is reestablished, provided the files have not been purged by the NE.

By default, accounting files are retained on the NE after processing by NSP.

The following prerequisites apply for accounting telemetry.

- The `networkInfrastructureManagement-accountingTelemetry` installation option must be present on the NSP.
By default, the accounting telemetry installation option enables collection of accounting files from model-driven NEs only. To enable processing of accounting files from classic NEs, the `collectFromClassicNes` flag must be set to `true`.
- If accounting files will be processed from classic NEs selective accounting file processing must be configured; see “To configure selective accounting file processing” in the *NSP Classic Management User Guide*.
- A file transfer mediation policy must be part of the discovery rule used to discover the NE.
 - If the unified discovery rule includes both a file transfer mediation policy for MDM and a classic discovery rule, the classic security access mediation policy is used for accounting telemetry for classic NEs, and the MDM file transfer policy is used for model-driven NEs.
 - Accounting telemetry is supported for model-driven NEs using SFTP only. SFTP and FTP are supported for classic NEs.
- The NE must be configured to allow parallel SSH sessions; see the NE documentation.
- An accounting policy must be configured on the NE. The accounting policy can be configured via CLI; see the NE documentation, or in the **Device Management, Device Configuration** views using an accounting intent type; see the *NSP Device Management Guide*.
- A file purge policy in the NSP is recommended; see “How do I configure file purge policies?” in the *NSP System Administrator Guide*.

1.2.4 Accounting telemetry catchup

If accounting files are already stored on the NE when NSP starts accounting telemetry collection, NSP performs a data backfill to pull and process these files. NSP supports data backfill of up to two hours of data.

If accounting telemetry data is output to the database and/or Kafka during data backfill, it may result in delayed catch up or data loss.

You can monitor the data backfill process in Grafana.

Use the **Grafana** button on the System Health dashboard to open Grafana, or access the tool at the following URL:

`https://NSP_address/grafana`

where *NSP_address* is the public NSP address

In Grafana, open the Telemetry Accounting Processor Metrics dashboard. When the Skipped File Pulls metric is zero, the data backfill is complete.

It is highly recommended to enable only the file output option in accounting telemetry subscriptions during the data backfill phase; see [2.8 “How do I manage subscriptions?” \(p. 36\)](#). After backfill is complete, the subscription can be edited to add additional output options as needed.

1.2.5 Accounting telemetry subscription best practices

An accounting telemetry subscription is configured to collect the accounting data from the NE and process it via NSP in up to three forms of outputs: file, Kafka, and database.

The following best practices are recommended to manage the data:

- Enable CN-Telemetry accounting subscriptions and use appropriate filters in the subscriptions to separate out the outputs.
Use NDI or NSP filters (instead of Null filters) for NE-based filtering.
- If needed, use the **Directory Path for File Output** parameter in the Create Subscription form to separate data collected by the subscription to a file for that subscription.
- Choose Kafka or File based output based on your needs:
 - Scale: an accounting processor supports 6 250 records/s for file output or 1 500 records/s for Kafka output
 - Data loss risk upon OSS disconnect from NSP:
Kafka output is subject to age-out - 24 h or 60 MBs of data per topic (subscription), whichever comes first. Risk of data loss increases with the data collection rate .
A file purge policy can be configured in the File Server settings when NSP manages file storage; see “How do I configure file purge policies?” in the *NSP System Administrator Guide*. The default is 24 h or 24 GBs, whichever comes first.
 - Use case:
If the only client is the OSS then either approach can be taken.
If the accounting data will be used for NSP applications, such as telemetry charting, or for event-driven third-party applications, Kafka output is required.
- Avoid editing, enabling, disabling, or deleting subscriptions where possible, because modification of the subscription causes disconnection and reconnection of SFTP sessions to the NE.
Frequent modification of the subscription in the scale environment may lead to performance degradation as it takes time to apply the modified configuration on all the applicable NEs.

Bulk operations are recommended for accounting file management in production deployments; see the Bulk Operations information on the [Network Developer Portal](#).

Nokia recommends the following for efficient and reliable bulk operations:

- Parallel requests: Up to 20 can be initiated for bulk operations.
- NEs per request: Use 3 000 to 5 000 NEs per request for balanced performance. When running parallel requests, ensure the combined total across all requests does not exceed 50 000 NEs.

-
- Purge policy: Set to a minimum of 3 hours for safe and consistent cleanup; see “How do I configure file purge policies?” in the *NSP System Administrator Guide*
 - Bulk operation timing: Trigger 20–30 minutes past the last accounting file collection hour to avoid overlap.
 - Storage maintenance: Remove the collated zip file of accounting files after each bulk data execution to prevent storage buildup.

1.2.6 Data available for collection

When NSP collects device telemetry, only data in statistics containers is available for collection. Furthermore, NSP may only make a subset of the attributes in the statistics containers available in its telemetry types. If you cannot find the counter you are looking for in the NSP telemetry types, verify that it is available in a statistics container in the relevant NE state YANG path.

For example, the SR Linux search tool can be found here: (<https://yang.srlinux.dev/>). Click on the NE release of interest and filter for the paths you need, as shown in the following example:

NOKIA

SR Linux v24.3.1 YANG Model

Tree Browser | Tree | Paths | Source

Path Browser

ethernet

AllStateConfig

Show prefixFilters

Platform

Search

7220-IXR-D5T

7220-IXR-H2

7220-IXR-H3

7220-IXR-H4

7250-IXR-10

7250-IXR-10E

7250-IXR-6

7250-IXR-6E

Reset

Features

Search

gnsi-nautiz

gnsi-certz

grpc-server

ingress-subif-policing

interface-evpn-internal-tags

interface-references

internal-tags

ip-acl-dscp-set

ip-qos-multifield

Reset

Note: Platform selection overrides already selected Features (if any)

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State	Path	Type
true	/interface[name=*/ethernet/statistics/in-1024b-to-1518b-frames	zero-based-counter64
true	/interface[name=*/ethernet/statistics/in-128b-to-255b-frames	zero-based-counter64
true	/interface[name=*/ethernet/statistics/in-1519b-or-longer-frames	zero-based-counter64
true	/interface[name=*/ethernet/statistics/in-256b-to-511b-frames	zero-based-counter64

For more information about the supported telemetry attributes for your NE family, see the artifact guide for the NE adaptor artifact bundle. For more information about telemetry definitions, see the Telemetry collection API documentation on the [Network Developer Portal](#).

1.2.7 Automatic subscriptions

Subscriptions are automatically created for Baseline Analytics, NSP Indicators, and OAM testing. To collect statistics for another purpose, [create a subscription](#) and set its state to enabled.

RESTCONF APIs are available for telemetry collection and aggregation; see the Network Assurance API documentation on the [Network Developer Portal](#).

1.3 What is an object filter?

1.3.1 Object filters

Object filters use XML Path (XPath) expressions against the NSP object YANG model to determine which objects a subscription, baseline, indicator, or chart applies to.

The object filter is combined with the telemetry type to create the list of relevant statistics data.

1.4 How do object filters work?

1.4.1 Creating an object filter

You need to enter an object filter during creation of [subscriptions](#), [baselines](#), and [indicators](#).

As you type in the Object Filter field, the field provides suggestions for available filters to match your input and identifies incorrect syntax.

For example, if you type `/nsp`, a dialog displays showing `nsp-equipment:network/network-element` and `nsp-service:services/service-layer`. Choose a suggestion and continue typing to generate more suggested elements to continue building the filter.

i **Tip:** Where available, you can use the plot utilization statistics option in the NE Inventory view to filter to NE objects.

From the NE Inventory page, expand the equipment tree to the object of interest. From the Table row actions menu () choose **Plot utilization statistics**. Data Collection and Analysis Visualizations opens in a new tab, showing a New Chart form. Click **Configure** in the New Chart form to see the object filter for the object. You can copy and paste this filter into your form.

1.4.2 Object filter components

An object filter is an XPath expression representing one or more network objects. The object filter indicates the objects to collect telemetry data from, such as an NE or a list of ports.

An XPath is composed of a series of XPath elements. Each element represents a step on the YANG model tree from left (root) to right (leaf or container). An element contains a name and a series of optional predicates.

Example: `network-element[ne-id='10.1.0.0']`

i **Note:** For accounting telemetry types, filtering is supported to the NE level; for example `network-element[ne-id='10.1.0.0']`

If an object filter including objects below the NE level is applied to a subscription for accounting telemetry, identifiers below the NE level are ignored.

1.4.3 Object filter evaluation

Object filters are evaluated every 15 min, based on object creation, deletion, or modification. The NSP discovers the change within 15 min, and telemetry data collection starts or stops accordingly.

1.4.4 Predicates

A predicate is an expression within square brackets that identifies the instance of each XPath element. Keys and attributes of the object (as defined in the NSP YANG model) may be included.

Predicates can contain the following operators:

- =
- >
- <
- >=
- <=
- !=
- and
- or

Subexpressions can be enclosed in parentheses to enforce the order of operations.

 **Note:** Comparison operators use string comparators, so “value2” is greater than “value1”.

Examples:

```
[ne-id='10.1.0.0' or ne-id='10.1.1.1']
```

```
[(type='7750 SR-12' and product='7750 SR') and (version='TiMOS-B 19.0.R1'  
or contains('version', '15.0'))]
```

1.4.5 Predicate functions

The following functions are supported within a predicate:

- **contains('attribute', 'search-text')**
This function searches for all objects where the named attribute contains the search text.
- **containsIgnoreCase('attribute-name', 'search-text')**
This function is the same as contains() but conducts a case-insensitive search.
- **not(<binary-expression>|<function-expression>)**
This function negates the result of the argument expression.
- **boolean ()**
A boolean function may be used to find objects by searching through an object's children and references.
The boolean function returns true if at least one of the specified object references or children satisfies the condition.
If a boolean function includes an instance-identifier object, the following requirements apply:
 - the instance-identifier must be the only object in the predicate
 - the operator in the predicate must be =
 - if there are multiple predicates, the format must be [predicate] [predicate]

- **equalsIgnoreCase(attribute, 'comparison-text')**
This function uses a case insensitive equality comparison between attribute value and comparison-text.
- **startsWith(attribute, 'prefix-text')**
This function matches the prefix-text against the start of the attribute value.
- **endsWith(attribute, 'suffix-text')**
This function matches the suffix-text against the end of the attribute value.
- **startsWithIgnoreCase(attribute, 'prefix-text')**
This function matches the prefix-text against the start of the attribute value, ignoring case.
- **endsWithIgnoreCase(attribute, 'suffix-text')**
This function matches the suffix-text against the end of the attribute value, ignoring case.

Examples:

- `[contains('component-id', 'port=1')]`
- `[not(contains('component-id', 'port=1'))]`
- `[contains('component-id', 'port=1') and not(contains('component-id', 'breakout'))]`
- To find all ports with ID 1/1/1/1 where mtu-value in the port-details has value 888:
`/nsp-equipment:network/network-element/equipment/port[contains(component-id, 'shelf=1/slot=1/card=1/port=1') and boolean(port-details[mtu-value = 888])]`
In this example, the boolean function returns `true` only if an mtu-value of 888 is found in the port details.
- To find UBT-T channel 1A ,port 6/1 on a specified Wavence NE:
`/network-device-mgr:network-devices/network-device[name='10.1.1.1']
/root/nokia-nsp-source:fdn[id='fdn:realm:sam:network: 10.1.1.1:shelf-1:cardSlot-1:card:port-6:radio-1']`
- To find a specified port on a specified Wavence NE:
`/network-device-mgr:network-devices/network-device[name='110.10.1.1']
/root/nokia-nsp-source:fdn[id='fdn:realm:sam:network: 10.10.1.1:shelf-1:cardSlot-1:card:port-3']`

1.4.6 XPath expressions

An XPath expression can be evaluated to produce a list of XPath identifiers. For the purpose of an object filter, the XPath expression must resolve to a NodeSet of NSP model identifiers.

XPath expressions can be chained together using a union “|” operator, which evaluates each subexpression independently into NodeSets and performs a set union operation to combine the NodeSets into a single NodeSet result.

Expressions can be static expressions, which resolve to the same result every time, or wildcard expressions, which can produce different results. The dynamic effect is created by using syntax that does not specifically identify a set of NSP model objects; there is no wildcard character.



Attention: The use of wildcards in a filter can lead to resource exhaustion and telemetry collection failure.

Use extreme caution when applying network-level wildcard filters, or use NE-level wildcard filters instead.

Example of a static expression:

```
/nsp-equipment:network/network-element[ne-id='10.1.0.0']  
/hardware-component/port[component-id='shelf=1/cardSlot=1/card=1/mdaSlot=  
1/mda=1/port=1/1/1' or component-id='shelf=1/cardSlot=1/card=1/mdaSlot=  
1/mda=1/port=1/1/2'] | /nsp-equipment:network/network-element[ne-id='10.  
1.0.1']/hardware-component/port[component-id='shelf=1/cardSlot=1/card=  
1/mdaSlot=1/mda=1/port=1/1/3' or component-id='shelf=1/cardSlot=1/card=  
1/mdaSlot=1/mda=1/port=1/1/4']
```

Examples of wildcard expressions:

Create a wildcard expression by not including all keys in the predicates.

- The following expression chooses all objects of type `port` on NE 1.1.1.1:

```
/nsp-equipment:network/network-element[ne-id='1.1.1.1']  
/hardware-component/port
```

- The following expression chooses all objects of type `port` for all NEs:

```
/nsp-equipment:network/network-element/hardware-component/port
```

- The following expression returns all child objects that can respond to the selected telemetry type and counters:

```
/nsp-equipment:network/network-element[ne-id='10.1.0.0']
```

1.4.7 Filtering models

Paths for MDM-managed NEs are based on the NE YANG models.

Equipment model

The **nsp-equipment:network** model is a normalized NSP YANG model that provides a view of the equipment for all network elements, abstracting the device specifics. Using NSP equipment model identifiers is recommended because the models are supported by NSP applications. In addition, filters based on the NSP equipment model can use operators such as AND or CONTAINS.

For example, the following path represents an individual port: `/nsp-equipment:`

```
network/network-element[ne-id='10.1.0.0']/hardware-component/port[name=  
'1/1/1']
```

Equipment model information can be found using the Network Inventory API; see the Postman collection on the [API Documentation page](#).

To assemble a filter, find the relevant model details and add predicates and predicate functions as needed. For example, a `get port` operation provides the path to the port.

Device model

The NE YANG model used by NSP is **network-device-mgr**.

- This model uses device model Xpath notation—this means that the filter can support all the paths as implemented by the node. Creating a path using the device model is useful when the object is not supported by, or not defined in, the NSP model.
- The device model provides a view of the device-specific paths. For example, an SR OS NE defines an LSP object using a path similar to `/network-device-mgr:network-devices/network-device[name='10.1.0.0']/root/nokia-state:state/router[router-name='Base']/mpls/statistics/lsp-egress[lsp-name='LSP1']`.
The telemetry mapping artifact must contain a definition for the device path for telemetry to be collected.

The following is an example of a **network-device-mgr** model filter type for an NFM-P-managed device. This filter targets a specific object instance by its FDN:

```
/network-device-mgr:network-devices/network-device[name='10.1.0.0']  
/root/nokia-nsp-source:fdn[id='fdn:realm:sam:lsp:from-11.50.150.30-id-2']
```

Service model

The **nsp-service:services** model is a normalized NSP YANG model that provides a view of the services for all network elements, abstracting the device specifics. Using NSP model identifiers is recommended because the models are supported by NSP applications. In addition, filters based on the NSP service model can use operators such as AND or CONTAINS.

See the following examples.

- ELAN:
 - The following expression chooses all endpoint objects belonging to an ELAN service with ID `my_Vpls`:
`/nsp-service:services/service-layer/elan[service-id='my_Vpls']
/endpoint`
 - The following expression chooses a specific endpoint belonging to an ELAN service with ID `my_Vpls`:
service endpoint: `/nsp-service:services/service-layer/elan[service-id='my_Vpls']/endpoint[endpoint-id='92.168.96.26-1/1/9:1']`
- Eline
 - The following expression chooses all endpoint objects belonging to an Eline service with ID `my_Epipe`:
`/nsp-service:services/service-layer/eline[service-id='my_Epipe']
/endpoint`
 - The following expression chooses a specific endpoint belonging to an Eline service with ID `my_Epipe`:
`/nsp-service:services/service-layer/eline[service-id='my_Epipe']
/endpoint[endpoint-id='92.168.96.26-1/1/9:2']`
- IES
 - The following expression chooses all endpoint objects belonging to an IES service with ID `my_ies`:
`/nsp-service:services/service-layer/ies[service-id='my_ies']/endpoint`

- The following expression chooses a specific endpoint belonging to an IES service with ID `my_ies`:
service endpoint: `/nsp-service:services/service-layer/ies[service-id='my_ies']/endpoint[endpoint-id='92.168.96.26-my_ies-intmy_nodeC']`
- L3VPN
 - The following expression chooses all endpoint objects belonging to an L3VPN service with ID `my_Vprn`:
`/nsp-service:services/service-layer/l3vpn[service-id='my_Vprn']`
`/endpoint`
 - The following expression chooses a specific endpoint belonging to an L3VPN service with ID `my_Vprn`:
`/nsp-service:services/service-layer/l3vpn[service-id='my_Vprn']`
`/endpoint[endpoint-id='92.168.96.26-my_Vprn-intmy_nodeC']`
- The following expression chooses all endpoint objects belonging to an Eline service with ID `my_Epipe` on a particular NE. Note that this relies on the endpoint-ID following the convention that it begins with the site ID followed by a dash.
`/nsp-service:services/service-layer/eline[service-id='my_Epipe']`
`/endpoint[startsWith(endpoint-id, '92.168.96.26-')]`
- The following expression chooses all endpoint objects belonging to all Eline services on a particular NE. Note that this relies on the endpoint-ID following the convention that it begins with the site ID, followed by a dash.
`/nsp-service:services/service-layer/eline/endpoint[startsWith(endpoint-id, '92.168.96.26-')]`



Important! Nokia recommends that telemetry subscription filters referencing service objects should not evaluate to more than 10 000 objects.

If no filter is included in the telemetry subscription, all service objects for all services on all sites would be selected. This is a bulk subscription that may overburden the NSP collector and should be used with discretion.

1.4.8 NE mode considerations

Equipment modeling may differ between MD and classic equipment. You may need to use `contains` or `or` statements to create mediation-agnostic filters.

Examples:

SR OS NE with chassis type SR 12 on an MD-managed NE: `type = '7750 SR-12'`

SR OS NE with chassis type SR 12 on a classic managed NE: `type = '7750-SR12'`

Port, on an MD-managed NE: `component-id='shelf=1/cardSlot=1/card=1/mdaSlot=1/mda=1/port=1/1/1'`

Port, on a classic managed NE: `component-id='shelf=1/slot=1/card=1/slot=1/card=1/port=1'`

1.4.9 Object filter examples

The following table shows some example object filters. You can use these as templates to create a similar filter.

Object filter	Description
/nsp-equipment:network/network-element[type = '7750 SR-12' or type = '7750-SR12']	All 7750 SR OS network elements with chassis type SR-12, regardless of NE mode.
/nsp-equipment:network/network-element[contains(ne-name,'SS-MD-INT-VDR') and (contains(ne-id,'22.8'))]	An NE with a name containing "SS-MD-INT-VDR" and an NE ID containing "22.8"
/nsp-equipment:network/network-element[contains(ne-name,'SIMPLESFT-7250IXRE-RG') and (startsWith(ne-id,'2.1.') or startsWith(ne-id,'2.2.') or startsWith(ne-id,'2.3.') or startsWith(ne-id,'2.4.') or startsWith(ne-id,'2.5.'))]/hardware-component/port[endsWith(component-id,'port=1')]	A port with a port number that begins with a 1, on an NE that meets the following criteria: - NE name contains "SIMPLESFT-7250IXRE-RG" - NE ID starts with 2.1, 2.2, 2.3, or 2.4
/nsp-equipment:network/network-element[ne-id='2.5.230.5']/hardware-component/port[component-id='shelf=1/slot=1/card=1/slot=1/card=1/port=1']	Port 1 on shelf 1, slot 1, card 1, slot 1, card 1, on the NE with NE ID 2.5.230.5
/nsp-equipment:network/network-element[contains(ne-name,'SIMPLESFT-7250IXRE-RG') and (startsWith(ne-id,'2.1.') or startsWith(ne-id,'2.2.') or startsWith(ne-id,'2.3.') or startsWith(ne-id,'2.4.') or startsWith(ne-id,'2.5.'))]/hardware-component/port[component-id='shelf=1/slot=1/card=1/slot=1/card=1/port=1']	Port 1 on shelf 1, slot 1, card 1, slot 1, card 1, on an NE that meets the following criteria: - NE name contains "SIMPLESFT-7250IXRE-RG" - NE ID starts with 2.1, 2.2, 2.3, or 2.4
/nsp-equipment:network/network-element[(contains(ne-name,'VSR') and (ne-id >= '10.231.0.3' and ne-id <= '10.231.0.9' or ne-id >= '10.52.12.192' and ne-id <= '10.52.140.192')) or (contains(ne-name,'-RR-') and ne-id >= '10.231.0.3' and ne-id <= '10.231.0.9')]	An NE that meets one of the following criteria: - NE name contains "VSR" AND - NE ID is in one of the specified ranges: 10.231.0.3 to 10.231.0.9 10.52.12.192 to 10.52.140.192 OR - NE name contains "-RR-" AND - NE ID is in the range 10.231.0.3 to 10.231.0.9
/nsp-equipment:network/network-element[ne-id='42.1.2.3']/hardware-component/port[component-id='shelf=1/cardSlot=1/card=1/mdaSlot=1/mda=1/port=1/1/4' or component-id='shelf=1/cardSlot=1/card=1/mdaSlot=1/mda=1/port=1/1/7']	One of the following ports on the NE with NE ID 42.1.2.3: - Port 1/1/4 on shelf 1, cardSlot 1, card 1, mdaSlot 1, mda 1 OR - Port 1/1/7 on shelf 1, cardSlot 1, card 1, mdaSlot 1, mda 1
/nsp-equipment:network/network-element[contains(ne-name,'SS-MD-INT-VDR')]/hardware-component/port[component-id='shelf=1/cardSlot=1/card=1/mdaSlot=1/mda=1/port=1/1/4']	Port 1/1/4 on shelf 1, cardSlot 1, card 1, mdaSlot 1, mda 1, on an NE with a name that contains "SS-MD-INT-VDR"
/nsp-equipment:network/network-element[contains(ne-name,'SS-MD-INT') and ((startsWith(ne-id,'62.') or startsWith(ne-id,'42.')) and contains('.1.2.'))]/hardware-component/port[component-id='shelf=1/cardSlot=1/card=1/mdaSlot=1/mda=1/port=1/1/11']	Port 1/1/11 on shelf 1, cardSlot 1, card 1, mdaSlot 1, mda 1, on an NE that meets the following criteria: - NE name contains "SS-MD-INT" AND - NE ID starts with 62 or 42, AND contains .1.2.

Object filter	Description
/nsp-equipment:network/network-element[not(contains(ne-name,'ist_'))]	Filters out NEs that contain "ist_" in their NE Name

1.5 What information can I find in the Telemetry Statistic Search Tool?

1.5.1 Telemetry Statistic Search Tool

The Telemetry Statistic Search Tool lists telemetry types available for statistics collection. There are two versions of the tool, one available offline which contains a complete database of telemetry statistics, and one available on-product which is dynamically updated and only contains statistics relevant to your network.

The offline tool is available for download on the Nokia Documentation Center as a part of the user documentation collection, and is hosted on the online documentation suite under the Additional Resources heading.

The on-product tool is found in the NSP Help Center. From the Help Center home screen, navigate to **Reference, Statistic Search, Telemetry Statistic Search**. Telemetry types are added to the tool when the required feature packages, adaptor files, and telemetry artifacts are installed on the NSP.

Telemetry types are added to the on-product tool when the required feature packages, adaptor files, and telemetry artifacts are installed on the NSP. Click the About tab in the tool to see which artifacts have contributed statistics to the tool.

Installation of the following files may be required for a telemetry type to appear in the tool, depending on the telemetry type:

- adaptation artifact bundles for an NE type
- device telemetry definition YANG bundles
- device telemetry (ct) adaptation bundles

1.6 What charts can I create in NSP?

1.6.1 Data Collection and Analysis Visualizations

NSP charts allow you to see telemetry data collection in real time.

The following views are available, depending on the feature packages installed. You can navigate from one view to another by choosing from the drop-down list at the top left.

The chart views display saved charts. You can plot a saved chart or create a new one.

The views display information in columns. Drag a column header to reorder the columns, drag a column header separator to resize columns, or click on a column header to sort the data in the column. An icon appears when the column is sorted.

Data Collection and Analysis Visualizations, Telemetry

- Telemetry Charts

Data Collection and Analysis Visualizations, Baseline Analytics

- Baseline Charts

-
- Anomalies

Data Collection and Analysis Visualizations, NSP Indicators

- Indicator Charts
- Threshold Crossing Events

You can click  (Table settings & actions) in the column header on any view to manage the display or export data.

Filter the list using the column header filters. To remove a filter, delete information from the filter field or choose **Clear filters** from the  (Table settings & actions) menu in the column header.

For information about charts created from Analytics reports, see the *NSP Analytics Report Catalog*.

2 Telemetry

2.1 What artifacts support telemetry?

2.1.1 Telemetry artifacts

Telemetry artifacts include vendor agnostic device telemetry bundles, device telemetry bundles, accounting telemetry bundles, and telemetry YANG definition files. Vendor agnostic and accounting telemetry artifacts are included in the adaptation artifact bundle for an NE type.

There is no dependency between vendor agnostic device telemetry bundles and device telemetry bundles, and no requirement to install both. If the telemetry counters you need to collect are vendor agnostic, you can install only the adaptation artifact bundle.

 **Note:** Device telemetry artifacts are for gNMI-capable devices only.

Device telemetry files may have dependency on telemetry YANG files. Before you install a device telemetry bundle, verify the files in the telemetry YANG collection to see if the counters you need are present in any YANG file. If you are installing telemetry YANG files and a device telemetry bundle, the YANG files must be installed first.

2.2 How do I prepare for telemetry collection?

2.2.1 Purpose

Use this procedure to ensure that all the telemetry types and counters you need are available in your NSP.

This procedure assumes that devices you want to collect telemetry from are already managed.

2.2.2 Telemetry artifacts

Telemetry artifacts include vendor agnostic (that is, NSP) device telemetry artifacts, device telemetry adaptation bundles, and telemetry YANG definition files. There is no dependency between NSP and device-specific telemetry definitions, and no requirement to install both. If the telemetry counters you need to collect are present in the NSP telemetry definitions, you do not need device-specific telemetry.

Device telemetry adaptation is dependent on telemetry YANG definition files. Before you install a device telemetry adaptation bundle, review the telemetry definitions in the YANG collection to see if the counters you need are present. If the required counters are present in the YANG files, that is, if the counters are supported, the YANG files must be installed before the device adaptation bundle.

Nokia recommends that you only install the telemetry YANG files required for your use cases.

2.2.3 Steps

1

Verify that you have obtained all the artifact bundles required for telemetry for your NEs.

2

Starting with Release 25.8, vendor agnostic telemetry artifacts are packaged in adaptation artifact bundles, therefore they are already present when you are managing the device.

However, in certain cases, specific vendor-agnostic telemetry adaptation bundles are available and may need to be installed individually; see the artifact guide. If a vendor agnostic telemetry bundle is required, install it; see “How do I install an artifact bundle?” in the *NSP Network Automation Guide*.

3

Use the Telemetry Statistic Search Tool to see if the telemetry types you need are installed with the NSP (vendor agnostic):

1. Open the NSP Help Center. From the Help Center home screen, navigate to **Reference, Statistic Search, Telemetry Statistic Search**.
2. Enter search terms in the Counter Name field to filter on the counters you need. Enter additional filters as needed to narrow the search.
3. Click on each row to see the Info panel. Verify whether the selected statistic is the one you need.
4. If the telemetry types you need are found, they are vendor-agnostic telemetry types and the required adaptation is present. This procedure is complete.
5. If the telemetry type is not found, proceed to the next step.

4

Extract the files from the YANG telemetry zip and look for the telemetry types:

1. View the list of telemetry files. The filenames provide information about the telemetry types included in the file.
2. Open the file and search for terms relevant to the counters you need. A counter, where present, will be found in a leaf element. A list of telemetry types is provided in the leaf.
3. If you find a telemetry type you need, install the YANG file using **yang-file.bash**; see “How do I manage MDM model definitions?” in the *NSP System Administrator Guide*.
4. If you do not find the definition or mapping you require, or you need to augment an existing telemetry type, contact your Nokia service representative.

5

Obtain any required device telemetry adaptation bundles from the Nokia [Support Portal](#) and install them from the **Artifacts, Artifact Bundles** view; see “How do I install an artifact bundle?” in the *NSP Network Automation Guide*.



Note: Device telemetry adaptation bundles are for gNMI-capable devices only.

6

Open the Telemetry Statistic Search Tool and search for your desired telemetry types to confirm that all are present.

END OF STEPS

2.3 What is a subscription?

2.3.1 Subscription

A subscription represents a request for statistics information management.

2.3.2 User-configured subscriptions

You can configure telemetry subscriptions.

For an NE managed by NFM-P, the statistics you need must be collected by the NFM-P.

2.3.3 Baseline and Indicator subscriptions

Baseline and indicator subscriptions are managed by Baseline Analytics and NSP Indicators respectively, and are read-only in the **Data Collection and Analysis Management, Subscriptions** view.

Baseline Analytics is available with the Network Operation Analytics feature package, and NSP Indicators is available with the Performance Indicators and Alerts installation option.



Note: To use Baseline Analytics and NSP Indicators with model-driven NEs, the gnmiTelemetry installation option is also required.

2.3.4 OAM subscriptions

NSP generates predefined OAM-related telemetry subscriptions in systems where MD-OAM is deployed. OAM subscriptions appear in the Telemetry subscriptions list, prefixed with **TestSuiteEx_OAM**.

OAM subscriptions are used by NSP OAM to receive and publish results for OAM tests. NSP creates these subscriptions when an appropriate type of test is executed, and also when an executing test is discovered in the network. These subscriptions should never be deleted. If they are deleted, OAM will encounter problems and may no longer produce or record OAM test results.

The following parameters may be edited:

- collection interval - if there is a need to change the collection interval for streaming results
- file prefix and file subscriptions - if there is a need to record results to a file

Do not edit other parameters in an OAM subscription.

NFM-P for OAM

NFM-P can also provide streaming results for nodes that support this. NFM-P can also produce results that are not streamed from the node. In this case, OAM obtains these results from the NFM-P result topic in kafka, and publishes them to the appropriate OAM subscription. There is a

NFM-P results mapping file that defines how this is done. Streaming results are obtained by subscribing to the node using GRPC.

In addition to streaming results, results are obtained by other mechanisms:

- Accounting results are obtained from accounting files on the node. These files sent to the NSP via FTP, parsed, and results are published to telemetry on the OAM subscriptions.
- CLI tests (loopback, linktrace) are executed and the results read from the CLI output. This output is parsed and results created and published to telemetry on the OAM subscriptions.

2.4 How do subscriptions work?

2.4.1 Statistics collection requirements

When you configure a telemetry subscription, you specify a set of statistics data and notification and storage options for the data. To configure the data set, select the type of statistics you are interested in, the frequency of collection, and configure filtering by NE and counters as needed.

Statistics collection requirements and behavior depend on the NE management type and statistics type, as described in the following table.

Telemetry type	Statistics configuration
MIB statistics for classic NEs	Statistics collection and database storage must be configured in NFM-P before creating a subscription in NSP. The subscription tells NFM-P to publish any statistics that match subscription requirements to Kafka as they are collected. ¹ The interval specified in the subscription request is ignored.
CN telemetry using gRPC	When the subscription is created, the NSP initiates collection of the telemetry data from the NE at the specified interval.
MDM telemetry for statistics not supported by CN telemetry, on model driven NEs	When the subscription is created, the NSP initiates collection of the telemetry data from the NE at the specified interval.
Accounting telemetry	Accounting telemetry is supported using CN telemetry.

Notes:

1. If statistics collection is set up in both NFM-P and NSP, the telemetry framework may receive the same information twice. The duplication could result in incorrect reports or duplicated indicator TCAs. Verify that the data being used for telemetry, baseline, or indicator subscriptions is only collected once. See “Troubleshooting duplicate data collection” in the *NSP Troubleshooting Guide* for more information.

2.4.2 Data storage

You can enable database storage for MDM- managed NEs as part of subscription creation. By default, the data collected is stored in Postgres unless there is an auxiliary database enabled, in which case all collected data is stored in the auxiliary database. For telemetry data that is collected by NFM-P, the database parameter in the subscription is ignored.

Historical data, that is, data that is stored in a database, is retained according to the age-out policy; see [2.15 “How do I edit an age-out policy?” \(p. 43\)](#).

i **Note:** For telemetry data to be available to the Analytics application for aggregated reports, it must be stored in the auxiliary database.

The Telemetry data API also provides the functionality to subscribe, stream, and plot historical and real-time data; see Use Case 5 in the Telemetry Collection tutorial on the [Network Developer Portal](#).

2.4.3 File subscriptions

You can enable file subscriptions to save accounting statistic information to a file. Files can be managed from the File Server; see [2.9 “How do I retrieve accounting files?”](#) (p. 37).

The directory path for file output is configured as part of subscription creation.

If the **Directory Path for File Output** parameter is left blank, accounting files will be saved to `accounting/output/ready/NE ID/date`.

If a path is entered, the path folders are created in the `accounting/output/ready` directory. For example, if you enter `new/files`, the path to the files is `accounting/output/ready/new/files/NE ID/date`.

The path can include any number of subdirectories.

i **Note:** Accounting files that are migrated from Release 25.8 or earlier are placed in `accounting/output/ready/`. These files are not organized by NE ID and date.

2.4.4 Kafka notifications

You can enable notifications in the subscription to send notifications to a Kafka topic. You can configure a custom topic name if needed, or allow a topic name to be generated automatically. Custom topic names always begin with `ns-eg-tel-`. For example, if you enter `my-subscription` in the custom notification parameter, the topic will be `ns-eg-tel-my-subscription`.

If a subscription with a custom topic name is deleted, the topic name can be used again by a new subscription.

i **Note:** Custom topic names can only be configured when the subscription is created, and cannot be edited.

2.4.5 Persistent Kafka Topics

Once a Kafka topic has been associated with a telemetry subscription, it remains associated until the telemetry subscription is deleted. If the subscription is disabled, the topic continues to appear in the NSP UI. However, to preserve resources, if the subscription has been disabled for more than five min, the notifications and information related to the topic are deleted from Kafka.

Telemetry subscriptions are disabled automatically by the system under a number of scenarios, including YANG file deletion, Kafka being unavailable or another issue that results in loss of NBI topics.

If a topic is deleted while a consumer, such as an OSS, is consuming it, an exception is thrown. To avoid this error, ensure that no consumers are monitoring the topic before disabling or deleting the subscription.

2.5 What is a telemetry chart?

2.5.1 Telemetry charts

A telemetry chart plots historical information if available, then streams real-time data on the same chart.

For example, you can configure a chart to plot data from the last 12 hours, followed by current data. This information can be useful in assessing changes.

i **Note:** Historical data must be available, meaning data must have been collected and retention policies must allow for the data to remain in the database.

2.6 How does telemetry charting work?

2.6.1 Telemetry charting

When you configure a telemetry chart in Data Collection and Analysis Visualizations, a new telemetry data subscription is created to request historical and/or real-time telemetry, as required by the chart configuration.

If historical telemetry data is requested, the data is retrieved from the database. Historical data must already be available in the database.

i **Note:** The telemetry data subscription requests data from the database based on the requirements configured for the chart, not on the subscription that collected the data.

If real-time telemetry data is requested, a new telemetry subscription is automatically created to request the data from the NEs. The telemetry chart plots data from the subscriptions created for plotting.

If you open Data Collection and Analysis Visualizations by cross-launching from a telemetry subscription in the **Data Collection and Analysis Management, Subscriptions** view, the New Chart form displays the parameters configured in the telemetry subscription you cross-launched from. When you click **Plot**, Data Collection and Analysis Visualizations creates telemetry subscriptions as needed to collect the data, and plots the data from the subscriptions it creates.

The subscriptions created for plotting are not saved to the database and do not create a file output.

2.7 How do I configure accounting collection settings?

2.7.1 Purpose

Use this procedure to configure collection of accounting files in NSP. Perform this procedure as needed before [creating a subscription](#) for accounting telemetry.

Accounting telemetry data provides packet and octet throughput information for queues that are associated with the following objects:

- SAPs or SDPs, which provide service accounting telemetry data
- network ports, which provide network accounting telemetry data
- subscriber profiles, which provide subscriber accounting telemetry data



Note: Updates to accounting collection settings may take up to 5 min to take effect.

2.7.2 Queue filters

In the accounting collection settings, you can configure up to 60 queue filters. The NEs generate accounting data files that include all of the configured queues. When a queue filter is configured, the statistic records for the queues that do not match the filter are discarded and not processed.

2.7.3 Policy ID insertion

You can configure the insertion of the accounting policy ID in the accounting file name. Policy ID insertion may help prevent duplication of accounting filenames. Insertion is disabled by default.

- If the Policy ID insertion option is disabled the filename format is:

```
<NE ID>_<dateTimeStamp>_<sequence number>.json.gz
```

- If the Policy ID insertion option is enabled the filename format is:

```
<NE ID>_<dateTimeStamp>_accountingPolicyId<ID>_<sequence number>.  
json.gz
```

The sequence number is a incremental number within the context of an accounting processor pod. The sequence number is reset to 0 on pod restart.

2.7.4 Settings to configure for accounting collection from classic NEs

If you will be collecting accounting telemetry data for classic NEs, selective accounting file processing must be configured; see “To configure selective accounting file processing” in the *NSP Classic Management User Guide*.

2.7.5 Steps

1

Open **Data Collection and Analysis Management, Subscriptions**.

2

Click (More Actions), **Accounting collection settings**

3

In the form that opens, choose the **Policy ID insertion** check box to include the accounting policy ID in the accounting files.

4

To add queue filters:

1. Click **+Queue Filter**.
2. In the form that opens, enter information in the **Telemetry type** field. As you type, the field filters for available telemetry types to match your input.
3. In the Queue numbers field, enter up to 60 queue numbers, separated by commas.
4. Click **Add**.

5

Click **Update**.

END OF STEPS

2.8 How do I manage subscriptions?



CAUTION

Service Disruption

The name of an object, including subscriptions, baselines, indicators, templates, and chart profiles, cannot contain a semicolon (;) or backslash (\).

The use of these characters in an object identifier will result in corrupted data that must be deleted by Nokia support.

2.8.1 Steps

1

Open **Data Collection and Analysis Management, Subscriptions**.

2

To create a subscription:

1. Click **+Subscription**.
2. In the Create Subscription form that opens, configure the General parameters as needed.
 - Enable database (DB) subscriptions as needed to save telemetry data to the NSP database.

For NSP deployments without an auxiliary database, telemetry data is stored in the NSP database. For NSP deployments with an auxiliary database, telemetry data is stored in the auxiliary database. For telemetry data to be available to Analytics, the auxiliary database must be deployed.
 - Enable file subscriptions as needed to save accounting statistic information to a file.
 - In the **Directory Path for File Output** field, enter a path for file output if needed.

If the field is left blank, accounting files will be saved to `accounting/output/ready/NE ID/date`.

If a path is entered, the path folders are created in the `accounting/output/ready` directory. For example, if you enter `new/files`, the path to the files is `accounting/output/ready/new/files`.

The path can include any number of subdirectories.
 - The subscription is enabled by default. It will start running immediately.

Choose **Disabled** in the **State** field if you want to enable your subscription later.
3. In the **Object Filter** field, enter filtering information as needed to filter the collected data.

As you type, the field provides suggestions for available filters to match your input and identifies incorrect syntax.

4. Enter information in the Telemetry Type field. As you type, the field filters for available telemetry types to match your input.
Choose the telemetry type you need from the list of matches.
When you enter a telemetry type, all counters are enabled by default.
5. To customize the counters, enable the **Enable notifications and notification counters** check box.
Click **Remove**  to remove a counter.
Click **+Counters** to add a counter that was removed.
Note: If database subscriptions are enabled, all counters are collected and persisted in the database, regardless of counter selection in the Counters list.
6. To customize the topic name for Kafka notifications, enable the **Specify a custom notification topic** check box.
Enter the name for your topic in the field. The name of the topic will be ns-eg-tel-*entry*, where *entry* is the text you entered.
7. Click **Create**.
The subscription begins collection immediately, unless it was disabled in step 2.

3

To edit a subscription:

1. Choose a subscription and click  (Table row actions), **Edit**.
2. In the form that opens, update the parameters as needed and click **Update**.



Note: Custom notification topic names cannot be edited.

4

To delete a subscription:

Choose a subscription and click  (Table row actions), **Delete**.

The subscription is removed immediately.



Note: Removing a subscription does not remove data from the database. The data collected by the subscription is retained according to the database retention policy. If a custom notification topic is in use, deleting the subscription deletes the topic, allowing the topic name to be used by a new subscription.

END OF STEPS

2.9 How do I retrieve accounting files?

2.9.1 Purpose

Use this procedure to locate accounting files in the NSP UI.

To prevent excessive disc consumption, a file purge policy in the NSP is recommended; see “How do I configure file purge policies?” in the *NSP System Administrator Guide*.

2.9.2 Kafka

To retrieve accounting files based on Kafka notifications, monitor the `accounting-file-ready` topic. See the following sample notification:

```
{
  "data":
  {
    "ietf-restconf:notification":
    {
      "eventTime": "2024-02-22T07:08:50Z",
      "telemetry:file-available":
      {
        "input-file-name": "act5566_20240112-094501.xml.gz",
        "ne-Id": "fded:fade:face:ffff::223",
        "output-file-name": "pathname/fded:fade:face:ffff::
223-20240222-070850.json.gz",
        "telemetry-subscription-name": "dmmbin"
      }
    }
  }
}
```

2.9.3 Steps

- 1 _____
Open **File Server**.
- 2 _____
Click on the accounting folder in the Root Directory List. Double-click on each subfolder to navigate through the path.
- 3 _____
To download a file, select the file and click  (Table row actions), **Download File**.

END OF STEPS _____

2.10 How do I plot a telemetry chart?



CAUTION

Service Disruption

The name of an object, including subscriptions, baselines, indicators, templates, and chart profiles, cannot contain a semicolon (;) or backslash (\).

The use of these characters in an object identifier will result in corrupted data that must be deleted by Nokia support.

2.10.1 Before you begin

When you create a telemetry chart, you configure a telemetry filter. For historical data to be displayed, the data must be available in the database; see [2.8 “How do I manage subscriptions?” \(p. 36\)](#).

Charts are created by streaming to the plotter. Historical data is queried and streamed to the plotter, then real-time telemetry subscriptions are created and the data from these subscriptions is streamed to the plotter.

Data Collection and Analysis Visualizations times out if telemetry data is not received. The time-out limit is either double the collection interval or two min, whichever is greater.

Chart limit

Up to 10 objects can be charted at a time. The number of objects is the number of resources returned by the object filter, multiplied by the number of counters.

If your object filter returns one resource, for example, one NE, you can chart up to 10 counters for the resource.

2.10.2 Steps

Create a chart

1

Open the New Chart window:

- From **Data Collection and Analysis Visualizations, Telemetry Charts**, click **+ Chart**.
- From **Data Collection and Analysis Management, Subscriptions**, choose a subscription and click  (Table row actions), **Open in Data Collection and Analysis Visualizations**.

2

In the window that opens, configure the parameters in the top panel:

1. Configure the **Collection Interval** parameter. If you are using NFM-P telemetry data, verify that the collection interval is long enough to allow time for Visualizations to receive the data before timing out.

2. From the **Time Range** drop-down list, choose the amount of historical data to display.
3. Click **Combine charts** to plot data from multiple data series on the same chart.

3

Click **+Definition**.

The telemetry and resource filter definition panels are displayed.

4

Enter information in the **Telemetry Type** field. As you type, the field filters for available telemetry types to match your input.

Choose the telemetry type from the list of matches.

5

Choose counters to display from the **Counters** drop-down list.

6

In the **Object Filter** field, enter filtering information to filter the collected data.

7

If you need to save the configuration as a chart:

1. Click **Save As**.
2. In the window that opens, enter a name for the chart and add a description if needed.
3. Click **Save**.

The chart is added to the list.

8

Click **Plot**.

END OF STEPS

2.10.3 Steps

Plot an existing chart

1

To plot an existing chart with no changes:

1. Open **Data Collection and Analysis Visualizations, Telemetry Charts**.
2. Choose a chart and click  (Table row actions), **Chart**.

2

To edit a chart and plot it, choose the chart and click  (Table row actions), **Edit**.

3

Edit the parameters as needed and click **Plot**.

END OF STEPS

Result

Visualizations displays a chart view showing the streaming data. While data is streaming, you can configure the **Group by** parameter in the upper left of the chart view to change how the data is grouped or click **Configure** in the upper right to view or change the configuration of the chart.

Click ⓘ(Chart Details) to open the Chart Details panel on the right side of the chart view to show details about the resources.

2.11 What is aggregation?

2.11.1 Aggregation

Aggregation is the practice of combining measurements for analysis. In the NSP, telemetry data is aggregated based on the time period of collection: hourly, daily, weekly, or monthly. When a report is run in Analytics, the granularity parameter in the report determines which aggregated data set is analyzed. For example, if you select hourly granularity, the report shows the hourly aggregated data. Some reports can use raw data, that is, data that is not aggregated, but this is not available for all reports.

2.11.2 What is an aggregation rule?

An aggregation rule defines how the data for a specified telemetry type is aggregated and how long each aggregated data set is retained.

Aggregation rules for model-driven telemetry are configured on the NSP server. They can be [viewed and edited](#) in NSP.

For aggregation information to be available, the auxiliary database must be configured and running.

Use the NFM-P to configure aggregation rules for data collected by the NFM-P; see the *NSP Analytics Report Catalog*.

2.11.3 What is the aggregation time zone?

The aggregation time zone is the time zone that is used to determine the start and end time of data to be aggregated:

- hourly aggregation: from xx:00 to xy:00
- daily aggregation: from midnight to midnight
- weekly aggregation: from midnight Monday to midnight Monday
- monthly: from midnight on the first of the month until midnight on the first of the following month

2.12 How do I set the aggregation time zone?

NOTICE

Changing the aggregation time zone changes the start and end times of all future aggregations. The first aggregation following the change may miss data or count data twice due to the difference in time zones.

2.12.1 Steps

- 1 _____
Open **Data Collection and Analysis Management, Aggregation**.
- 2 _____
Click  (More Actions), **Time Zone Setting**.
- 3 _____
In the form that opens, enter a time zone in the **Aggregation Time Zone** field.
As you type, the list of available time zones is refined to match your input.
- 4 _____
Choose the time zone from the list.
- 5 _____
Click **Ok** in the confirmation dialog.
- 6 _____
Click **Save**.

END OF STEPS _____

2.13 How do I edit aggregation information?



Note: For the Aggregation view to be available, the auxiliary database must be configured and running.

2.13.1 Steps

- 1 _____
Open **Data Collection and Analysis Management, Aggregation**.
- 2 _____
Choose an aggregation rule and click  (Edit).

3

In the form that opens, edit the parameters as needed and click **Update**.

END OF STEPS

2.14 What is an age-out policy?

2.14.1 Age-out policy

An age-out policy is a retention policy for stored data. Data older than the retention period is purged from the database.

NSP creates a policy for each telemetry type, a policy for baselines, and a policy for indicators.

The following table describes the default and maximum age-out values for different policy types. For the baseline and indicator age-out policies, the default and maximum values depend on the data storage type, Postgres or AuxDb.

Age-out policy type	Data storage type	Default age-out value	Maximum age-out value
Telemetry policies	Postgres	4 h	9672 h (403 days)
	AuxDb	24 h	9672 h (403 days)
Baseline policy and Indicator policy	Postgres	35 days	35 days
	AuxDb	90 days	403 days

2.15 How do I edit an age-out policy?

2.15.1 Steps

1

Open **Data Collection and Analysis Management, Age-Out Policy**.

2

Open the Edit form:

- To edit a telemetry policy, choose Telemetry from the drop-down list, choose a policy and click  (Edit).
- To edit the baseline or indicator policy, choose the policy type from the drop-down list.

3

Edit the retention period and click **Update**.

END OF STEPS

3 Flow statistics collection

3.1 Flow statistics collection in the NSP

3.1.1 Introduction

The NSP can collect flow statistics from managed NEs. Although other criteria may apply, a flow is basically a series of IP packets that share a common source, destination, and type of payload, for example, traffic that is specific to an application.

When an NE is configured to collect flow statistics, the NE monitors the traffic on one or more interfaces to identify the flows, aggregates the flow data, and regularly exports flow data records to an external system.

The NSP uses NSP Flow Collectors to collect the following flow statistics from managed NEs:

- IPFIX v10 (RFC 7011), with the following Nokia variants:
 - System Cflowd
 - CGNAT
- Netflow v5



Note: CGNAT support remains restricted to the 7750 SR, for Format 1 and Format 2 CSV records only. See [3.6 "Flow statistics" \(p. 54\)](#).



Note: The Flow Collector API can be used to collect AA Cflowd records. AA configuration, such as AA applications, groups, or subscriber types, must be performed using NFM-P; see the *NFM-P Statistics Management Guide*.

3.1.2 Functional description

IPFIX data is sent from routers, packaged in IPFIX messages that contain IPFIX records. Each record is a collection of fields determined by an IPFIX template. These templates are periodically sent by the routers to the collectors. IPFIX templates can be modified (fields can be removed from or added to the records). A flow collector uses these templates to decode the IPFIX records.

The fields contained in a record may include - but are not limited to - the following:

- counters
- timestamps
- delays
- quantities
- labels
- IDs

The Flow Collector generates "data types" using sets of counters that are found in IPFIX records and/or values computed based on IPFIX records. The Flow Collector collects these data types at

configured intervals, then exports the collected data in various formats (CSV text files, Kafka notifications, auxiliary database records).

Each data type can be aggregated in one or more ways by applying an "aggregation type" to it. Each aggregation type is defined by a set of fields from the IPFIX record (usually IDs). When no aggregation is applied the Flow Collector refers to the data as "raw" (raw aggregation). During each interval, aggregated data is collected in a memory buffer, and is exported in one of the supported formats when the interval ends.

Collected data can be exported in the following formats:

- System Cflowd
 - auxiliary database
- CGNAT
 - CSV text files
- Netflow v5
 - auxiliary database

3.1.3 Flow Collector flow record reference

Documentation is now available for all record types collected by the Flow Collector. This documentation consists of a file that describes all supported flow record formats. Users can download the file from the [Network Developer Portal](#).

3.2 Pathway: configure flow statistics collection

3.2.1 Stages

1

Specify the data and aggregation types that must be turned on in the aggregation policy. Turn on only what is needed to minimize resource consumption (CPU, memory, network bandwidth, disk IO, and database storage space) and specify the collection interval length for each data type. Note that the size of the interval has an impact on memory resources (longer intervals consume more memory). See [3.3 "How do I configure flow statistics aggregation?" \(p. 47\)](#).

2

Specify the destination of the collected data. For some data types, the destination is predetermined. For others it can be selected. Specify the upload target for each file format in the upload policy, and for CSV file formats, configure FTP or SFTP server upload targets. See [3.4 "How do I set upload targets and policies?" \(p. 51\)](#).

3

Add routers to the collection policy, specifying the management IP (source of UDP data), the system IP (the collector translates the management IP to the system ID, which can be IPv4 or IPv6), and the data protocol. See [3.5 "How do I specify the NEs for flow statistics collection?" \(p. 53\)](#).

4

Configure the routers (NEs) to collect desired flow information (according to their capabilities) and send the collected flow information to the NSP Flow Forwarder Service. The Flow Forwarder exposes an NSP VIP that has to be used by the routers as the target for the IPFIX data. The Flow Forwarder uses specific UDP ports for each data type and the routers have to be configured to send the data to the proper UDP port. They are as follows:

- System Cflowd: 4741
- CGNAT: 4742
- Netflow V5: 2055

In DR deployments, NEs must be configured to send flows to both the primary and standby sites

3.3 How do I configure flow statistics aggregation?

3.3.1 Steps

1

Log in to a station that has access to the NSP system.

2

Open a console window.

3

Issue the following RESTCONF API call to specify what data/aggregation types are enabled for collection:



Note: In order to issue a RESTCONF API call, you require a token; see the My First NSP API Client tutorial on the [Network Developer Portal](#) for information.

PATCH `https://address:443/restconf/data/flowconf:config/policies/aggregation=data_type,aggregation_type`

where

address is the NSP cluster advertised address

data_type is a data type listed in the table below

aggregation_type is an aggregation type listed in the table below

The message body is the following:

```
{
  "aggregation" : [
    {
      "enabled" : "true",
    }
  ]
}
```

The following table shows the data and aggregation types that can be enabled, as well as their default state and protocol.

Table 3-1 Data types and aggregation types

Data type	Aggregation type	Default	Mode	Protocol	Template
BASIC_IPV4	AGG_RAW	false	SYS	FLOW_INFO_10_NOKIASYS	BASIC_IPV4
CONVERSATION_TRAFFIC	AGG_CONVERSATION_IPV4	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV4
CONVERSATION_TRAFFIC	AGG_CONVERSATION_IPV6	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV6
CONVERSATION_TRAFFIC	AGG_INTERFACE_CONVERSATION_IPV4	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV4
CONVERSATION_TRAFFIC	AGG_INTERFACE_CONVERSATION_IPV6	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV6
ESTIMATED_BITRATE	AGG_LSP_STACK	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IP
HOST_TRAFFIC	AGG_HOST_IPV4	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV4
HOST_TRAFFIC	AGG_HOST_IPV6	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV6
HOST_TRAFFIC	AGG_INTERFACE_HOST_IPV4	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV4
HOST_TRAFFIC	AGG_INTERFACE_HOST_IPV6	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV6
L2_IP	AGG_RAW	false	SYS	FLOW_INFO_10_NOKIASYS	L2_IP
MPLS	AGG_RAW	false	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV4, MPLS_IPV6
NAT_SESSION_END	AGG_RAW	false	SYS	FLOW_INFO_10_NOKIABBNAT	NAT_END
NAT_SESSION_END_SHORT	AGG_RAW	false	SYS	FLOW_INFO_10_NOKIABBNAT	NAT_END

Table 3-1 Data types and aggregation types (continued)

Data type	Aggregation type	Default	Mode	Protocol	Template
NAT_SESSION_START	AGG_RAW	false	SYS	FLOW_INFO_10_NOKIABBNAT	NAT_START
NETFLOW_V5	AGG_RAW	false	SYS	FLOW_INFO_5_DEFAULT	NETFLOW_V5
OBSERVED_TRAFFIC	AGG_INTERFACE	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV4, MPLS_IPV6
OBSERVED_TRAFFIC	AGG_TOS	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV4, MPLS_IPV6
PROTOCOL_TRAFFIC	AGG_PROTOCOL	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV4, MPLS_IPV6
TRAFFIC_VOLUME	AGG_SRC_AS_DEST_AS	true	SYS	FLOW_INFO_10_NOKIASYS	BASIC_IPV4
TRAFFIC_VOLUME	AGG_VPN_IPV4	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV4
TRAFFIC_VOLUME	AGG_VPN_IPV6	true	SYS	FLOW_INFO_10_NOKIASYS	MPLS_IPV6
TRAFFIC_VOLUME	AGG_VPN_IPV4, AGG_VPN_IPV6	true	SYS	FLOW_INFO_10_NOKIASYS	L2_IP

4

Issue the following RESTCONF API call to specify the length of the aggregation intervals for each data type:



Note: In order to issue a RESTCONF API call, you require a token; see the My First NSP API Client tutorial on the [Network Developer Portal](#) for information.

PATCH `https://address:443/restconf/data/flowconf:config/policies/agg-interval=entry`

where

address is the NSP cluster advertised address

entry is an entry listed in the table below

The message body is the following:

```
{
  "agg-interval" : [
    {
      "duration" : "3",
    }
  ]
}
```

i **Note:** The settings in the above example represent the default values.

i **Note:** The statistics collection interval affects NSP Flow Collector performance. A larger interval results in proportionally larger files, which take longer to store and transfer.

i **Note:** For CGNAT statistics, you must set the collection interval no higher than the following, based on the expected flow rate:

- 350 000 flows/s—1 minute
- 50 000 flows/s—5 minutes
- 25 000 flows/s—15 minutes

The following table shows the aggregation intervals that can be configured, as well as their default state and protocol.

Table 3-2 Aggregation intervals

Entry	Default	Mode	Protocol
BASIC_IPV4	15	SYS	FLOW_INFO_10_NOKIASYS
CONVERSATION_TRAFFIC	15	SYS	FLOW_INFO_10_NOKIASYS
ESTIMATED_BITRATE	1	SYS	FLOW_INFO_10_NOKIASYS
HOST_TRAFFIC	15	SYS	FLOW_INFO_10_NOKIASYS
L2_IP	15	SYS	FLOW_INFO_10_NOKIASYS
MPLS	15	SYS	FLOW_INFO_10_NOKIASYS
NAT_SESSION_END	1	SYS	FLOW_INFO_10_NOKIABBNAT
NAT_SESSION_END_SHORT	1	SYS	FLOW_INFO_10_NOKIABBNAT
NAT_SESSION_START	1	SYS	FLOW_INFO_10_NOKIABBNAT
NETFLOW_V5	15	SYS	FLOW_INFO_5_DEFAULT
OBSERVED_TRAFFIC	15	SYS	FLOW_INFO_10_NOKIASYS
PROTOCOL_TRAFFIC	15	SYS	FLOW_INFO_10_NOKIASYS
TRAFFIC_VOLUME	15	SYS	FLOW_INFO_10_NOKIASYS

-
- 5 _____
Close the console window.

END OF STEPS _____

3.4 How do I set upload targets and policies?

3.4.1 Steps

- 1 _____
Log in to a station that has access to the NSP system.

- 2 _____
Open a console window.

- 3 _____
Issue the following RESTCONF API call to configure a policy that defines the FTP/SFTP servers that can be used to upload the data collected in files:

 **Note:** In order to issue a RESTCONF API call, you require a token; see the My First NSP API Client tutorial on the [Network Developer Portal](#) for information.

**POST https://address:443/restconf/data/flowconf:
config/policies/upload-target**

where *address* is the NSP cluster advertised address

The message body is the following:

```
{
  "upload-target" : [
    {
      "name" : "name",
      "protocol" : "protocol",
      "ip-address" : "IP address",
      "port" : "port",
      "username" : "username",
      "password" : "password",
      "descr" : "description",
    }
  ]
}
```

where

name is the server name

protocol is one of the following:

- FTP
- SFTP

IP address is the server IP address

port is the server port

username is the FTP or SFTP username

password is the FTP or SFTP password

description is an optional description

4

Issue the following RESTCONF API call to configure a policy that defines the FTP/SFTP servers that can be used to upload the data collected in files:

i **Note:** In order to issue a RESTCONF API call, you require a token; see the My First NSP API Client tutorial on the [Network Developer Portal](#) for information.

POST `https://address:443/restconf/data/flowconf:config/policies/upload`

where *address* is the NSP cluster advertised address

The message body is the following:

```
{
  "upload" : [
    {
      "file-type" : "file type",
      "server" : "server",
      "enabled" : "true",
    }
  ]
}
```

where

file type is NAT or CSV

server is the server IP address

i **Note:** The settings in the above example represent the default values.

5

Close the console window.

END OF STEPS

3.5 How do I specify the NEs for flow statistics collection?

3.5.1 Steps

1 _____
Log in to a station that has access to the NSP system.

2 _____
Open a console window.

3 _____
Issue the following RESTCONF API call for each NE:

i **Note:** In order to issue a RESTCONF API call, you require a token; see the My First NSP API Client tutorial on the [Network Developer Portal](#) for information.

POST `https://address:443/restconf/data/flowconf:config/policies/collection`

where *address* is the NSP cluster advertised address

The message body is the following:

```
{
  "collection" : [
    {
      "mgmt-ip" : "mgmt_IP",
      "sys-ip" : "sys_IP",
      "descr" : "description",
      "protocol" : "protocol"
    }
  ]
}
```

i **Note:** The *sys_IP* value must match the System ID that the NSP associates with the NE, for example, as shown on the NE properties form in the GUI. System ID can be IPv4 or IPv6.

where

mgmt_IP is the management IP address

sys_IP is the system IP address

description is an optional description

protocol is one of the following:

- FLOW_INFO_10_NOKIASYS
- FLOW_INFO_10_NOKIABBNAT
- FLOW_INFO_5_DEFAULT

4

Close the console window.

END OF STEPS

3.6 Flow statistics

3.6.1 Overview

This section describes the format of the CGNAT Cflowd records that an NSP Flow Collector generates using 7750 SR Cflowd IPFIX data as input.

For each IPFIX CGNAT session record, an NSP Flow Collector produces a raw output record in CSV format. The input IPFIX records have varying numbers of fields, depending on the format; a CSV-formatted output record includes all fields from all formats. A field that is absent from an input record is treated as follows in the associated CSV record:

- A numerical field is populated with a value of zero.
- A text field is populated with an empty string.

CSV file naming convention

A CGNAT Cflowd CSV file is named as follows, using hyphen-separated elements:

MODE-NAT_SESSION_TYPE-AGG_RAW-Nsource_IP-internal_ID-template-IPv-linterval-Timestamp-V7.csv

where

MODE is the Flow Collector mode, which is SYS

TYPE is the NAT data type, which is START or END

source_IP is the system IP address of the source NE

internal_ID is an internal identifier that is not relevant to third-party processing

template is the template ID

v is the flow IP version

interval is the collection interval, in seconds

timestamp is the UNIX time of the collection interval start, in seconds

Filename examples

The following is a session-start filename example:

SYS-NAT_SESSION_START-AGG_RAW-N203.0.113.3-Og0s0m0x2-X256-IP4-I60-T1555440480-V7.csv

The following is a session-end filename example:

SYS-NAT_SESSION_END-AGG_RAW-N203.0.113.3-Og0s0m0x2-X257-IP4-I60-T1555440480-V7.csv

3.6.2 CSV record format

The following subsections describe the NAT session-start and session-end CSV record formats, which are the following:

- Format 1—identical on the 7750 SR
- Format 2—identical on the 7750 SR

Each record includes all fields from each format. A Comment entry states which formats include a field; no comment means that the field is present in each format.

i **Note:** A numeric field that is absent from a format is populated with a zero value, and a string field that is absent contains no value.

CGNAT session-start record

The following table lists, in sequential order, the fields in a CGNAT session-start CSV record.

Field	IPFIX field	Data type	Comment
1	FLOW_ID	Long integer	—
2	SOURCE_IPV4_ADDRESS	IPv4 address	Format 1
3	POST_NAT_SOURCE_IPV4_ADDRESS	IPv4 address	Format 2; replaces Format 1 SOURCE_IPV4_ADDRESS
4	DESTINATION_IPV4_ADDRESS	IPv4 address	—
5	SOURCE_TRANSPORT_PORT	Unsigned short integer	—
6	POST_NAPT_SOURCE_TRANSPORT_PORT	Unsigned short integer	Format 2
7	DESTINATION_TRANSPORT_PORT	Unsigned short integer	—
8	FLOW_START_MILLISECONDS	Long integer	—
9	PROTOCOL_IDENTIFIER	Unsigned byte	—
10	NAT_INSIDE_SVC_ID	Unsigned short integer	—
11	NAT_OUTSIDE_SVC_ID	Unsigned short integer	—
12	NAT_SUBSTRING	String	—

CGNAT session-end record

The following table lists, in sequential order, the fields in a CGNAT session-end CSV record.

Field	IPFIX field	Data type	Comment
1	FLOW_ID	Long integer	—
2	SOURCE_IPV4_ADDRESS	IPv4 address	Format 1
3	POST_NAT_SOURCE_IPV4_ADDRESS	IPv4 address	Format 2; replaces Format 1 SOURCE_IPV4_ADDRESS

Field	IPFIX field	Data type	Comment
4	DESTINATION_IPV4_ADDRESS	IPv4 address	—
5	SOURCE_TRANSPORT_PORT	Unsigned short integer	—
6	POST_NAPT_SOURCE_TRANSPORT_PORT	Unsigned short integer	Format 2
7	DESTINATION_TRANSPORT_PORT	Unsigned short integer	—
8	FLOW_END_MILLISECONDS	Long integer	—
9	PROTOCOL_IDENTIFIER	Unsigned byte	—
10	FLOW_END_REASON	Unsigned byte	—
11	NAT_INSIDE_SVC_ID	Unsigned short integer	—
12	NAT_OUTSIDE_SVC_ID	Unsigned short integer	—
13	NAT_SUBSTRING	String	—

4 Baseline Analytics

4.1 What is a baseline?

4.1.1 Baselines

A baseline provides the logic for the collection of baseline statistics, and for the detection of anomalies, through detecting a historical trend and generating expected values based on the trend.

An anomaly is a value that deviates enough from a baseline's expected value to be flagged for attention. You can configure anomaly thresholds as part of baseline creation.

Baselines and anomaly detection are included with NSP Baseline Analytics.

When you create baselines, you specify a resource or group of resources to collect a set of statistics over a defined time window. Information collected during that window is used to calculate a data point for the baseline. You also define a season, which is the length of time statistics need to be measured to assess trends. For example, to assess network traffic, you could set up a 15-minute window with a one-week season, which provides values calculated every 15 minutes over a one-week period.

Creation of a baseline creates a baseline subscription to collect the required data.

i **Note:** Baseline subscriptions and telemetry subscriptions are separate. A baseline cannot be generated from data collected by a telemetry subscription.
On-demand NFM-P statistics cannot be used to create baselines.

A baseline consists of the following components. The components appear in the Create and Edit forms:

- [General parameters](#)
- [Filter & Counters](#)
- [Anomaly Detectors](#)

Baselines are created on a per-resource basis. A resource is an entity that can collect the desired statistics. In the Create Baselines form, configure the required parameters and choose the resources to collect the statistics.

If the NE is managed using MDM, configuration of a baseline initiates statistics collection. If the NE is managed by NFM-P, statistics collection must be configured on the NFM-P and the resource must already be collecting the desired statistics for a baseline to be created.

i **Note:** Baseline Analytics is different from the NSP Analytics application.
Baseline Analytics provides near-real-time baseline and anomaly detection from telemetry counters, for example, received octets for the `/telemetry:base/interfaces/interface` telemetry type.
The Analytics application computes a baseline for data configured for reporting, for example, utilization and throughput for a port in a Port LAG Details report, or bandwidth and data for an

application group in a Router Level Usage Summary report with Baseline. See the *Analytics Report Catalog* and the *NSP User Guide* for more information about Analytics.

4.1.2 General parameters

The general parameters of a baseline include the following:

- **Description**
Add an optional description for filtering on the Baselines view.
- **Collection interval**
For MDM-managed NEs, this represents the interval at which to collect the statistics, for example, every 30 seconds. For NEs managed by the NFM-P, this value is ignored and statistics are collected according to the settings configured in the NFM-P.
- **Season**
A season is the length of time statistics must be collected for a pattern to be seen. For example, for network traffic you can expect the data pattern to repeat on a weekly basis.
- **Window duration**
Window duration is the size of the data bucket for telemetry calculation. For example, a counter calculates the change between the first and last values taken during the window. The calculation used depends on the counter type parameter in the Filters & Counters panel.
- **Admin State and Training Status**
These parameters are enabled by default when a baseline is created. They can be changed in the Edit form.
 - **Admin State**
If the Admin State of a baseline is Enabled, NSP is monitoring the statistics.
 - **Training Status**
If the Training Status is Active, NSP is incorporating new information into the baseline's model. If the Training Status is Paused, future anomalies will be detected against the expected values that are already calculated.

If you are monitoring for error counters, such as packet loss, you can pause learning after a season with no errors, which sets the expected number of errors to zero, while continuing to monitor.

For example, if you create a baseline and set the Collection Interval to 30, the Season to 1 week, and the Window Duration to 15 minutes, the baseline subscription collects the statistics values every 30 seconds, calculates a baseline data point every 15 minutes, and assesses trends based on one week of data.

4.1.3 Filter & Counters parameters

As you proceed through baseline creation, more parameters become available on the UI. The Filter & Counters parameters declare the telemetry values to be collected, the counter types, and the resources of interest.

After you select a telemetry type, the **+Counters** button becomes available.

You can configure one of the following counter types:

- **Counter:** a counter takes input counter values and calculates the change in value over the

window. For example, if the counter represents the number of transmitted octets and windows are 15 minutes, a counter baseline value is the number of octets transmitted over 15 minutes.

- **Gauge:** a gauge takes input values and calculates the mean value over the window. For example, if the input value is octets per second over a 30 s period and windows are 15 minutes, a gauge baseline value is the mean octets per second over 15 minutes.
An example of a gauge is CPU usage; it is a bounded value between 1 and 100%.
- **Sampled:** a sampled baseline takes sampled values and calculates the sample mean value over the window. Sampled values represent the value at the exact time the sample was taken, not the value since the last sample was taken. For example, if CPU % is sampled every 2 minutes and windows are 15 minutes, a sampled baseline value is the sampled mean over the samples collected in the 15 minutes.

An example of a sampled value is latency.

Configure an object filter as needed to filter the available resources; see [1.4 “How do object filters work?”](#) (p. 19).

When at least one counter is added and a counter type is specified, the **Verify Resources** button becomes available.

4.1.4 Detectors

A detector defines the rules for anomaly detection. The detector rule provides an acceptable range of expected values. If a detected value exceeds the range, it is marked anomalous.

Anomaly detection is optional.

A detector rule is composed of the following:

- **algorithm** — the formula to use to compare the expected and measured values
- **comparison** — greater than or less than
- **evaluate what** — value, rate, or bandwidth

The measured values may be converted to a rate or bandwidth to perform the evaluation:

- **rate** — $\text{value} / \text{window}$
- **bandwidth** — $(\text{value} * 8) / \text{window}$

- **threshold** — the end of the acceptable range

The Comparison and Threshold parameters define the range of acceptable values. For example, a rule could state that a value with an absolute Z-score greater than 2 is an anomaly.

Algorithms

You can define a rule based on an algorithm.

The following algorithms are suitable for most purposes:

- **Z-score**
This refers to the Z-score (number of standard deviations) of the measured value against the expected values. In this case, the expected value is the mean.
Formula: $(\text{measured} - \text{expected}) / \text{stddev}$
- **Z-score absolute**

This refers to the absolute value of the Z-score of the measured value against the expected values. In this case, the expected value is the mean.

Formula: $|(\text{measured} - \text{expected}) / \text{stddev}|$

The Z-score algorithms are useful because they incorporate the standard deviation. In addition to recording how far the current value is from the mean, the algorithm also factors in the variability of the values. This can be very important when deciding if a value is anomalous. If your values are highly variable, that is, the standard deviation is high, it is important to choose a Z-score algorithm.

You can also use one of the following:

- **relative difference mean**

This is the relative difference using the absolute value of the arithmetic mean of the measured and expected values.

Formula: $|(\text{measured} - \text{expected})| / (|\text{measured} + \text{expected}| * 0.5)$

This algorithm could be suitable if the standard deviation is very small, that is, if there is very little variation in the values.

- **relative change signed**

This is the relative change (including the positive or negative sign) between the measured and expected values.

Formula: $(\text{measured} - \text{expected}) / |\text{expected}|$

- **relative change**

This is the relative change (with no sign) between the measured and expected values.

Formula: $|(\text{measured} - \text{expected})| / |\text{expected}|$

- **change over mean**

This is the change of the measured and expected values over the absolute value of the arithmetic mean.

Formula: $(\text{measured} - \text{expected}) / (|\text{measured} + \text{expected}| * 0.5)$

- **change max score**

This is a score that becomes more sensitive as the measured or expected value approaches +/-100. This detector algorithm works well with percentages although it may have use with other types of values.

Formula: $(\text{sign}(\text{measured} - \text{expected}) * (|\text{measured} - \text{expected}| + \max(\text{measured}, \text{expected}))) / 200$

4.1.5 Baseline charts

A baseline chart plots captured telemetry data for a specified period, along with baseline expected values and the range relative to the expected value that is considered normal by the anomaly detectors, that is, the range of values that are not anomalies.

Anomalies are indicated where they are detected.



Note: A baseline must be trained for at least one season before expected values can be charted. Detection of anomalies requires one or more seasons depending on the detector.

Baseline and anomaly charts and the list of anomalies are available in Data Collection and Analysis Visualizations.

To open Data Collection and Analysis Visualizations for a baseline, choose the baseline and click  (Table row actions), **Open in Data Collection and Analysis Visualizations**. The view opens to a New Chart form with the baseline selected.

RESTCONF APIs are available for Baseline Analytics; see the Network Infrastructure Management API documentation on the [Network Developer Portal](#).

To plot a baseline chart, see [4.5 “How do I plot a baseline chart?”](#) (p. 65).

To chart an anomaly, see [4.6 “How do I chart an anomaly?”](#) (p. 67).

4.1.6 Baseline Analytics data storage

Baseline data is stored in Postgres, unless there is an auxiliary database enabled, in which case all collected data is stored in the auxiliary database.

The following data is stored:

- statistics data collected during the configured window; see [General parameters](#)
- the calculated baseline for the window
- anomalies

By default, data is stored in Postgres for 35 days and in the auxiliary database for 90 days. The maximum storage time in Postgres is 35 days; the maximum in the auxiliary database is 403 days.

These values can be changed using the RESTCONF API or by updating the age-out policy; see [2.15 “How do I edit an age-out policy?”](#) (p. 43).

4.2 What are the best practices for baseline and anomaly detector creation?

4.2.1 Baseline best practices

When baselines are configured, NSP creates a baseline with the parameter values for each resource selected. For example, if you want to collect baseline values for 20 ports with the same telemetry type, configure the Create Baselines parameters and the NSP will create 20 baselines.

Bulk creation of baselines can take significant time and resources to complete. If the bulk creation process is interrupted by a network disruption or user action, it results in failed or partial creation.

If a baseline creation process has been interrupted by a network problem or user action (such as a page refresh), incorrect entries may exist in the database. Contact Nokia support.

Keep these suggestions in mind:

- The collection interval configured in NSP is ignored by resources that are managed by NFM-P. Instead, the collection interval of resources managed by NFM-P is determined by the corresponding NFM-P MIB entry policy.
- Use object filters in the Create Baselines form to target the resources you need; see [1.4 “How do object filters work?”](#) (p. 19).
- Avoid unfiltered baseline creation requests unless the network is limited in size.
- If you configure the Convert to Bitrate parameter for a baseline counter and want to detect anomalies, set the Evaluate What parameter in the detector rule to Bandwidth. Using a different

evaluation setting results in a mismatch between the Anomalies list in Visualizations and the baseline and anomaly chart generated for the baseline.

- Ensure that you have enough time to wait for the creation operation to complete before configuring baseline creation.

The following table shows some average time frames for bulk baseline creation.

Number of baselines	Average maximum time required
100	30 s
1000	3 min
5000	15 min
10 000	30 min



Note: The baseline subscriptions are created last; therefore, no statistics are collected for any baseline in the group until the bulk creation is completed.

4.3 How do I create baselines?



CAUTION

Service Disruption

The name of an object, including subscriptions, baselines, indicators, templates, and chart profiles, cannot contain a semicolon (;) or backslash (\).

The use of these characters in an object identifier will result in corrupted data that must be deleted by Nokia support.



Important! If you are creating multiple baselines, the create operation may take a long time and cannot be stopped. Ensure that you are able to wait for the operation to complete before proceeding. For example, the average maximum time required for the creation of 100 baselines is 30 s; the average maximum time required for the creation of 10 000 baselines is 30 min.

4.3.1 Steps

1

Open **Data Collection and Analysis Management, Baselines**.

2

Click **+Baseline**.

3

In the Create Baselines form that opens, configure the General parameters as needed.

-
- 4

Enter information in the Telemetry Type field. As you type, the field filters for available telemetry types to match your input.

Choose the telemetry type you need from the list of matches.
 - 5

Click **+Counters**.
 - 6

In the **Add Counters** form that opens, choose the required counters and click **Add**.
 - 7

Configure the Counter parameters as needed.
 - 8

Click **+Counters** to add additional counters.
 - 9

In the **Object Filter** field, enter filtering information as needed to filter the collected data.

As you type, the field provides suggestions for available filters to match your input and identifies incorrect syntax.
 - 10

Click **Verify Resources**.

NSP searches for resources that are collecting statistics to match the telemetry type and object filter.

 **Note:** NFM-P resources appear in the list after the NFM-P collects the statistics. Therefore, you need to wait for the current collection interval to end before the NFM-P resource appears in the list.

If needed, you can click **Stop Verification** at any time and **Edit Filters & Counters** to update the configuration and restart the verification process.
 - 11

When the resources you need appear in the resources list, click **Stop Verification**.
 - 12

Select the resources you need in the resources list.
 - 13

In the Detectors panel, add a rule for each counter as needed:

 1. Click **+Rule**.
-

-
2. Configure the rule parameters.
- Repeat steps 1 and 2 to add detector rules for other counters.

14

Click **Create**.

END OF STEPS

4.4 How do I edit or delete baselines?

4.4.1 Steps

- 1

Open **Data Collection and Analysis Management, Baselines**.
- 2

Choose a baseline and click  (Table row actions), **Edit**.
- 3

To stop or restart monitoring, change the **Admin State** parameter.
If the admin state is Disabled, NSP stops monitoring the statistics.
- 4

To stop or restart training, change the **Training Status** parameter.
If the training status is Paused, new data will no longer affect the calculated baseline.
- 5

To reset training:
 1. If the Admin State is Enabled, select Disabled from the drop-down list and click **Update**.
 2. Click **Reset Baseline Windows** and click **Apply** to confirm.
 3. Wait a minimum of 15 min.
You can close the form while you are waiting, if needed.
 4. Click **Reset Baseline Windows** again and click **Apply** to confirm.
 5. Set the Admin State to Enabled and click **Update**.
- 6

To edit detectors:
 1. Click  (Delete) to remove the existing detector.
 2. Click **+Rule**.
 3. In the Detector Rule form that opens, configure the new parameters.

4. Click **Test** to preview how your changes will impact the range of expected values.
5. Click **Add**.

7 _____
Click **Update**.

- 8 _____
- To delete baselines:
- a. Select one or more baselines and click  (Table row actions), **Delete**.
In the form that opens, confirm the deletion.
 - b. From **Data Collection and Analysis Management, Subscriptions**, select a baseline subscription and click  (Table row actions), **Delete**.
Deleting the subscription also deletes all associated baselines.

END OF STEPS _____

4.5 How do I plot a baseline chart?



CAUTION

Service Disruption

The name of an object, including subscriptions, baselines, indicators, templates, and chart profiles, cannot contain a semicolon (;) or backslash (\).

The use of these characters in an object identifier will result in corrupted data that must be deleted by Nokia support.

4.5.1 Steps

Create a chart

- 1 _____
Open the **New Chart** form:
 - From **Data Collection and Analysis Visualizations, Baseline Charts**, click **+ Chart**.
 - From **Data Collection and Analysis Management, Baselines**, choose a baseline and click  (Table row actions), **Open in Data Collection and Analysis Visualizations**.
- 2 _____
In the form that opens, configure the parameters in the top panel as needed:
 - From the **Time Range** drop-down list, choose the amount of historical data you need to display.
 - Click **Combine charts** to plot data from multiple data series on the same chart.

3

Click **+Baselines**.

The list of available baselines is populated.

4

Choose up to 10 baselines from the list and click **Add**.

5

To save the chart:

1. Click **Save As**.

2. In the window that opens, enter a name for the chart and add a description if needed.

3. Click **Save**.

The chart is added to the list.

6

Click **Plot**.

END OF STEPS

4.5.2 Steps

Plot an existing chart

1

To edit a chart before you plot it, choose the chart and click  (Table row actions), **Edit**.

2

Edit the parameters as needed and click **Plot**.

END OF STEPS

Result

Visualizations displays a chart view showing the baseline and anomaly data. From the chart view, you can change the information displayed: choose a different time range, click **Configure** to change the list of baselines or click  (More actions), **Combine charts** to turn combined charts on or off.

Click  (Chart Details) to open the Chart Details panel on the right side of the chart view to show details about the resources.

Hover over any point on the chart to see the measured values, expected values, and range at the indicated point.

4.6 How do I chart an anomaly?

4.6.1 Purpose

The Anomalies view shows the table of anomalies that have been detected and saved to the database. The table provides information about each anomalous value and why the value was flagged as an anomaly. You can filter the table according to the baseline name, resource, counter, or telemetry type.

4.6.2 Procedure

You can chart an anomaly from the **Data Collection and Analysis Visualizations, Anomalies** view.

1. Select the anomaly and click **Chart** .

The chart displays, showing the three hours before and after the selected anomaly.

2. From the chart form, click **Configure** to change the time range or add additional baselines if needed.

5 NSP Indicators

5.1 What is an indicator?

5.1.1 Definition and purpose

An indicator is a subscription to a customized metric. Indicators allow you to define and track KPIs using a combination of telemetry data, arithmetic operators, and aggregation functions. Indicator functionality is enabled with the Performance Indicators and Alerts installation option.

When you create an indicator, NSP creates an indicator subscription to collect the required data.

Note: Indicator subscriptions and telemetry subscriptions are separate. An indicator cannot be generated from data collected by a telemetry subscription.

On-demand NFM-P statistics cannot be used to create indicators.

5.1.2 Sample indicator

A sample indicator will be referenced throughout this chapter to demonstrate how users can configure and utilize an indicator.

This sample indicator will be configured to track the average utilization of a three-node ring, sampling from four ports and a LAG in the ring. Alarms will be set to generate if the average utilization is increasing above 50% or decreasing below 5%.

Figure 5-1, “Sample indicator in the Indicators view” (p. 69) shows the sample indicator in the **Data Collection and Analysis Management, Indicators** view.

Figure 5-1 Sample indicator in the Indicators view

Admin State	Name	Description	Telemetry Type	Counters	Units	Thresholds	Actions	Multiple Output	Collect (seconds)
Enabled	ArithPort2_bps	tx + rx on each port on ...	telemetry/base/interf...	received-octets-period...	bps	✓	✓	✓	60
Enabled	cpu-avg		telemetry/base/syste...	cpu-usage	%	✓	✓	✓	60
Enabled	MaxTempNfmp9814	max temp on 150.33	telemetry/base/hardw...	temperature	deg C	✓	✓		60
Enabled	RingIngressUtil	avg input util on ingres...	telemetry/base/interf...	input-utilization	%	✓			30

5.1.3 Indicator components

An indicator consists of the following components:

- [General parameters](#)
- [Filter & Counters](#)
- [Thresholds](#)

The components appear in the Create and Edit forms.

5.1.4 General parameters

An indicator is defined, in part, through the configuration of the following general parameters:

- unique name and optional description

- **collection interval**

For NEs managed by MDM, this represents the interval for collecting the statistics (for example, every 30 s), while for NEs managed by the NFM-P, this value is ignored and statistics are collected according to the settings configured in the NFM-P.

- **window duration**

Window duration is the size of the data bucket for telemetry calculation.



Note: For simple indicators, that is, indicators with only one counter and no formula, there is no data bucket. The window duration is ignored; KPI values are calculated at the collection interval.

5.1.5 Counters & Formula parameters

As you proceed through indicator creation, more parameters become available on the UI. The Counters & Formula parameters of an indicator declare the telemetry values to be collected and the operations and aggregation functions to apply. An object filter ensures that the data is collected on the resources of interest.

Counters

After you select a telemetry type, the **+Add Counters** button becomes available, allowing you to select the specific counters to include.

For the sample indicator, selecting the `/telemetry:base/interfaces/utilization` telemetry type allows for the collection of the input-utilization counter; see [Figure 5-2, “Sample Indicator General parameters”](#) (p. 71).

Figure 5-2 Sample Indicator General parameters

The screenshot shows the 'Edit Indicator' configuration page for 'RingIngressUtil' in the Nokia Network Services Platform. The interface includes a sidebar with 'General', 'Configuration', and 'Thresholds' tabs. The 'General' tab is selected, displaying various configuration fields. The 'Admin State' is set to 'Enabled'. The 'Name' is 'RingIngressUtil' and the 'Description' is 'avg input util on ingress ports to ring'. The 'Collection Interval' is '30' seconds and the 'Window Duration' is '1 minute'. The 'Units' are set to '%'. Under 'Counters & Formula', the 'Telemetry Type' is 'telemetry/base/interfaces/utilization', the 'Selected Counter(s)' is 'input-utilization', and the 'Formula' is '1 avg(input-utilization_avg)'. There are 'Cancel' and 'Update' buttons at the bottom right.

Formula

An indicator formula defines the calculation of a KPI value. A formula is optional if only one counter is configured.

You can select a formula to apply operations and functions to the counters. Formulas are composed of the following components:

Component	Definition	Notes
Counter functions	Aggregation functions performed on a per-counter basis with one value output per counter for each resource. The output of the counter function is then used in the rest of the formula calculation. Counter functions are expressed as <i>counter_function</i>, for example, <i>input-utilization_avg</i>.	The following function types are available on a counter: • sum: adds up all samples received for the counter during the window duration. Sum is used for counted statistics such as octets or packets. • avg: the average of all samples received for the counter during the window duration. Avg is used for rate or value statistics such as percentage or temperature. • min/max: the minimum or maximum value of all samples received for the counter during the window duration.

Component	Definition	Notes
Arithmetic operations	Simple arithmetic operations using one or more counters received from each unique resource. These operations produce n KPI outputs (one per resource), per window period. Operations can include counter functions and numbers. Negative numbers must be in parentheses. An example of an arithmetic expression is: $ (\{\text{received-octets-periodic_sum}\} - \{\text{transmitted-octets-periodic_sum}\}) * 8 / 60$	The following arithmetic operations can be performed: • addition (+) • subtraction (-) • multiplication (*) • division (/) • absolute value ()
Aggregation functions	A formula can perform aggregations on one or more counters received from every resource. This aggregation produces a single indicator output (one KPI value for all resources) per window period. An aggregation operation can perform multiple aggregation operations and can include arithmetic operations. For example, a formula for average utilization in the sample indicator is: $\text{avg}(\{\text{input-utilization_avg}\})$	The following aggregation operations are available: • min • max • sum • avg

For the sample indicator, the formula is configured to provide an average of the counter value data, see [Figure 5-2, “Sample Indicator General parameters” \(p. 71\)](#).

Formulas can also combine these components as needed.

For example, a formula can combine the components: $\text{max}(\{\{\text{transmitted-octets-periodic_sum}\} + \{\text{received-octets-periodic_sum}\} * 8 / 60\} / \text{sum}(\{\text{speed_avg}\}) * 100$. This formula creates a single mixed-expression-operator that has two aggregation operations: the first outputs the port with the maximum value of $(\{\text{tx} + \text{rx}\} * 8 / 60)$, the second sums the speed of all ports. These two outputs are then divided and the result is multiplied by 100.

Object filter

You can configure an object filter as needed to define a subset of available resources to which the indicator applies; see [1.4 “How do object filters work?” \(p. 19\)](#).

When at least one counter has been selected, the **Verify Resources** button becomes available. Check the contents of the resource list to ensure that your object filter is retrieving the correct objects.

For the sample indicator, the object filter determines the ports and LAGs to collect information from, see [Figure 5-2, “Sample Indicator General parameters” \(p. 71\)](#).

5.1.6 Thresholds

Thresholds define the preferred range of data values for the indicator. Thresholds are optional; you can create zero, one, or multiple thresholds, for increasing or decreasing values.

When you configure thresholds for an indicator, you can also define the actions to be triggered by threshold violations.

When the indicator is plotted in Visualizations, thresholds appear on the plot and threshold crossing events are indicated.

A **threshold crossing event (TCE)** occurs when the data range for a defined threshold is violated; that is, a value outside the preferred range for the indicator has been detected.

Each threshold can be configured to trigger one or more actions if a TCE occurs:

- **raise a threshold crossing alarm (TCA)**

The TCA is raised in Current Alarms.

Note: Alarms related to indicators might still be raised shortly after a resource (for example, an NE) is unmanaged in NSP if statistics messages are already in the system waiting to be processed.

- **generate email**

An email can be generated for each threshold crossing event or aggregated by time period or number of events.

Email server settings must be configured in NSP Settings by an administrator.

- **output a message to a Kafka topic**

The Kafka topic must already be created. The default topic is **nsp-act-action-event**.

For the sample indicator, thresholds are configured with the following, as shown in [Figure 5-3, “Sample Indicator, increasing threshold parameters”](#) (p. 74) and [Figure 5-4, “Sample Indicator, decreasing threshold parameters”](#) (p. 74):

- an increasing threshold for a value of 50
- an alarm action with major severity
- a falling threshold for a value of 5 with a Kafka message showing low utilization

This configuration allows for high and low utilization to be handled differently according to different levels of urgency. If needed, another increasing threshold could be set at 40, for example, with warning severity, to tell operators that the ring may need attention.

Figure 5-3 Sample Indicator, increasing threshold parameters

RingIngressUtil Edit Indicator

General Configuration Thresholds

Thresholds

Threshold	Direction	Alarm	Email	Kafka
50	Increasing	✓	✓	✓
5	Decreasing	✓	✓	✓

Threshold Setting

Threshold* 50 Direction* Increasing

Actions

Severity* Major

Kafka

Topic Name* nsp-act-action-event

Content High Ring Utilization detected

Figure 5-4 Sample Indicator, decreasing threshold parameters

RingIngressUtil Edit Indicator

General Configuration Thresholds

Thresholds

Threshold	Direction	Alarm	Email	Kafka
50	Increasing	✓	✓	✓
5	Decreasing	✓	✓	✓

Threshold Setting

Threshold* 5 Direction* Decreasing

Actions

Alarm

Severity* Warning

Kafka

Topic Name* nsp-act-action-event

Content Low Ring Utilization detected

Threshold crossing alarms (TCAs)

Indicator TCAs appear in Current Alarms as alarms with Source System “fdn:app:nsp-indicator” and Alarmed Object Type “nsp-indicator:rta-indicator-rules/rule”. The Alarmed Object ID contains the indicator rule name and the Additional Text contains the threshold violation details.

The Object Full Name field contains the indicator name and the specific object instance. For aggregate TCAs, where the aggregation functions such as avg() or sum() are used, the threshold violation may be based on statistics from multiple objects, and therefore cannot be attributed to a single object instance. In this case, the Object Full Name field contains the name of the indicator rule.

The visibility of TCAs depends on user access control settings.

- Aggregate TCAs do not contain a Site ID or Site name, and are considered system level alarms. To view these alarms, the user must either have the Administrator role, or have both **Access to all Equipment** and **Access to all Services** enabled.
- If the user has access to certain NEs only, the user sees indicator TCAs for those NEs, but no aggregate TCAs.

5.1.7 Simple and complex indicators

Simple indicators use a single counter and no formula. A simple indicator is evaluated every collection interval, for every resource returned by the object filter. If a window duration parameter is configured but no formula is used, the window duration is ignored.

Complex indicators use one or more counters and a formula. A complex indicator uses counter statistics received during each window duration to aggregate each incoming counter from each resource into a counter function. This output is then, depending on the formula, used in arithmetic or aggregation operations, or both, to calculate the KPI.

5.1.8 Indicator charts

An indicator chart plots calculated KPI data for a specified period along with configured thresholds. If at least one threshold has been defined, meaning the Thresholds Configured parameter is True, the threshold values are displayed when hovering on the far right-hand edge of the indicator chart.

Rising and falling thresholds appear on the chart as dotted lines. Threshold crossing events are indicated where they are detected.

Indicator charts and a list of threshold crossing events are available in Data Collection and Analysis Visualizations.

To open Data Collection and Analysis Visualizations for a selected indicator, choose the indicator and click  (Table row actions), **Open in Data Collection and Analysis Visualizations**. If the indicator has multiple outputs, choose the resources in the form that opens, click **+** (Add) and click  **Open in Data Collection and Analysis Visualizations**. The view opens to a New Chart form with the indicator resources selected.

RESTCONF APIs are available for NSP Indicators; see the Network Infrastructure Management API documentation on the [Network Developer Portal](#).

To plot an indicator chart, see [5.5 “How do I plot an indicator chart?”](#) (p. 80).

To chart a threshold crossing event, see [5.6 “How do I chart a threshold crossing event?”](#) (p. 82).

5.1.9 Indicator data storage

Indicator data is stored in Postgres unless there is an auxiliary database enabled, in which case all collected data is stored in the auxiliary database.

By default, data is stored in Postgres for 35 days and in the auxiliary database for 90 days. The maximum storage time in Postgres is 35 days; the maximum in the auxiliary database is 403 days.

These values can be changed using the RESTCONF API or by updating the age-out policy; see [2.15 “How do I edit an age-out policy?” \(p. 43\)](#).

5.2 What are the best practices for indicator creation and management?

5.2.1 Indicator best practices

Keep the following notes in mind when creating and managing indicators:

- Consider the following when you set the collection interval and window duration:
 - The collection interval configured in NSP is ignored by resources that are managed by NFM-P. Instead, the collection interval of resources managed by NFM-P is determined by the corresponding NFM-P MIB entry policy.
 - When configuring an indicator with a formula for resources managed by NFM-P, the window duration must be at least twice as long as, and be a multiple of, the NFM-P MIB policy. An example is a 5 min MIB policy and a 15 min window duration.
 - When configuring an indicator with a formula for resources managed by MDM, the window duration must be at least twice as long as, and be a multiple of, the collection interval.
- Avoid divide-by-zero scenarios when constructing formulas. Arithmetic operations that are divided by zero produce no output, while aggregation operations output zero from the operation that is divided by zero.

- Before completing indicator creation, it is a best practice to verify resources to ensure you see a reasonable value for each counter.

Certain counters may not be fully supported in an NE managed by NFM-P and they default to zero, which causes an unexpected calculated KPI value. Other counters may have an actual value of zero. For example, a value of zero for a speed counter can indicate a down port—that is, an actual value of zero speed—or a counter that does not appear in the relevant mapping artifact.

- Clicking **Verify Resources** in an indicator shows the network device identifiers for the resources that respond to the object filter. The network device identifier is the unique identifier included by the NE in the telemetry messages.

The network device identifier is also the NE identifier used for indicator charting and threshold events in Visualizations.

- If an indicator configuration is updated, the indicator restarts, causing an expected gap in data until it can start collecting data with the new window duration.
- Indicator data points are time stamped to the start of the window duration. For example, for an indicator with a window duration of 15 min, the data point or threshold event at 16:15 on the chart represents the window from 16:15 to 16:30.

- Data points for simple indicators are output immediately.

Output from indicators with formulas

The data required by the formula must be received and processed after the window duration ends before generating the indicator value.

The output of a window is delayed until a new message is received at the collection interval in the next window.

Example: Indicators with a 5-minute collection interval and 15-minute window duration. The 15-minute collection window starts at midnight (00:00 to 00:15).

After the window closes, the indicator waits until it receives a message at the 5-minute collection interval in the next window, before proceeding to calculate the indicator value for the previous window.

The delay for calculation depends on the formula:

- Output from an indicator with an arithmetic formula requires an additional collection interval. For this example, the delay is 5 min.

Example formula: `{received-octets-periodic_sum}+{transmitted-octets-periodic_sum}`

The data point for the 00:00 to 00:15 window is processed when the next collection occurs at 00:20.

- Output from an indicator with an aggregate formula requires an additional collection interval, plus the window duration (for this example, 20 minutes).

Example formulas:

- `sum({received-octets-periodic_sum}+{transmitted-octets-periodic_sum})`
- `max({received-octets-periodic_sum})`

The data point for the 00:00 to 00:15 window is processed at 00:35.

The example formula is a sum of values which are collected at an interval. Each of these values is processed when the next collection occurs, at 00:20. The aggregated sum is processed when another sum of values message for the 00:15-00:30 window is received, at 00:35.

- A more complex formula requiring multiple levels of aggregation requires additional delays.

Example formula, with two layers of aggregation: `max(({transmitted-octets-periodic_sum}+{received-octets-periodic_sum})/sum({speed_avg}))`

The delay for this formula is one collection interval plus one window duration per level of aggregation, or 35 min.

5.3 How do I create an indicator?



CAUTION

Service Disruption

The name of an object, including subscriptions, baselines, indicators, templates, and chart profiles, cannot contain a semicolon (;) or backslash (\).

The use of these characters in an object identifier will result in corrupted data that must be deleted by Nokia support.

5.3.1 Steps

- 1 _____
Open **Data Collection and Analysis Management, Indicators**.
- 2 _____
Click **+Indicator**.
- 3 _____
In the **Create Indicator** form that opens, enter a unique name for the indicator.
- 4 _____
Configure the collection interval and window duration parameters.
- 5 _____
If needed, click **Apply Template** at the top right of the window, select a template from the list, and click **Apply**.
The parameters configured in the template appear in the **Create Indicator** form. These parameters can be changed.
- 6 _____
Configure the counters and formula as needed:
 1. Enter information in the **Telemetry Type** field. As you type, the field filters for available telemetry types to match your input
Choose the telemetry type from the list of matches
 2. Click **+Counters**
 3. If you have chosen more than one counter, a formula is required; see [5.1.5 “Counters & Formula parameters” \(p. 70\)](#)
Enter information in the **Formula** field. As you type, the field identifies incorrect syntax
- 7 _____
In the **Object Filter** field, enter filtering information as needed to filter the collected data; see [1.4 “How do object filters work?” \(p. 19\)](#).
As you type, the field provides suggestions for available filters to match your input and identifies incorrect syntax.
- 8 _____
Click **Verify Resources** to ensure that the resources found by the object filter are correct for your requirements and that counter values are returned.

9

Configure thresholds as needed:

1. Click **+Threshold**.
2. In the **Add Threshold** window that opens, enter a threshold value in the **Threshold** field.
3. Choose a direction from the drop-down list.
4. Click **Add**.
5. Add additional thresholds as needed.

10

Configure threshold actions as needed:

1. Click on a threshold you created in [Step 9](#).
2. Click **+Action** and choose **Alarm**, **Email**, or **Kafka** from the drop-down list.
 - To configure an alarm, choose the severity from the drop-down list.
 - To configure an email action, enter the recipient email addresses, separated with commas, and enter the boilerplate information in the Subject and Content fields.
To aggregate emails, enable the Aggregation toggle and configure the parameters. For example, the default value of 5 in the Period and Number of Events fields will send an email every five min, and for every five threshold crossing events, whichever comes sooner.
 - To configure a Kafka message, enter an existing topic name and enter the message information in the Content field.
3. Add additional actions for each threshold as needed.
4. Click **Save**.

11

Click **Create**.



Note: If your formula does not include all the counters you selected, the **Create** button is dimmed. If you cannot create the indicator, check the Formula field to ensure all counters you selected are part of the formula and spelled correctly.

END OF STEPS

5.4 How do I edit or delete an indicator?



Important! When an indicator is changed, the current data window becomes invalid and a new data bucket is created starting from the next interval. Depending on the timing, up to two window periods of data could be missed.

5.4.1 Steps

- 1 _____
Open **Data Collection and Analysis Management, Indicators**.
- 2 _____
Choose an indicator and click  (Table row actions), **Edit**.
- 3 _____
In the form that opens, update the parameters as needed and click **Update**.
- 4 _____
To delete an indicator:
Choose an indicator and click  (Table row actions), **Delete**.

END OF STEPS _____

5.5 How do I plot an indicator chart?



CAUTION

Service Disruption

The name of an object, including subscriptions, baselines, indicators, templates, and chart profiles, cannot contain a semicolon (;) or backslash (\).

The use of these characters in an object identifier will result in corrupted data that must be deleted by Nokia support.

5.5.1 Steps

Create a chart

- 1 _____
Open the **New Chart** form:
 - From **Data Collection and Analysis Visualizations, Indicator Charts**, click **+ Chart**.
 - From **Data Collection and Analysis Management, Indicators**, choose an indicator and click  (Table row actions), **Open in Data Collection and Analysis Visualizations**.
 - If the indicator has multiple outputs, choose resources in the form that opens and click  **Open in Data Collection and Analysis Visualizations**.
- 2 _____
In the window that opens, configure the parameters in the top panel as needed:

- From the **Time Range** drop-down list, choose the amount of historical data to display.
- Click **Combine charts** to plot data from multiple data series on the same chart.

3

Add indicator resources:

- a. Click **+Indicators**.
- b. If a file icon appears in the Multiple column, the indicator resources are aggregated for the calculated KPI. Click **+** to add it to the chart.
- c. If a folder icon appears in the Multiple column, the KPI is calculated at the resource level for the indicator.

Add resources:

1. Choose an indicator and click **Select Resources** .
2. Choose a resource and click **Add** **+**.

Choose up to 10 indicator resources and click **Add** at the bottom right of the window.

4

If you need to save the configuration as a chart:

1. Click **Save As**.
2. In the window that opens, enter a name for the chart and add a description if needed.
3. Click **Save**.

The chart is added to the list.

5

Click **Plot**.

END OF STEPS

5.5.2 Steps

Plot an existing chart

1

- a. To plot an existing chart with no changes, choose a chart and click  (Table row actions), **Chart**.
- b. To edit a chart and plot it, choose the chart and click  (Table row actions), **Edit**.

2

Edit the parameters as needed and click **Plot**.

END OF STEPS

Result

Visualizations displays a chart view showing the KPI data. From the chart view, you can change the information displayed: choose a different time range, click **Configure** to change the list of indicators, or click **More** , **Combine charts** to toggle between a combined chart for all selected indicators and separate charts.

Click  (Chart Details) to open the Chart Details panel on the right side of the chart view to show details about the resources.

Hover over any point on the chart to see measured values at the indicated point. Hover over the far right edge of the chart to show the configured thresholds.

5.6 How do I chart a threshold crossing event?

5.6.1 Purpose

The **Data Collection and Analysis Visualizations, Threshold Crossing Events** view shows the table of threshold crossing events that have been detected and saved to the database. The table provides information about each threshold crossing value and the indicator the threshold is configured on.

5.6.2 Procedure

You can chart a threshold crossing event from the **Data Collection and Analysis Visualizations, Threshold Crossing Events** view.

1. Open **Data Collection and Analysis Visualizations, Threshold Crossing Events**.
2. Select the threshold crossing event and click **Chart** .
3. From the chart, click **Configure** to change the time range or add additional indicators if needed.

5.7 What is an indicator template?

5.7.1 Indicator templates

A template can be defined to assist users creating indicators through providing a common set of parameters that can then be applied to different target resources.

When a template is applied, the configuration information in the template is copied to the indicator configuration. The indicator parameters provide the object filter, collection interval, and window. That is, the KPI is defined in the template and the KPI object and collection details are defined in the indicator. Thresholds are optional and can be configured in the template or the indicator.

For example, a template can be created for average port utilization. An indicator created from the template specifies the port and configures the data collection.

5.8 How do I create an indicator template?



CAUTION

Service Disruption

The name of an object, including subscriptions, baselines, indicators, templates, and chart profiles, cannot contain a semicolon (;) or backslash (\).

The use of these characters in an object identifier will result in corrupted data that must be deleted by Nokia support.

5.8.1 Steps

1

From the **Data Collection and Analysis Management, Indicator Templates** view, click **+Template**.

2

In the **Create Indicator Template** form that opens, enter a name for the template.
The name must be unique.

3

Enter a description and units as needed.

4

Enter information in the **Telemetry Type** field. As you type, the field filters for available telemetry types to match your input.
Choose the telemetry type from the list of matches.

5

Click **+Counters**.

6

If you have chosen more than one counter, a formula is required; see [5.1.5 "Counters & Formula parameters" \(p. 70\)](#).
Enter information in the **Formula** field. As you type, the field identifies incorrect syntax.

7

Configure thresholds as needed:

1. Click **+Threshold**.
2. In the Add Threshold window that opens, enter a threshold value in the Threshold field.
3. Choose a direction from the drop-down list.

4. Click **Add**.
5. Add additional thresholds as needed.

8

Configure threshold actions as needed:

1. Click on a threshold you created in [Step 7](#).
2. Click **+Action** and choose Alarm, Email, or Kafka from the drop-down list.
 - To configure an alarm, choose the severity from the drop-down list.
 - To configure an email action, enter the recipient email addresses, separated with commas, and enter the boilerplate information in the Subject and Content fields.
To aggregate emails, enable the Aggregation toggle and configure the parameters. For example, the default value of 5 in the Period and Number of Events fields sends an email every 5 min and for every five threshold crossing events, whichever comes sooner.
 - To configure a Kafka message, enter an existing topic name and enter the message information in the Content field.
3. Add additional actions for each threshold as needed.
4. Click **Save**.

9

Click **Create**.



Note: If your formula does not include all the counters you selected, the **Create** button is dimmed. If you cannot create the template, check the Formula field to ensure all counters you selected are part of the formula and spelled correctly.

END OF STEPS

5.9 How do I edit an indicator template?

5.9.1 Steps

1

From the **Data Collection and Analysis Management, Indicator Templates** view, choose a template.

2

Click (Table row actions), **Edit**.

3

In the form that opens, update the parameters as needed and click **Update**.

END OF STEPS

5.10 How do I delete an indicator template?

5.10.1 Steps

1

From the **Data Collection and Analysis Management, Indicator Templates** view, choose a template.

2

Click  (Table row actions), **Delete**.

END OF STEPS

6 OAM tests

6.1 What is OAM in NSP?

6.1.1 OAM functions

OAM testing functions in the NSP UI are provided from the following views in Data Collection and Analysis Management:

- **Data Collection and Analysis Management, Tests:** Enables the creation and listing of tests along with their corresponding results. OAM tests can target model-driven and classically managed SR OS NEs unless any limitation is documented.
- **Data Collection and Analysis Management, Test Suites:** A test suite groups tests to be executed in parallel or sequentially, allowing them to be run as a collective set.
- **Data Collection and Analysis Management, Test Templates:** A test template is a predefined OAM test configuration that serve as a template for creating consistent testing. These templates are used to create OAM tests.
- **Data Collection and Analysis Management, Config Objects:** The Config Objects view lists configuration objects associated with the OAM tests.

6.2 How is OAM data collected?

6.2.1 Data collection for OAM results

OAM test results can be collected via the following methods:

- **RESTCONF API:** RESTCONF APIs are available for OAM information retrieval and testing; see the Network and Service Assurance API documentation on the [Network Developer Portal](#).
- **Kafka:** OAM test results are exported to the Kafka topic `oam.test_execution`. Users can read this topic to collect OAM results for any network element. NFM-P SNMP test results are collected from Kafka by NSP. Kafka must be enabled in the NFM-P for NSP to receive the results. Kafka is enabled by default for most test types except NFM-P resynched tests. You must enable Kafka in the NFM-P for NSP to receive the results of these tests.
- **Database:** By default, OAM test data is stored in Postgres unless there is an auxiliary database enabled, in which case result data is stored in the auxiliary database.

6.3 What are OAM tests?

6.3.1 Connectivity Fault Management

PM session CFM testing is based on Metro Ethernet Forum Specification 35 - Service OAM Performance Monitoring Implementation Agreement, which details a standardized method to test and report network delay and loss using CFM messaging. This testing is performed in Layer 2 networks.

Ethernet Connectivity Fault Management in NSP is implemented based on the IEEE Y.1731 and 802.1ag OAM standards. The standards describe protocols for detecting, isolating, and reporting connectivity faults in an Ethernet network. CFM is a Layer 2 OAM object infrastructure.

To perform CFM testing in NSP, the `md-index-range` and `ma-index-range` attributes must be configured on the NE; see the NE documentation.

You can use Ethernet CFM for the following:

- path discovery
- fault detection
- fault isolation
- fault notification

The following table describes terms relevant to CFM in NSP.

Term	Expansion	Notes
MD	Maintenance domain	MD is the administrative container that defines the scope, reach and boundary for CFM message handling. It is typically the area of ownership and management responsibility. MD levels are in the range from 0 to 7, where the larger the domain, the larger the MD level assigned.
MA	Maintenance association	MA is the construct where the different management entities such as MEPs are contained. An MA is created within an MD. There is also an administrative context where a linkage is made between the domain and the service using the MA bridging-identifier configuration option.
MEG	Maintenance entity group	Typically, an MEG represents one service and consists of a group of MEPs. Only one MEG can be associated with a service, but one service can be associated with multiple MEGs. MDs and MEGs are distributed to NEs using policy distribution.
MEP	Maintenance endpoint	MEPs are the entities at the edge of a CFM MD and define the boundary for the domain. MEPs are responsible for initiating, processing, and terminating CFM messages. Each MEP with the same MD and MA represents endpoints for a single service.

CFM DMM

For CFM Delay Measurement Message (DMM) tests, calculations are made to report on three criteria: frame delay, frame delay range, and interframe delay variation. DMM test frames are issued at regular intervals from a source MEP. Delay measurement information for the forward, backward, and round trip paths is determined from the DMR frames received from the destination MEP.

CFM linktrace

CFM linktrace messages that contain a target unicast MAC address are sent to multicast destination MAC addresses. Each MIP at the same MD level replies with a linktrace response. Messages are forwarded to the next hop until they reach the destination MAC address.

Results of each hop are displayed in the NSP.

Note: Round trip time, TTL, and probe details are not displayed in the NSP for linktrace tests.

CFM LMM

The CFM Loss Measurement Message (LMM) single-ended session test is a method of exchanging transmit and receive counters between peer MEPs to determine the exact loss on a point-to-point Ethernet virtual circuit.

The following validations are performed:

- The LMM test suite tests can only be created or executed for MEG subgroups with exactly two MEPs.
- The LMM test suite tests can only be executed if the source and target MEPs are not already the source or target MEPs (respectively) of a currently running LMM test.

CFM loopback

CFM loopback messages are sent to a unicast destination MAC address. The MEP at the destination responds to the loopback message with a loopback reply. An MEP or MIP can reply to a loopback message if the destination MAC address matches the MAC address of the MEP or MIP. CFM loopback tests verify connectivity to a specific MEP or MIP.

You can also perform multicast loopbacks by providing a multicast address (class 1 multicast destination) that aligns with the level that the originating MEP is configured on. Only one multicast test can be run at a time per NE, and results from the previous test are deleted when a new test is started. The stored values include the responding MEP MAC address, the sequence number, and a locally-assigned Rx index (allowing you to detect out-of-order responses).

CFM SLM

The CFM Synthetic Loss Measurement (SLM) session test is an extension of the Y.1731 standard that provides a method of exchanging transmit and receive counters to determine frame loss between an MEP and the destination MAC address or remote MEP ID of another node in the network. This test is used to verify MEP-MEP connectivity in the network and can be used to approximate the frame loss of actual data traffic. CFM SLM session tests measure frame loss using synthetic frames rather than data traffic. Frame loss is measured by calculating the difference between the number of synthetic frames that are sent and received.

6.3.2 TWAMP Light

The PM session testing framework can also be used in the IP domain to perform TWAMP IP level monitoring over IPv4 and IPv6 addresses.

TWAMP is a delay/loss measurement protocol that uses both the TCP connection service and the UDP session service. TWAMP Light tests target Layer 3 interfaces, providing an option to monitor IP SLA performance as related to KPIs.

TWAMP Light tests can operate on a base router or on routing instances belonging to L3VPNs. The target, or destination entity, requires configuration of a TWAMP Light reflector with a prefix associated with the TWAMP Light test source IP address and a specified UDP listening port in the range specified for the NE type. For example, the 7750 SR TWAMP Light reflector destination port range is 864, 64364-64373. A target node requires a configured reflector for the base router instance or for each of the tested L3VPN-based virtual routing instances.

TWAMP Light tests require a UDP source port in range undefined or configured in the NE range and a UDP destination port that matches the target device reflector UDP listening port.

Twamp reflectors, Twamp servers, and streaming templates cannot be configured in NSP. They can be discovered from the router configuration or configured using RESTCONF; see the OAM Tests tutorial on the [Network Developer Portal](#) for RESTCONF information.

See the **Data Collection and Analysis Management, Config Objects** view for the list of configured Twamp reflectors.

TWAMP Light test types

You can configure a TWAMP Light test to record delay, loss, or delay and loss statistics. Delay tests provide streaming results, accounting-session, and accounting-bin type results; loss tests provide accounting-loss-session type results. Streaming results require configuration of a streaming delay template. The streaming template defines the frequency at which real-time streaming results are received.

Accounting type results require configuration of an accounting policy, a file policy, and associated TWAMP Light measurement interval configuration.

6.4 What OAM test types are available in NSP?

6.4.1 Available test types

You can create the following OAM tests:

- CFM DMM
- CFM Linktrace
- CFM LMM
- CFM Loopback
- CFM SLM
- TWAMP Light

Tests are created using test templates. Before a test can be created, at least one test template must exist for the test type.

Tests can be created at any time. No additional configuration is required on the tested objects for testing to be performed.

RESTCONF APIs are available for MD-OAM configuration and testing; see the Network and Service Assurance API documentation on the [Network Developer Portal](#).

6.5 What test types are available for a specific NE?

6.5.1 Test support by NE type

The ability to support OAM testing varies by NE type due to hardware capabilities. The following table shows available test types by NE type, and provides additional information if applicable.

A check mark in a cell indicates that the test type is supported for the NE type.

For model-driven NEs, OAM support may vary by release as well as by NE type. For details about OAM support for model-driven NEs, see the artifact guide for the NE type and release.

OAM testing on 210 WBX and Omniswitch NEs is not supported. For information about other NE types, contact Nokia.

Test type	NE type				
	7750 SR, 7450 ESS, 7950 XRS	7250 IXR	7705 SAR	7210 SAS	Wavence
CFM Loopback	✓	✓	✓	✓	✓
CFM LinkTrace	✓	✓	✓	✓	✓
OAM-PM CFM DMM Session	✓	✓	Not supported by the NE	✓ Streaming statistics are not supported by the NE.	✓
OAM-PM CFM SLM Session	✓ Streaming statistics are not supported by the NE.	✓ Streaming statistics are not supported by the NE.	Not supported by the NE	✓ Streaming statistics are not supported by the NE.	✓ eth-cfm-slm-loss-session telemetry is supported.
OAM-PM CFM LMM Session	✓ Streaming statistics are not supported by the NE.	✓ Streaming statistics are not supported by the NE.	Not supported by the NE	Not supported by the NE	Not supported by the NE

Test type	NE type				
	7750 SR, 7450 ESS, 7950 XRS	7250 IXR	7705 SAR	7210 SAS	Wavence
OAM-PM TWAMP Light Session	✓	✓	Not supported by the NE The NE does not support the test, however, a 7705 SAR NE can act as a TWAMP Light Reflector.	✓ For test or test suite creation on an L3VPN service, the NE Service ID must be provided instead of the service name in the Routing Instance.	Not supported by the NE The NE does not support the test, however, a Wavence NE can act as a TWAMP Light Reflector.

6.6 How do I create an OAM test?

6.6.1 Purpose

Use this procedure to configure test parameters. See [6.7 “How do I execute an OAM test?” \(p. 94\)](#) to run the tests.

6.6.2 Test creation with IP Edge NEs

For tests involving IP Edge NEs, that is, tests where at least one entity involves a 7705 SAR or 7210 SAS NE, the destination MAC address used by the test varies based on the use case. The correct value must be entered to ensure a valid test. The following table describes the behavior.

Entity type	Input	NSP behavior
MEP	User enters a MAC address in the Destination-mac-address field. The Destination entity field is not configured.	The destination MAC address provided by the user is used as the target MAC address. The user must enter a valid destination MAC address.
	The Destination entity field is configured. The Destination-mac-address field can be configured or not.	The operational MAC address of the remote MEP selected as the destination entity is used as the target MAC address. The destination MAC address entered is ignored.
ELAN or ELINE -EndPoint or Tunnel-binding	No MEP currently exists	The destination MAC address provided by the user is used as the target MAC address. The user must enter a valid destination MAC address.
	An MEP already exists	The operational MAC address of the remote MEP in the destination entity is used as the target MAC address. The destination MAC address entered is ignored.

6.6.3 Steps

- 1 _____
Open **Data Collection and Analysis Management, Tests**.
- 2 _____
Click **+Test**.
- 3 _____
In the Create OAM Test form that opens, select the Test type and Entity type.
- 4 _____
Click on the Service field if applicable:
 1. Select an attribute in the drop-down list, then enter values for that attribute in the field. As you type, the list is filtered for services that match your input.
 2. Click  as required to add additional filter criteria.
 3. Choose an entity from the list and click **Select**.
- 5 _____
Select an entity reference type if applicable.
- 6 _____
Click on the Source test entity field:
 1. Select an attribute in the drop-down list, then enter values for that attribute in the field. As you type, the list is filtered for entities that match your input.
 2. Click  as required to add additional filter criteria.
 3. Choose an entity from the list and click **Select**.
- 7 _____
Click on the Destination test entity field:
 1. Select an attribute in the drop-down list, then enter values for that attribute in the field. As you type, the list is filtered for entities that match your input.
 2. Click  as required to add additional filter criteria.
 3. Choose an entity from the list and click **Select**.
- 8 _____
Choose a test template. The test parameters are displayed.

-
- 9 _____
Configure the test parameters.
- 10 _____
Verify the parameters in the Advanced field.
- 11 _____
Click **Create**. The test appears in the list of tests.

END OF STEPS

6.7 How do I execute an OAM test?

6.7.1 Steps

- 1 _____
Open **Data Collection and Analysis Management, Tests**.
- 2 _____
In the **Filter** column at the left of the view, configure a filter and click **Retrieve**. The list of tests is populated and filtered according to your input.
-  **Note:** You can sort the list by clicking on a column header; however, the sorting function is not available for all columns.
- 3 _____
Choose a test and click ⓘ to open the **Test Details** panel to verify that the test parameters meet your needs. If the **Deployed state** parameter is Deployed, the NSP has received confirmation from the MDM server that mediation was successful.
- 4 _____
From your selected test row, click ⋮ (Table row actions), **Execute**. The Execute dialogue opens.
- 5 _____
Configure the required parameters and click **Execute**.
- The Sync mode parameter specifies whether or not execution requests generate notifications in the UI. The **Sync-execute** option (default) provides a notification if the execution fails.
 - The Result Classifier parameter specifies the name of the result classifier used to determine test success or failure. Result classifiers are configurable using a REST API.
Check the **Perform result classification** check box to apply the classifications.
 - The **Publish results** check box publishes results to Kafka.

- The **Save results to database** check box makes results available to the NSP UI. If this box is not checked, the UI does not display the test results.

6

To stop a test that is running, choose the test in the list and click  (Table row actions), **Stop**.

END OF STEPS

6.8 How do I edit an OAM test?

6.8.1 Purpose

Use this procedure to update attributes of an existing test. To update a test attribute, the following criteria must be met:

- The test must have been created in NSP using a test template.
The Edit function is not available for tests that are discovered from the NE.
- The test must not be running.
- The attribute you need to change must be visible and editable in the test template.
See [6.20 “How do I edit an OAM test template?” \(p. 105\)](#) to update the template if needed.

6.8.2 Steps

1

Open **Data Collection and Analysis Management, Tests**.

2

In the **Filter** column at the left of the view, configure a filter and click **Retrieve**. The list of tests is populated and filtered according to your input.



Note: You can sort the list by clicking on a column header; however, the sorting function is not available for all columns.

3

Choose a test and click  (Table row actions), **Edit**.

4

In the form that opens, update the parameters as needed and click **Update**.

END OF STEPS

6.9 How do I delete an OAM test?

6.9.1 Steps

1

Open **Data Collection and Analysis Management, Tests**.

2

In the **Filter** column at the left of the view, configure a filter and click **Retrieve**. The list of tests is populated and filtered according to your input.



Note: You can sort the list by clicking on a column header; however, the sorting function is not available for all columns.

3

Choose a test and click (Table row actions), **Delete**.

END OF STEPS

6.10 How do I view OAM test results?

6.10.1 Purpose

Use this procedure to view results of tests. Test results are available in the NSP UI regardless of where the tests were originally configured, as long as the results are saved to the NSP database:

- For tests configured using the NFM-P UI, you must enable the **Publish to Kafka Topic** parameter when configuring the test in the STM.
The STM publishes results to the NFM-P Kafka topic, from which the MD-OAM app obtains the results and saves them to the database.
- For tests configured using the NFM-P REST APIs, you must ensure that **publishTopic=true** in **Sas.Test**. Results are published to the NFM-P Kafka topic, from which the MD-OAM app obtains the results and saves them to the database.
- For tests configured on the NE using CLI:
 - For OAM-PM tests (DMM/SLM/LLM/TWAMP Light), configured on the NE CLI, you must open the test in the NFM-P UI and enable the **Publish to Kafka Topic** parameter.
 - SAA based tests, such as Ethernet CFM, configured using CLI are not discovered by the NFM-P and not available to view in NSP.

See the NE documentation for details about creating the tests.

TWAMP Light delay results can also be provided in an accounting file, in the classes `telemetry:/base/oampm-accounting/twl-session-acc-stats` and `telemetry:/base/oampm-accounting/twl-bin-acc-stats`. TWAMP Light loss results are available in an accounting file in the class `telemetry:/base/oampm-accounting/twl-session-loss-acc-stats`

6.10.2 Steps

1

Open **Data Collection and Analysis Management, Tests**.

2

In the **Filter** column at the left of the view, configure a filter and click **Retrieve**. The list of tests is populated and filtered according to your input.



Note: You can sort the list by clicking on a column header; however, the sorting function is not available for all columns.

3

To view test information, choose a test and click **Test Details** ⓘ.

The Test Details panel shows the test attributes, and the Tested Entities panel shows the list of entities. The information provided depends on the test type.

4

To view results of a test, select the test and click ⋮ (Table row actions), **View Results**.

The Test executions page opens, showing the executions and their results.



Note: After a test has executed, there is a brief processing delay before results are available. For tests that have just finished running, Nokia recommends that you wait a minimum of 5 s before viewing the results.

5

Select a different time range for the test executions, if required. Choose a range from the drop-down list, or configure a customized time range.

6

If required, choose a telemetry type from the drop-down list.

7

To see additional information columns on the Test executions page, click ⋮ (Table row actions), **Manage Columns**. Select or de-select columns to suit your information needs.



Note: The sorting function is not available for columns on the Test executions page. By default, executions are sorted by Time Captured.

8

For more detailed information about a specific test, choose an execution and click **Test result details** ☑.

The information provided depends on the test type.



Note: Test result details are not available for child results, such as linktrace hops.

END OF STEPS

6.11 How can I troubleshoot TWAMP Light test issues?

6.11.1 Troubleshooting suggestions

The following guidelines apply to TWAMP Light tests, but may also be useful for other test types. Contact support if you need further assistance.

- The test does not start.
Usually, errors on test execution occur when there is a difference in state between the NSP and the NE or if test attributes are outside the accepted range of the NE. In this case, follow the guidance from the error or warning message returned when the test is started. Executing the test in sync-mode “sync-execute” ensures that the most complete error and warning messages are shown.
- The test starts and results are received, but the packet counts and delay values are all zero.
 - Confirm that the destination address is reachable from the NE executing the test.
 - Confirm that the TWAMP Light reflector is enabled, that the test dst-udp-port parameter matches the reflector listening port, and that the TWAMP Light reflector has a matching prefix for the test src IP address parameter.
- The test starts but no results are observed.
 - Check the execution status to confirm that the test is executing.
 - Confirm the test type and the expected result classes: verify from the results class (streaming or accounting) the expected delay until the first result is expected to arrive. Confirm that the correct class of result is selected, and click **Refresh Results** to trigger result retrieval from results stored to the database.
 - Confirm that the configuration is correct and complete: the deployment state is “deployed”, the streaming template is defined (for intended streaming delay results), the bulk result is true (for intended accounting results), and properties (IP addresses, UDP ports, measurement interval) are correct.
 - If streaming test results are intended, validate the configuration and status of the streaming delay template (especially the admin state and sample window period).
 - Validate telemetry subscriptions for the test classes: telemetry subscriptions are created automatically for tests created in the NSP. The subscription name format is `TestSuiteEx_OAM-PM-test_type-statistic-type`; for example, the subscription for TWAMP Light delay streaming statistics would be named `TestSuiteEx_OAM-PM-TWAMP-streaming`.
 - Validate the TWAMP Light reflector configuration:
 - Does the target node have a deployed, enabled reflector?
 - Does the UDP port match those of configured tests?
 - Is the reflector on the correct routing instance (router- or service-based)? TWAMP Light reflector prefixes can be viewed from the RESTCONF API.
 - Validate the bin group information for accounting type results: confirm that the bin group configured in the test is configured on the node and that admin state is enabled. Details of bin groups may be configured or retrieved using RESTCONF API or discovered from node configuration.

6.12 What is an OAM test suite?

6.12.1 OAM test suite

A test suite is a collection of tests that are grouped together to allow for multiple tests to be executed together or run in sequence. The suite includes both the tests and, where applicable, instructions for running tests sequentially or in parallel. Test suites can provide improved automation for OAM testing.

NSP provides a read-only list of test suites created in the MD-OAM RESTCONF API and in NSP.

See the OAM Tests tutorial on the [Network Developer Portal](#) for information about using the API.

 **Note:** On-demand test suites can be created and executed for LMM and SLM tests, but results may not be available to be shown.

6.13 How do I create an OAM test suite?

6.13.1 Steps

- 1

Open **Data Collection and Analysis Management, Test Suites**.
- 2

Click **+ Suite**.
- 3

In the Generate OAM Tests form that opens, choose a Test type.
The list of templates in the Template field is updated based on your selection.
- 4

Choose a test template if needed.
If a template is not selected, an appropriate system template is automatically selected based on the value of the execute type field. If a template is selected, the value of the execute type field is imported from the template and is read-only in the form.
- 5

Add one or more entities:
 1. Choose an entity type from the Entity type drop down.
 2. Click **+ Select** to open a selection form.
 3. Choose one or more entity objects from the list to add them to the Bin. Use the page selectors to navigate the list.
 4. Verify the list of entity objects in the Bin and click **Select**.
 5. To change the list of selected entities, repeat the previous steps to re-create the list.



Note: For test suites involving IP Edge NEs, 7705 SAR and 7210 SAS, the entity type must be MEP. See [6.5 "What test types are available for a specific NE?" \(p. 91\)](#) for supported test types on IP Edge NEs.

6

Click on the Service field if applicable:

1. Select an attribute in the drop-down list, then enter values for that attribute in the field. As you type, the list is filtered for entities that match your input.
2. Click  as required to add additional filter criteria.
3. Choose entities from the list and click **Select**.

7

Configure the test parameters as needed.

The test suite name must be unique.

8

Generate the test suite:

- a. To automatically execute the test suite after generation, enable Execute and click **Generate & Execute**.
- b. To create the test suite without automatically executing, disable Execute and click **Generate**.

END OF STEPS

6.14 How do I stop or start an OAM test suite?

6.14.1 Steps

1

Open **Data Collection and Analysis Management, Test Suites**.

2

Choose a test suite and click  (Table row actions), **Execute**.

3

In the form that opens, update the parameters if needed and click **Execute**. The execution status is updated to Running.



Tip: Disable the **Publish results** parameter if you do not need the results to be published to Kafka. This may reduce the processing impact.

4

To stop a test suite that is running, choose the test suite in the list and click  (Table row actions), **Stop**.

END OF STEPS

6.15 How do I edit or delete an OAM test suite?

6.15.1 Supported functions

Test suites can only be modified using the MD-OAM RESTCONF API.

To delete a test suite, choose a test suite and click  (Table row actions), **Delete**.

Deleting a test suite deletes all associated tests.

6.16 How do I view OAM test suite results?



Note: After a test has executed, there is a brief processing delay before the results are available. For tests that have just finished running, Nokia recommends that you wait a minimum of 5 s before viewing the results.

6.16.1 Steps

1

Open **Data Collection and Analysis Management, Test Suites**.

2

Choose a test suite and click  (Table row actions), **View Details**.

The View Test Suite Details page opens, showing the following:

Tab	Notes
Aggregated Results	Each row of the aggregated results table corresponds to an execution of the test suite. Whenever the test suite is successfully started, a new aggregated results row is added. To view the individual results for a specific test suite execution, select the aggregated results row corresponding to the execution you are interested in and click View individual results  .
Lifecycle Results	The Lifecycle Results table shows events from the execution of the test suite, such as stop and start timestamps and error events.

Tab	Notes
Individual Results	The page displays the results of each test executed. By default, the results from the most recent test suite execution are shown, that is, the execution ID from the first row of the aggregated results table is chosen automatically. You can view results for other test suite executions by specifying another execution ID in the Test suite execution ID field or by returning to the Aggregated Results tab and clicking View individual results . For more detailed information about a specific test, choose an execution and click View Results . Note: If a test suite was created from the NSP, the tests will all be the same type. If the test suite was created using RESTCONF, multiple test types could be included. To view results from a different type than is currently displayed, choose the telemetry type from the drop-down list. See the Tests tab for a list of tests in the suite.
Generation Log	The page displays log information from the generation of the suite and tests.
Tests	The page lists the test identifiers included in the suite. If the test suite is an on-demand suite, the Tests tab shows the included tests grouped by stage. Stages are executed sequentially, but tests within each stage will be executed either sequentially or in parallel depending on how the stage is configured. For more detailed information about a specific test, double click on an execution or choose an execution and click View Results .

END OF STEPS

6.17 What is an OAM test template?

6.17.1 OAM test template

An OAM test template is a set of parameters that a user can employ to create or customize a test for a particular use case. You can create a template for each test type, or create separate templates for different use cases. The test template becomes the form used to create and run the test in the future.

On-demand templates are for tests that run a single execution; proactive templates are for tests that execute repeatedly.

6.17.2 Template parameters

A test template includes all the parameters for a test. The template creator can set the parameters or leave them blank and can choose whether the parameter is visible or editable when the template is used.

When a test is created from the template, the parameters are applied as follows:

- If the template creator sets the parameter value in the template and does not set the parameter instance as editable, the test runs with the value set in the template.

- If the template creator sets the parameter instance as editable, the test runs with the value set by the user who creates the test. The template creator can also set the parameter value, creating a template default.
- If the template creator does not set the parameter value and does not set the parameter instance as editable, the test runs with the default value in the YANG model, if applicable.

Test templates can use variable substitution to auto-populate some values, based on entity mapping in the YANG model. The definition of a variable has the following syntax:

`${variable-name}`

For example, "`service:${service-name}`" is resolved from the current test object after any source/destination entities are resolved.

For more information about variable substitution, see the OAM Tests tutorial on the [Network Developer Portal](#).

i **Note:** Templates for use with IP Edge NEs, 7705 SAR and 7210 SAS, have specific requirements; see [6.19 "How do I create an OAM test template?" \(p. 103\)](#). A compliant template must be used for any test involving an IP Edge NE, including tests between an IP Edge NE and another NE type.

6.18 What is a system template?

6.18.1 System template

System templates are available to the NSP for test creation.

An operator can create tests using any template. However, when the NSP creates a test, it must use a system template. This requirement includes test suites created by an operator.

Default system templates are provided with installation of the NSP. You can modify these templates and add additional ones.

6.19 How do I create an OAM test template?



CAUTION

Service Disruption

The name of an object, including subscriptions, baselines, indicators, templates, and chart profiles, cannot contain a semicolon (;) or backslash (\).

The use of these characters in an object identifier will result in corrupted data that must be deleted by Nokia support.

i **Note:** Templates for use with IP Edge NEs, 7705 SAR and 7210 SAS, have specific requirements; see [Step 4](#). A compliant template must be used for any test involving an IP Edge NE, including tests between an IP Edge NE and another NE type.

6.19.1 Steps

- 1

Open **Data Collection and Analysis Management, Test Templates**.
- 2

Click **+Template**.
- 3

In the Create Test Template form that opens, configure the template settings in the panel on the left.

By default, the settings are not visible or editable at test creation. Enable visibility or editing for each setting as needed. When you enable editing, visibility is automatically enabled.
 - Configure the **Execute type** parameter to configure the template as On-demand or Proactive.
 - Select **System Template** to configure the template as a system template.If the template will be used in tests involving IP Edge NEs, proceed to [Step 4](#).
- 4

If you are configuring a template for use with tests involving 7705 SAR or 7210 SAS NEs, ensure that the requirements for IP Edge NE tests are met:
 - The **Destination Type** must be **Destination MAC address**, and the field must be editable.
 - The name of the configured MA must start with *AssociationNameType-*.
where:
 - *AssociationNameType-* is the configured Association name type followed by a hyphen, for example, String-test15.
 - If the Association name type is VPN ID, the format is VpnID-, for example, VpnId-00AABB:EEFFCC66.
 - If the Association name type is ICC-based, the format is lccBased-, for example, lccBased-ABCD12EFGH01.
 - The name of the configured MD must start with *DomainNameType-*.
where:
 - *DomainNameType-* is the configured Domain name type followed by a hyphen, for example, Name-test15.
 - DNS and MAC must be capitalized in MD names, for example, MAC-FA:13:3F:2F:80:AF-1234
 - If the Domain name type is Format, the MD name must include the level, for example, None-7.
- 5

View the test creation form layout in the panel on the right side of the page. Drag the fields to change the order if needed.

-
- 6 _____
Click **Create**.

END OF STEPS _____

6.20 How do I edit an OAM test template?

6.20.1 Steps

- 1 _____
Open **Data Collection and Analysis Management, Test Templates**.
- 2 _____
Choose a template and click  (Table row actions), **Edit**.
Use the **All Templates** drop-down list if required to filter the list to show only on-demand or proactive templates.
- 3 _____
In the form that opens, update the parameters as needed and click **Update**.

END OF STEPS _____

6.21 How do I delete an OAM test template?

6.21.1 Steps

- 1 _____
Open **Data Collection and Analysis Management, Test Templates**.
- 2 _____
Choose a template and click  (Table row actions), **Delete**.
- 3 _____
Click **Delete** to confirm the deletion.

END OF STEPS _____

6.22 What is a config object?

6.22.1 Definition

A config object represents an OAM configuration object that is used for testing, other than the test object itself.

NSP provides a list of configuration objects. Click on the object type drop-down list to switch to the list for a different object type.

6.23 What config objects are available in NSP?

6.23.1 Objects presented

The following table describes the available configuration objects.

Configuration object type	Description	Notes
Bin Group	A bin group is a template of Performance Measurement histogram measurement bin definitions. Bin groups contain measurement bins, which count instances of frame delay (FD), frame delay range (FDR), or interframe delay variation (IFDV) metrics within a defined range of lower to upper threshold values. Bin group definitions may also include definitions for delay-event thresholds and calculation of average FD, FDR, and IFDV values.	Bin groups define a service level agreement quality quantitatively. Service offerings with differentiated service level agreements may be measured using different bin groups.
CFM Domain	CFM Maintenance Domain (level 0 to 7)	Each MD can include multiple MAs.
CFM Association	CFM maintenance associations	Auto-configuration of MAs is supported when creating CFM-based tests.
CFM MEP	CFM maintenance entity point. An MEP is an endpoint in an MA, on which CFM tests can be performed. Each MEP with the same MD and MA represents endpoints for a single service.	Service MEPs, such as MEPs on SAPs, and facility MEPs, such as MEPs on LAG ports or network interfaces, are supported. Auto-configuration of some MEPs is supported when creating CFM-based tests. Facility MEPs on LAG ports or network interfaces cannot be auto-generated.
Streaming Delay Template	A template that defines the real-time measurement of FD and IFDV from delay-based Performance Management tests. Average values are calculated over 10-to-60-second intervals and stream in real time.	NSP supports delay streaming using gNMI, for 7750 SR NEs.

Configuration object type	Description	Notes
TWAMP Server	TWAMP is a delay/loss measurement protocol that uses both the TCP connection service and the UDP session service. A TWAMP server may be configured to accept TCP control connections from a specified range of IP networks and TCP ports. It will also accept requests for clients to initiate UDP test sessions for specific IP address or UDP port pairs on the service router.	—
TWAMP Light Reflector (ID)	A TWAMP Light reflector controls the set of IP network and UDP ports where the service router replies to TWAMP Light test messages. TWAMP Light reflectors may be configured for both base and service routing instances.	Auto-create/update of TWAMP Light reflectors is supported when starting TWAMP Light tests.
TWAMP Light Reflector (Router)		
TWAMP Light Reflector (Service)		

6.23.2 What config objects are supported for a specific NE?

The ability to support config objects varies by NE type. The following table shows available config objects by NE type.

A check mark in a cell indicates that the config object is supported for the NE type.

Config objects on 210 WBX and Omniswitch NEs are not supported. For information about other NE types, contact Nokia.

Config object	NE type				
	7750 SR, 7450 ESS, 7950 XRS	7250 IXR	7705 SAR	7210 SAS	Wavence
Session Reflectors (TWAMP Light)	✓	✓	✓	✓	✓
CFM Maintenance Domain (Name, Type, Level)	✓	✓	✓	✓	✓
CFM Maintenance Entity Group [Global & Sites] (Name, Type)	✓	✓	✓	✓	Not supported by the NE.
CFM Maintenance End Points (MEPs)	✓	✓	✓	✓	✓

Config object	NE type				
	7750 SR, 7450 ESS, 7950 XRS	7250 IXR	7705 SAR	7210 SAS	Wavence
Bin Groups	✓	✓	Not supported by the NE.	✓	✓
OAM-PM Measurement Intervals	✓	✓	Not supported by the NE.	✓	✓

6.24 What are the best practices for creating config objects for classic NEs?

6.24.1 Classic NE limitations and best practices

Some limitations apply when creating config objects, especially for classic NEs.

- TWAMP Light Reflector (for model-driven and classic nodes):
Valid values for the UDP port field in NSP are 0, 1 - 65 535.
- Bin Groups:
There is no bin count entry in NSP, and NSP does not create default bins for classic NEs. At least two bin entries must be added for each BinType (fd, fdr, ifdv) when a BinGroup is created, and one of the BinNumbers must be 0.
- CFM Domain and CFM Association:
The MA-Index and MD-Index fields must not be modified during editing of a CFM Domain or CFM Association flow.

6.25 How do I create or change a config object?

6.25.1 Purpose

Use this procedure to create or modify a config object from the NSP UI. Parameters in the Create OAM Config Object form depend on the type of config object being created. Some parameters may not be applicable to all NE types.

Config objects can also be created using RESTCONF, using classic IP mediation, or, for CFM objects in particular, can be auto-generated when CFM tests are created.



Note: Naming convention requirements apply for config objects that involve IP Edge NEs, that is, 7705 SAR and 7210 SAS. If a config object may be used in a test involving an IP Edge NE, the naming convention must be followed. This includes hybrid scenarios involving an IP Edge NE and another NE type. See the [Network Developer Portal](#) for more details.



CAUTION

Service Disruption

The name of an object, including config objects, cannot contain a semicolon (;) or backslash (\).

The use of these characters in an object identifier will result in corrupted data that must be deleted by Nokia support.

6.25.2 Steps

1

Open **Data Collection and Analysis Management, Config Objects**.

2

Select a type of config object from the selector below the **Data Collection and Analysis Management** breadcrumb.

The default is **CFM Domain**.

3

To create a config object:

1. Click **+Create**.
2. In the Create OAM Config Object form that opens, configure the parameters as needed.
3. Click **Create**.

The config object appears in the view.

4

To edit a config object:

1. Choose a config object and click  (Table row actions), **Edit**.
2. In the form that opens, update the parameters as needed and click **Update**.

5

To delete a config object:

Choose a config object and click  (Table row actions), **Delete**.

The config object is removed immediately.

END OF STEPS

6.26 What can I do with config objects in NSP?

6.26.1 Supported operations

You can create, view, edit, or delete a config object in the NSP UI.

Any other actions must be performed using RESTCONF or classic IP mediation, depending on how the object was created.



Note: Some config object parameters are not displayed in the **Data Collection and Analysis Management, Config Objects** view. All parameters are displayed in the **Edit OAM Config Object** form for a selected config object.

Part II: Sample procedures

Overview

Purpose

Describes sample procedures for Data Collection and Analysis Management and Data Collection and Analysis Visualizations.

Contents

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7 Telemetry management sample procedures

7.1 Pathway: install and locate specific telemetry types

7.1.1 Purpose

This process shows how to use NSP to set up and locate the telemetry types needed for a sample scenario.

In this scenario, we want to start monitoring the aggregate-dropped packet count for egress traffic on VPRN SAPs on SR OS 24.10 devices. We need to install the required files and confirm the telemetry types we'll need to create subscriptions for.

This scenario assumes that the adaptation artifact bundle for SR OS 24.10 with the current NSP release has been installed. This installation populates the vendor agnostic and accounting adaptation for SR OS.

This scenario also requires that SR OS device telemetry definitions and telemetry adaptation bundles are downloaded from the Nokia [Support Portal](#); see the artifact guide for the filenames and download paths.

For example, in the first iteration of Release 25.8, the filenames are `nsp-telemetry-yang-sros-25_8.zip` and `nsp-telemetry-cr-ct-sros-4.1.0-rel.0.zip`.

7.1.2 Stages

1

Check the Telemetry Statistic Search Tool:

1. Open the NSP Help Center. From the Help Center home screen, navigate to **Reference, Statistic Search, Telemetry Statistic Search**.
2. Enter `aggregate-dropped` in the Counter Name field. The search tool filters for telemetry types with `aggregate-dropped` in the counter name.
Enter additional filters as needed to narrow the search.
3. Click on each row to see the Info panel. Verify whether the selected telemetry type will monitor the counters you need.
4. If the telemetry type is found, you are ready to [set up your subscriptions](#).

For our example, the telemetry type we need is not in the NSP definition. Next, we'll need to see if it's defined in the device telemetry YANG definitions.

2

Extract the files from the telemetry YANG zip and look for the telemetry types:

1. View the list of telemetry files. The filename `sros-service-vprn-telemetry.yang` suggests it could contain the telemetry types we need.
2. Open the file and search for "aggregate". The desired counter is present in a leaf element.

The telemetry type appears in the list:

```
leaf aggregate-dropped-packets{
tel-ext:coll-format "intervalized";
type uint64 {
range "0..max";
} //type uint64
} //leaf aggregate-dropped-packets
leaf aggregate-dropped-octets{
tel-ext:coll-format "intervalized";
type uint64 {
range "0..max";
} //type uint64
} //leaf aggregate-dropped-octets
leaf last-cleared-time{
type string;
} //leaf last-cleared-time
container db-info {
leaf index-1 {
type string;
tel-ext:unique-index "true";
default "time-captured system-id object-id";
}
leaf partition-key {
type string;
default "time-captured";
}
leaf partition-interval {
type string;
default "hourly";
}
leaf order-by {
type string;
default "system-id";
}
leaf segmented-by {
type string;
default "system-id";
}
leaf segmentation {
type boolean;
default "true";
}
} //container db-info
} //list service_vprn_interface_sap_egress_qos_sap-egress_traffic-manager_statistics
```

3

Install the `sros-service-vprn-telemetry.yang` file using **yang-file.bash**; see “How do I manage MDM model definitions?” in the *NSP System Administrator Guide*.

4

Now that the YANG file we need has been installed, we can install device adaptation files. For this example, the device adaptation bundle is `nsp-telemetry-cr-ct-sros-4.1.0-rel.0.zip`.

Install the bundle from the **Artifacts, Artifact Bundles** view; see “How do I install an artifact bundle?” in the *NSP Network Automation Guide*.

5

Check the Telemetry Statistic Search Tool:

1. Open the NSP Help Center. From the Help Center home screen, navigate to **Reference, Statistic Search, Telemetry Statistic Search**.
2. Enter `aggregate-dropped-packets` in the Counter Name field to verify that the artifacts have been loaded successfully and are valid for SR OS 24.10.

3. Click on each row to see the Info panel and obtain further details about the counter.

7.1.3 We're done

The counters we need are available in the NSP, and we have the information we need to [configure a subscription](#).

7.2 Setting up statistics and aggregation for Port Throughput on the 7750 SR

7.2.1 Purpose

This sample procedure shows how to use NSP to set up the data collection and aggregation required to run a Port Throughput report in Analytics for 7750 SR NEs managed using MDM.

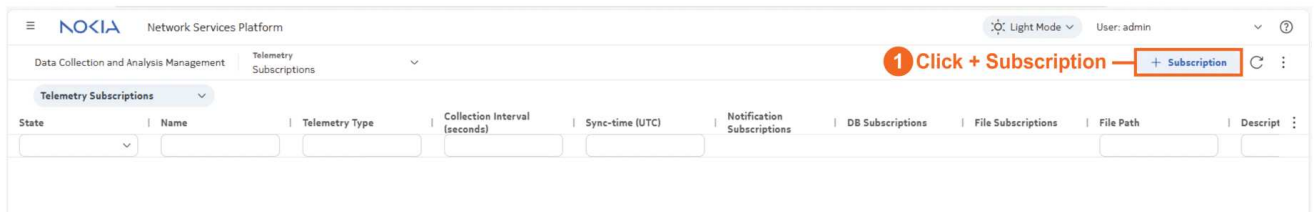
The following prerequisites must be completed:

- Telemetry types for port throughput have been installed.
- At least one 7750 SR NE has been discovered in the network and is reachable.
- Aggregation rules have been deployed.

7.2.2 Steps

1

From **Data Collection and Analysis Management, Subscriptions**, we'll create a subscription.



40485

We need interface utilization statistics for 7750 SR NEs. An object filter that contains “7750” ensures that we receive data whether the NE type says “7750 SR” or “7750 SR-12”.

Create Subscription

General

Filters & Counters

1 Enter a name for the subscription

Subscription Name: SRforReporting

Collection Interval (seconds): 900

Sync-Time (hh:mm): 00:00

State: Enabled

2 Set database storage to Enabled to store the data in the auxiliary database

DB Subscriptions: Enabled

File Subscriptions: Disabled

Directory Path for File Output: e.g. path/to/file

Filters & Counters

3 Filter the data to only store data from an NE with a product type containing 7750

Object Filter: 1 /nsp-equipment:network/network-element[contains('product', '7750')]

4 Enter the statistics type

Telemetry Type: telemetry:/base/interfaces/utilization

☐ Enable notifications and notification counters

+ Counters

Counter

- input-utilization
- input-utilization-periodic
- interface-id
- output-utilization
- output-utilization-periodic
- received-octets
- received-octets-periodic

5 Click Create

Cancel Create

40479

2

Let's look at the aggregation rule for interface data. From the Aggregation view, choose the aggregation rule and click **Edit Aggregation Rule** to see the details.

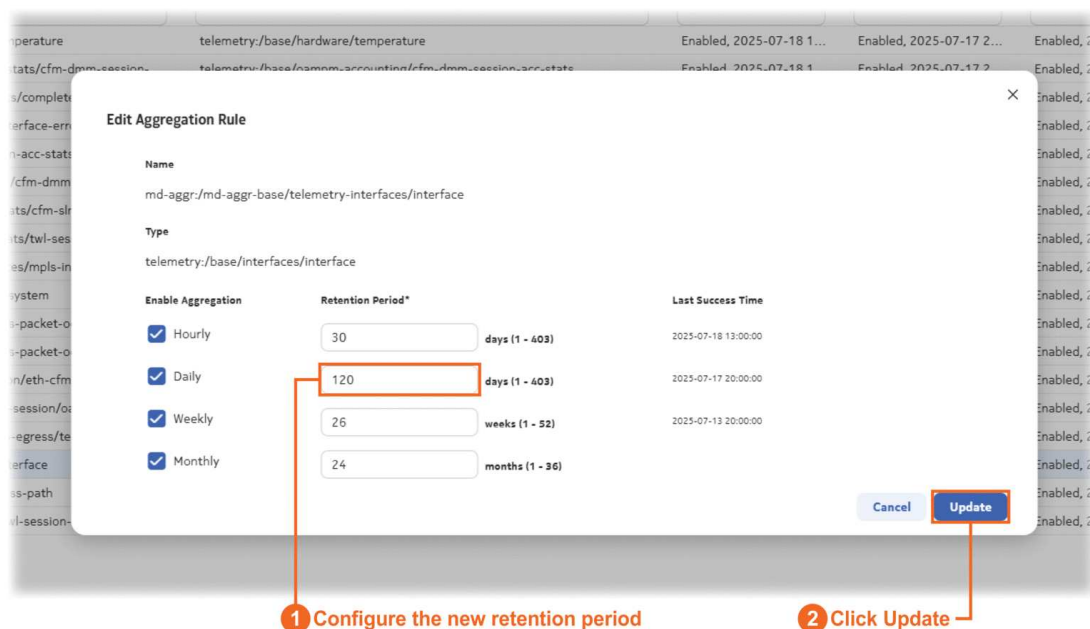
The **Retention Period** fields show how long aggregated data is stored in the auxiliary database. The default retention period for daily aggregation is 90 days. This means that daily granularity is available in Analytics for up to 90 days of data.

Enable Aggregation	Retention Period*	Last Success Time
☒ Hourly	30 days (1 - 403)	2025-07-18 13:00:00
☒ Daily	90 days (1 - 403)	2025-07-17 20:00:00
☒ Weekly	26 weeks (1 - 52)	2025-07-13 20:00:00
☒ Monthly	24 months (1 - 36)	

40487

3

Changing the retention period to 120 days makes daily granularity available for a longer time. Change the retention time and click **Update**.



40488

END OF STEPS

We're done

We've set up a subscription for interface utilization data, and aggregation is deployed. When data has been successfully collected and aggregated, we'll be able to run a Port Throughput Summary (NSP) report in the Analytics application.

7.3 Setting up baselines

7.3.1 Purpose

This sample procedure shows how to use NSP to create baselines to monitor transmitted and received octets on a network port. These baselines can be used to detect anomalies and to create baseline and anomaly charts.

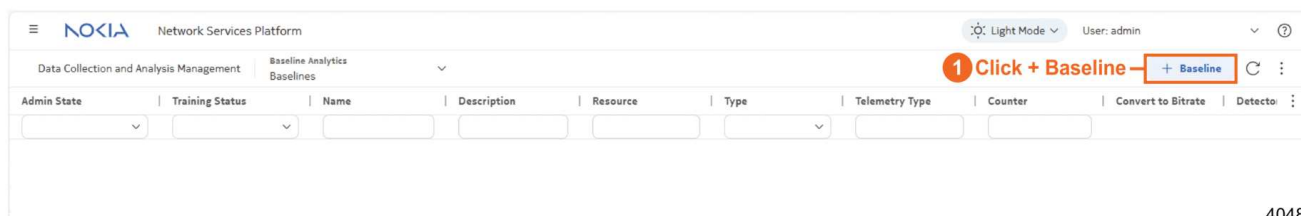
The following prerequisites must be completed:

- The NE we need to monitor has been discovered in the network and is reachable.
- If the NE we need to monitor is managed by NFM-P, statistics collection has been configured and started in the NFM-P.

7.3.2 Steps

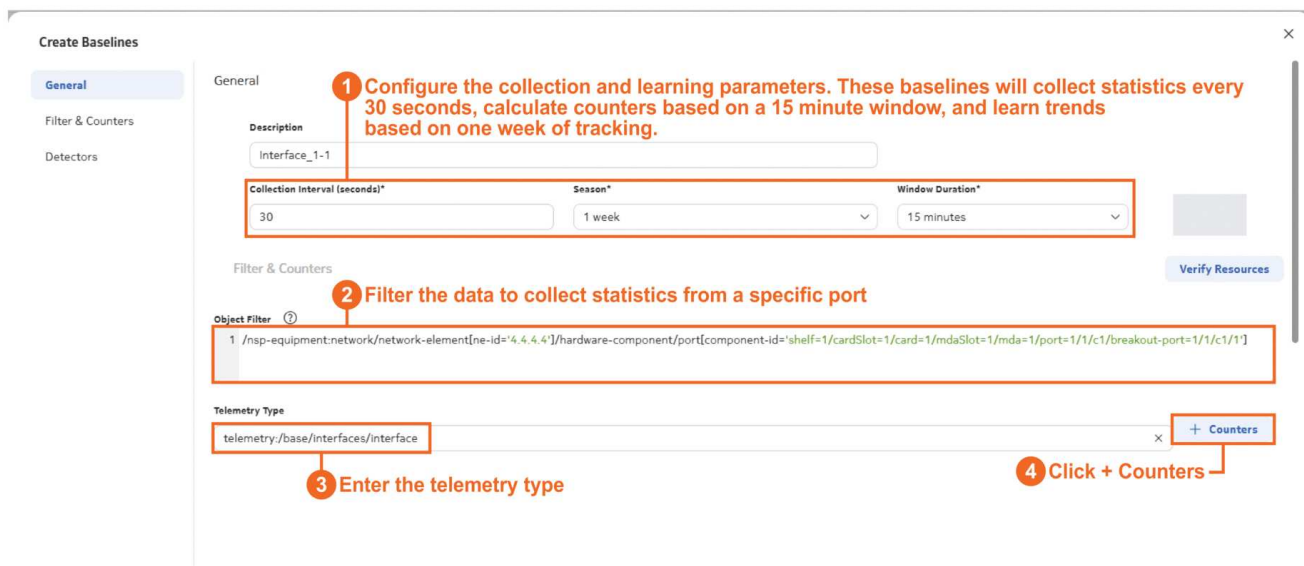
1

From **Data Collection and Analysis Management, Baselines**, we'll create baselines.



40482

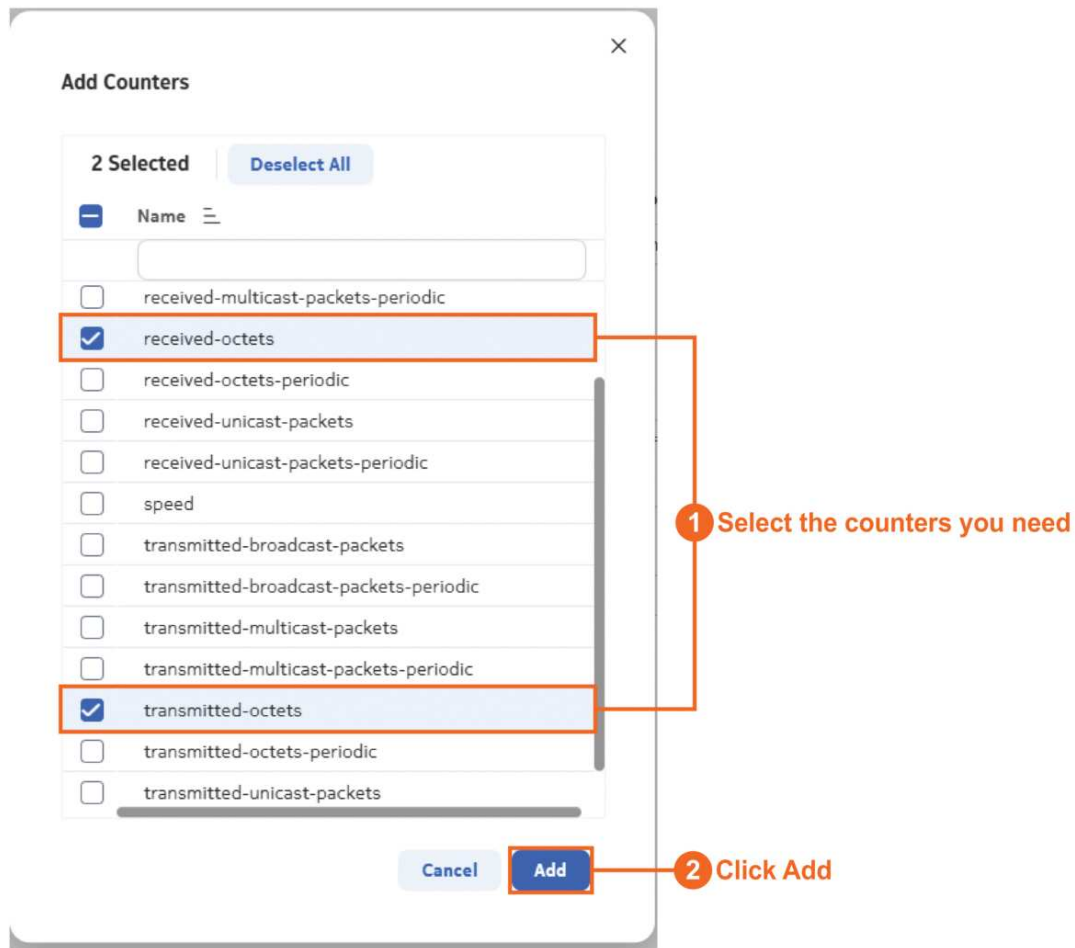
Creating an object filter to specify the NE and port makes finding the resources easier.



40489

2

A form opens, showing the available counters. We'll choose transmitted and received octets. Select the counters and click **Add**.



40497

3

We can configure the counter type or conversion to bitrate if needed. We'll use the defaults for these baselines.

Create Baselines

General

Filter & Counters

Detectors

General

Description

Interface_1-1

Collection Interval (seconds)*

300

Season*

1 week

Window Duration*

15 minutes

Filter & Counters

Object Filter

1 /nsp-equipment:network/network-element[ne-id='4.4.4.4']/hardware-component/port[component-id='shelf=1/cardSlot=1/card=1/mdaSlot=1/mda=1/port=1/1/c1/breakout-port=1/1/c1/1']

Telemetry Type

telemetry:/base/interfaces/interface

+ Counters

Counter	Type	Units Name	Convert to Bitrate
received-octets	Counter		☐
transmitted-octets	Counter		☐

2 Click Verify Resources

Verify Resources

1 Configure counter parameters as needed

40505

4

NSP checks for resources corresponding to the filters and counters specified. When the resources we need appear in the resources list, click **Stop Verification**. The verification does not stop on its own.

Create Baselines

General

Resources List

Detectors

Resources List

1 When the resources we need appear in the list, click Stop Verification

Stop Verification

Provider	Name	Counter	Type	Units Name	Convert to Bitrate	Resource
MDM	{4.4.4.4}, 1/1/c1/1	received-octets	Counter			/network-device-mgr:network-devi...
MDM	{4.4.4.4}, 1/1/c1/1	transmitted-octets	Counter			/network-device-mgr:network-devi...

40500

5

Select the resources in the resources list.

The screenshot shows the 'Create Baselines' window with the 'General' tab selected. The 'Resources List' section displays a table with the following data:

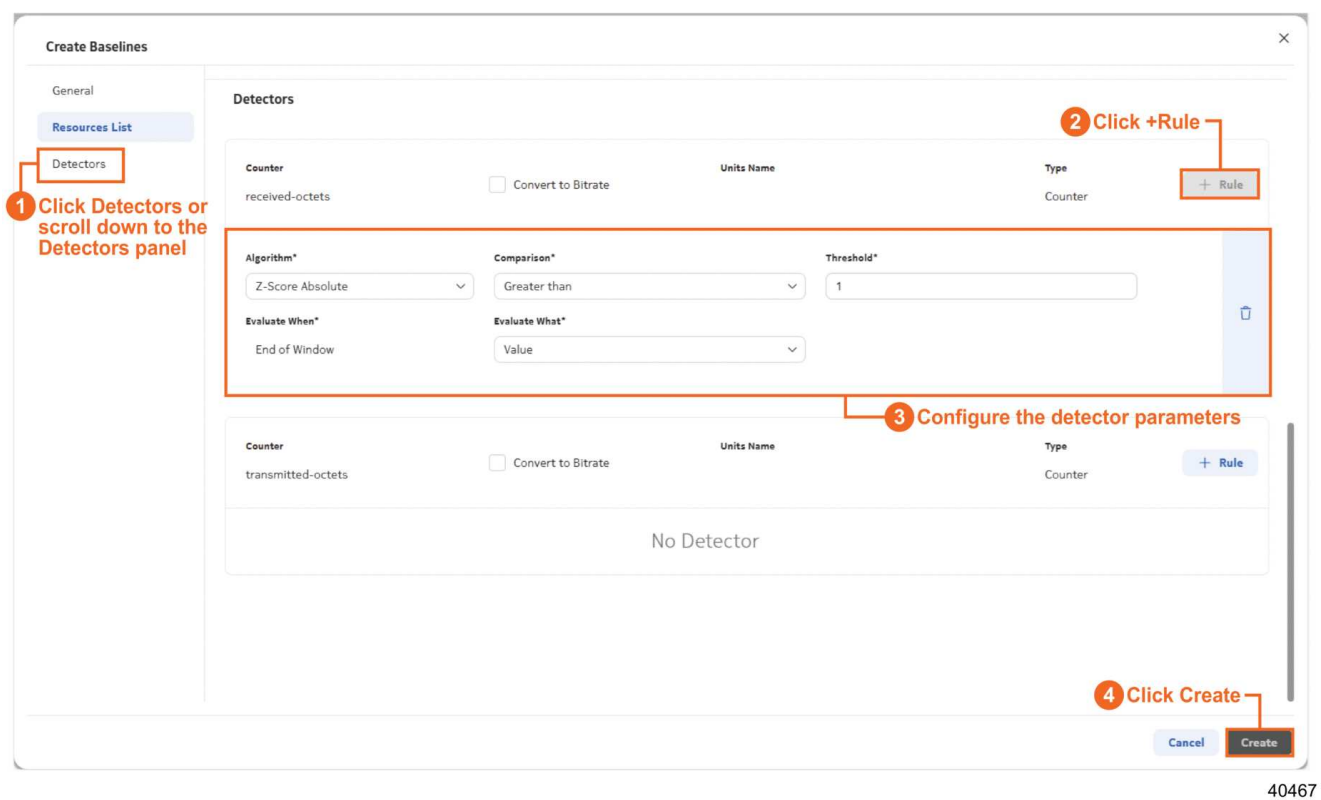
Provider	Name	Counter	Type	Units Name	Convert to Bitrate	Resource
☒	MDM	{4.4.4.4}, 1/1/c/1/1	received-octets	Counter		/network-device-mgr:network-devi...
☒	MDM	{4.4.4.4}, 1/1/c/1/1	transmitted-octets	Counter		/network-device-mgr:network-devi...

An annotation '1 Select the resources we need' points to the selection checkboxes in the first two rows of the table.

40498

6

The final step is configuring anomaly detection. Then, we can click **Create**.



40467

END OF STEPS

We're done

NSP creates a baseline for each resource. When complete, they appear in the list in **Data Collection and Analysis Management, Baselines**.

NOKIA Network Services Platform									
Data Collection and Analysis Management									
Baseline Analytics									
Baselines									
+ Baseline									
Admin State	Training Status	Name	Description	Resource	Type	Telemetry Type	Counter	Convert to Bitrate	Detector
Enabled	Active	nodeD (4.4.4.4), 1/1/c...	Interface_1-1	/network-device-mgr.n...	Counter	telemetry/base/interf...	received-octets		✓
Enabled	Active	nodeD (4.4.4.4), 1/1/c...	Interface_1-1	/network-device-mgr.n...	Counter	telemetry/base/interf...	transmitted-octets		✓

40472

7.4 Creating a simple indicator

7.4.1 Purpose

This sample procedure shows how to use NSP to create an indicator with a single counter and no formula. This indicator outputs the input utilization of each port on NE 4.4.4.4.

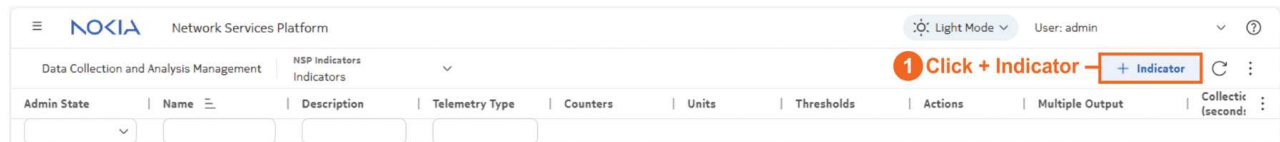
The following prerequisites must be completed:

- The NE we need to monitor has been discovered in the network and is reachable.
- If the NE we need to monitor is managed by NFM-P, statistics collection has been configured and started in the NFM-P.

7.4.2 Steps

1

From **Data Collection and Analysis Management, Indicators**, we'll create an indicator.



40469

Enter a name for the indicator, and configure the collection interval and window duration.

Create Indicator

General

Configuration

Thresholds

General

Configuration

Counters & Formula

Object Filter

1

Configure the name and collection parameters. This indicator will collect statistics every 900 seconds (15 minutes). The window duration is ignored for simple indicators.

Apply Template...

Name*

NE 4 input util

Description

Collection interval (seconds)*

300

Window Duration* ⓘ

15 minutes

Units

Telemetry Type*

Type to find telemetry type...

+ Counters

Selected Counter(s)*

No Counters Added

Formula ⓘ

Object Filter ⓘ

1

Cancel

Create

40490

2

Configure the telemetry type.

Create Indicator

General

Apply Template...

Name*

NE 4 input util

Description

Collection Interval (seconds)*

300

Window Duration* ⓘ

15 minutes

Configuration

Units

%

1 Enter the units to display the values

Counters & Formula

Telemetry Type*

telemetry:/base/interfaces/utilization

2 Enter the telemetry type

3 Click + Counters

+ Counters

Selected Counter(s)*

No Counters Added

Formula ⓘ

Object Filter ⓘ

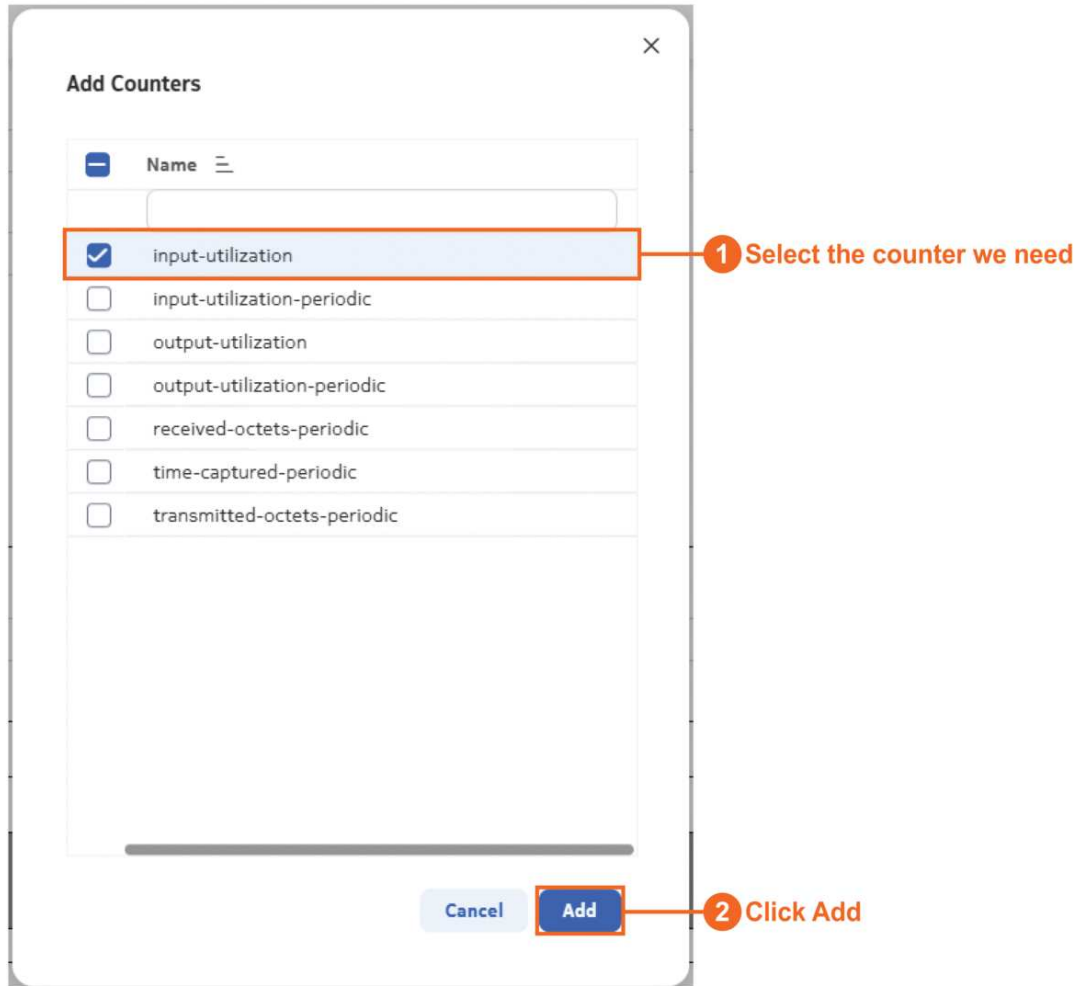
1

Cancel Create

40478

3

A form opens, showing the available counters. We'll choose input utilization.



40481

4

Configure an object filter to collect statistics from a specific NE, and click **Verify Resources** to make sure the filter is correct.

Create Indicator

General

Configuration

Thresholds

General

Collection Interval (seconds)*

300

Window Duration* ⓘ

15 minutes

Configuration

Units

%

Counters & Formula

Telemetry Type*

telemetry:/base/interfaces/utilization

+ Counters

Selected Counter(s)*

input-utilization

Formula ⓘ

1

Object Filter ⓘ

1 /nsp-equipment:network/network-element[ne-id='4.4.4.4']

1 Filter the data to collect data from a specific NE

Resource List

2 Click Verify Resources

Verify Resources

40496

5

When the resources we need appear in the resources list, click **Stop Verification**. The verification does not stop on its own.

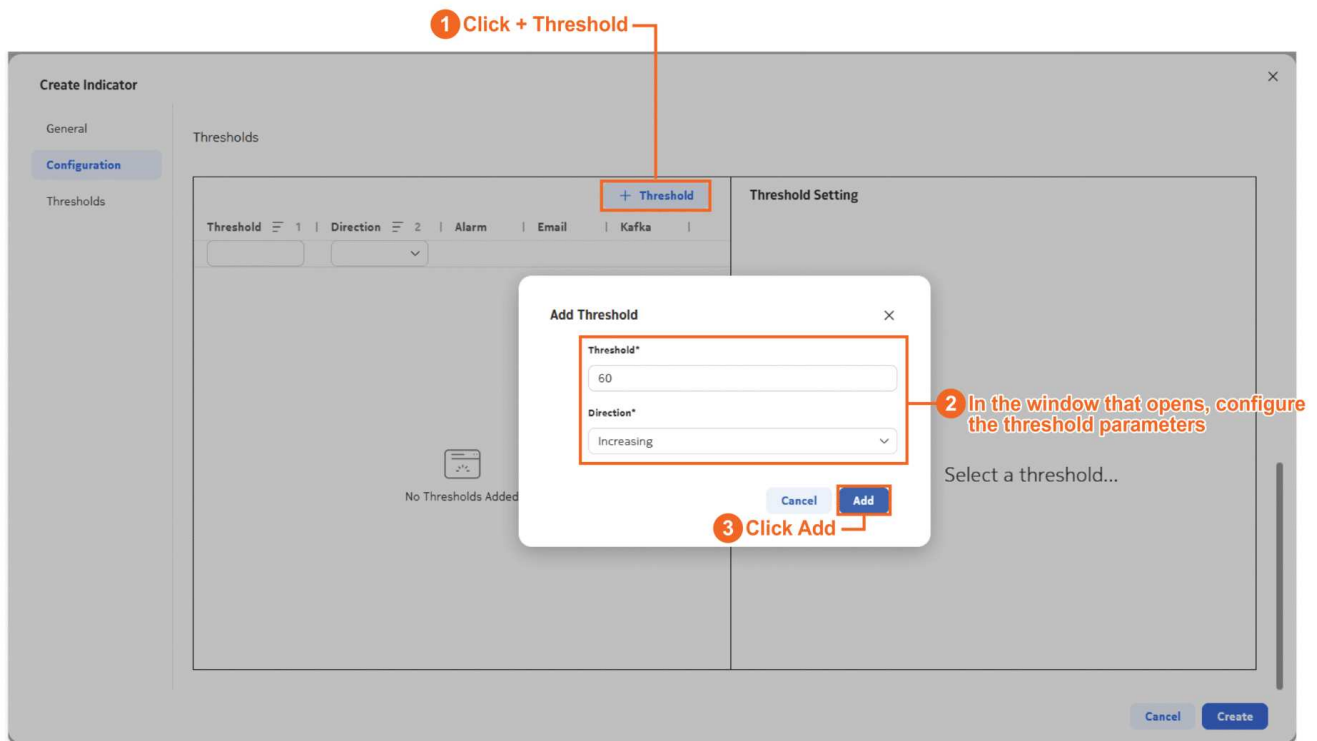
6

Create thresholds as needed. We'll create thresholds for utilization increasing past 60% and for decreasing below 5%.

128

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3HE-21451-AAAC-TQZZA

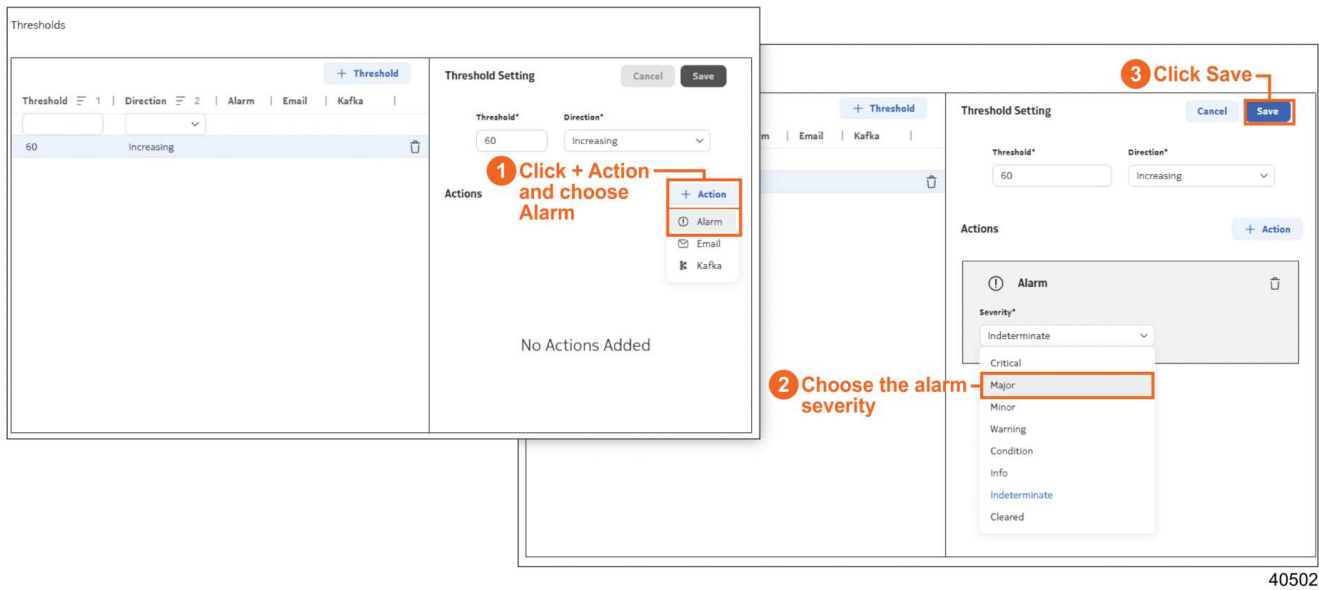
Release 25.11
July 2026
Issue 2



40501

7

Configure threshold actions. We'll configure a Threshold Crossing Alarm alarm to be generated for each threshold. The alarm for the increasing threshold has Major severity, and the alarm for the decreasing threshold has Minor severity.



8

When both thresholds are configured, we can click **Create** to create the indicator.

Create Indicator

General

Configuration

Thresholds

Thresholds

+ Threshold

Threshold	Direction	Alarm	Email	Kafka
60	Increasing	✓		
5	Decreasing			

Threshold Setting

Threshold*

5

Direction*

Decreasing

Actions

+ Action

Alarm

Severity*

Minor

Cancel

Create

3 Click Create

40484

END OF STEPS

We're done

The indicator appears in the list in **Data Collection and Analysis Management, Indicators**.

Network Services Platform

Light Mode

User: admin

Data Collection and Analysis Management

NSP Indicators

Indicators

+ Indicator

Admin State	Name	Description	Telemetry Type	Counters	Units	Thresholds	Actions	Multiple Output	Collectio (second
Enabled	NE 4 input util		telemetry:/base/interf...	input-utilization	%	✓	✓	✓	300

40499

7.5 Creating a complex indicator using a template

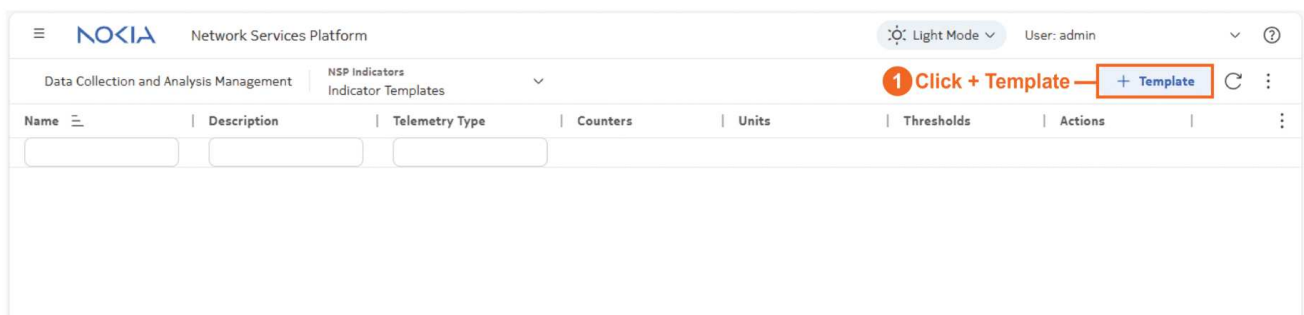
7.5.1 Purpose

This sample procedure shows how to use NSP to create an indicator template with a formula and apply the template to an indicator.

7.5.2 Steps

1

From **Data Collection and Analysis Management, Indicator Templates**, we'll create a template.



40486

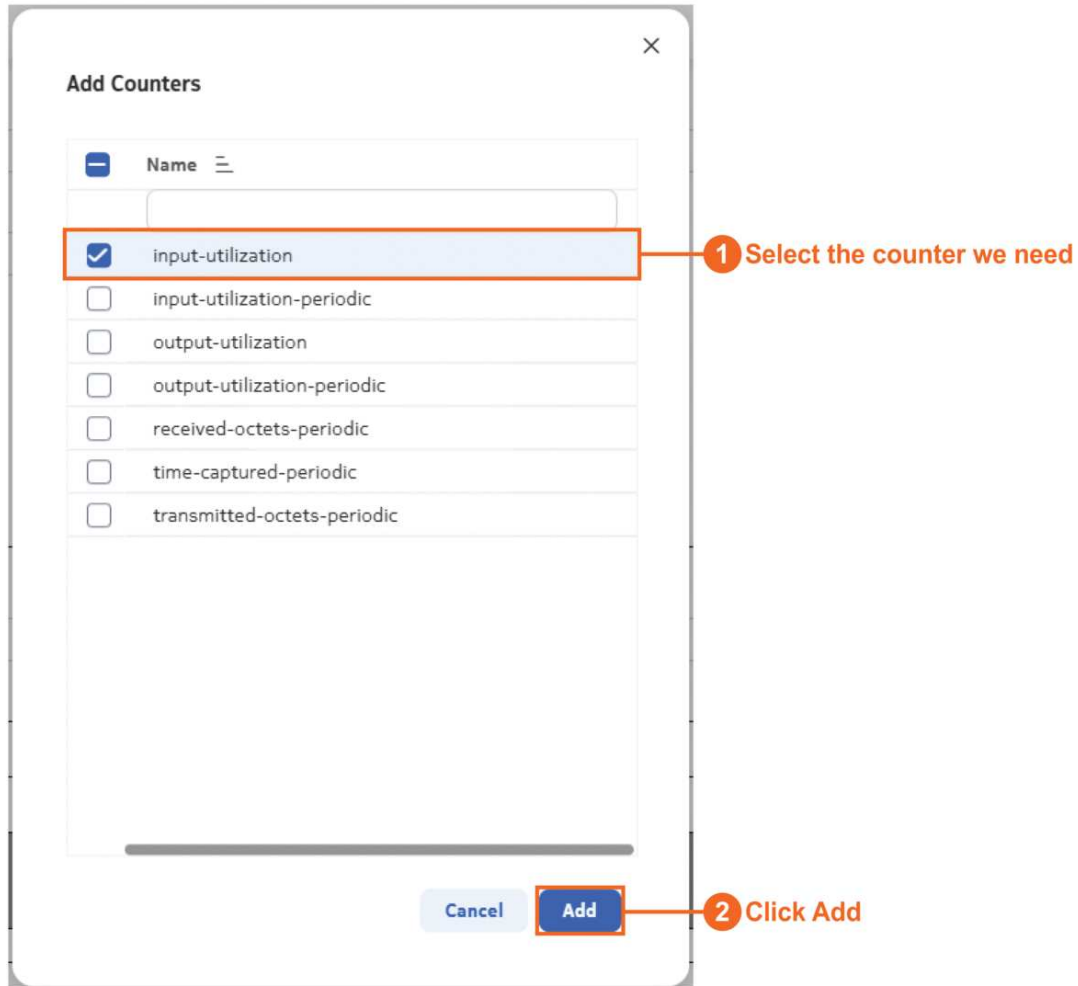
The screenshot shows the 'Create Indicator Template' dialog box with the following components and annotations:

- General Tab:** The 'Name*' field contains 'Ring Utilization' and is highlighted with a red box and annotation 1: '1 Configure a name for the template'.
- Configuration Section:**
 - The 'Units' field contains '%' and is highlighted with a red box and annotation 2: '2 Enter the units to display'.
 - The 'Counters & Formula' section has a 'Telemetry Type*' field containing 'telemetry:/base/interfaces/utilization', highlighted with a red box and annotation 3: '3 Enter the telemetry type'.
 - To the right of the 'Telemetry Type*' field is a '+ Counters' button, highlighted with a red box and annotation 5: '5 Click + Counters'.
 - Below the 'Telemetry Type*' field is a 'Selected Counter(s)*' area showing 'No Counters Added'.
 - At the bottom is a 'Formula' field with a question mark icon.

40480

2

A form opens, showing the available counters. We'll choose input utilization.



40481

3

Enter a formula for the indicator to use to calculate KPI values. Our formula calculates average input utilization.

Click **Create** to create the template.

Create Indicator Template

General

Name* Ring Utilization **Description**

Configuration

Units %

Counters & Formula

Telemetry Type* telemetry:/base/interfaces/utilization **+ Counters**

Selected Counter(s)* input-utilization

Formula* ? 1 avg([input-utilization_avg])

Thresholds

+ Threshold **Threshold Setting**

2 Click Create

Cancel Create

40477

4

Now we can create the indicator. From the Indicators view, click **+ Indicator**. Enter a name for the indicator and configure the collection settings.

Click **Apply Template**.

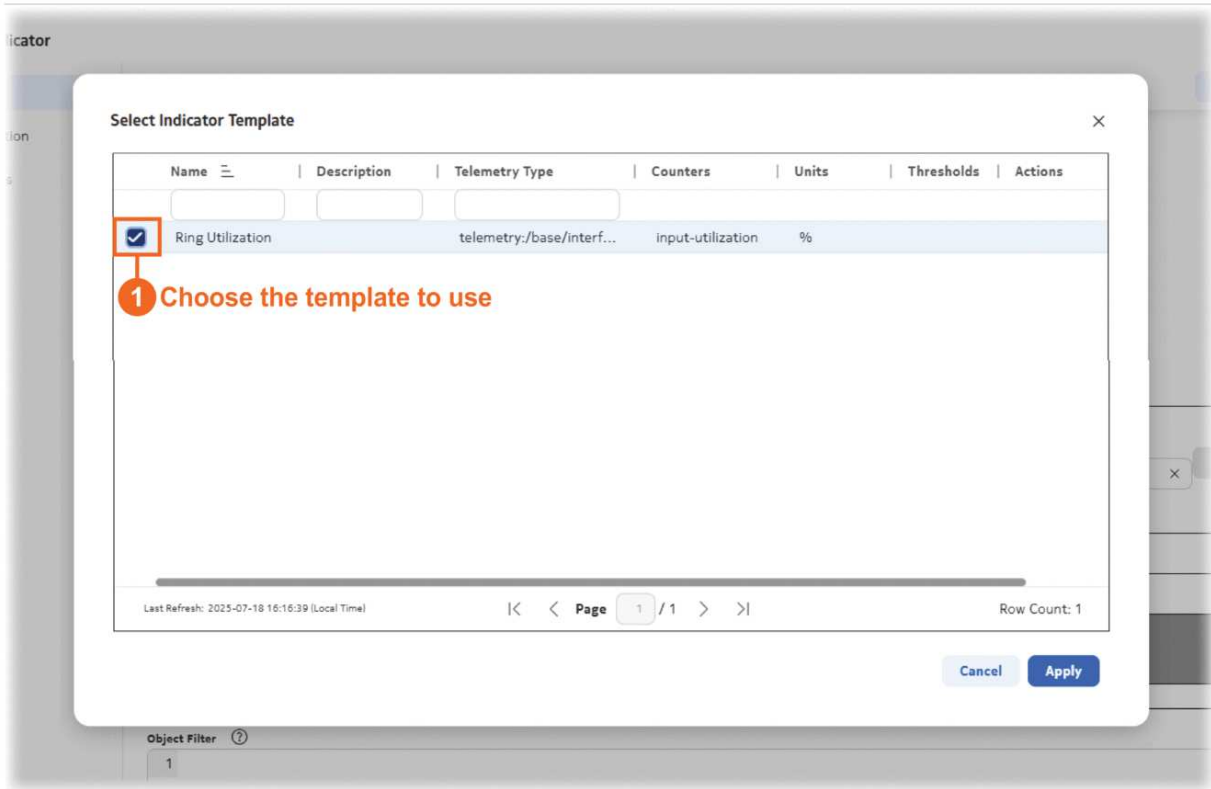
The screenshot shows the 'Create Indicator' dialog box with the following elements and annotations:

- General Tab:** Includes fields for Name, Description, Collection Interval (seconds), and Window Duration. An annotation '1 Enter a name for the indicator' points to the 'Name' field, which contains 'Ring Ingress Utilization'.
- Configuration Section:** Includes a 'Units' field and a 'Counters & Formula' section. An annotation '2 Configure collection parameters. The indicator will collect statistics every 60 seconds and output the aggregated counter function after every window duration to the rest of the indicator formula.' points to the 'Collection Interval (seconds)' field, which is set to '60'.
- Counters & Formula Section:** Includes a 'Telemetry Type' dropdown, a 'Selected Counter(s)' list, and a 'Formula' text area. An annotation '3 Click Apply Template' points to the 'Apply Template...' button in the top right corner.
- Object Filter:** A dropdown menu showing '1'.
- Buttons:** 'Cancel' and 'Create' buttons at the bottom right.

40483

5

In the window that opens, choose the template and click **Apply**.



40471

The template has provided most of the parameters for the indicator, including defining how we want to calculate ring utilization. Next, we can apply an object filter to define the resources to monitor.

6

Click **Verify Resources** to show the resources found by the filter and ensure that they are returning values.

Create Indicator

Configuration

telemetry:/base/interfaces/utilization

Selected Counter(s)*

input-utilization

Formula* ①

1 avg(input-utilization_avg)

Object Filter ②

1 /network-device-mgr:network-devices/network-device[name='11.50.150.33']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.50.150.33:shelf-1:cardSlot-1:card:daughterCardSlot-1:daughterCard:port-1']

2 /network-device-mgr:network-devices/network-device[name='11.50.150.33']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.50.150.33:shelf-1:cardSlot-1:card:daughterCardSlot-1:daughterCard:port-2']

3 /network-device-mgr:network-devices/network-device[name='11.50.150.33']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.50.150.33:shelf-1:cardSlot-1:card:daughterCardSlot-1:daughterCard:port-3']

4 /network-device-mgr:network-devices/network-device[name='11.50.150.35']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.50.150.35:shelf-1:cardSlot-1:card:daughterCardSlot-1:daughterCard:port-10']

5 /network-device-mgr:network-devices/network-device[name='11.50.150.35']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.50.150.35:shelf-1:cardSlot-1:card:daughterCardSlot-1:daughterCard:port-11']

6 /network-device-mgr:network-devices/network-device[name='11.50.150.35']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.50.150.35:shelf-1:cardSlot-1:card:daughterCardSlot-1:daughterCard:port-12']

7

Resource List

Verify Resources

① Enter an object filter. This filter finds ports and LAGs in a specified ring.

② Click Verify Resources

40468

7

When the resources of interest appear in the resources list, click **Stop Verification**. The verification does not stop on its own.

Create Indicator

Configuration

telemetry:/base/interfaces/utilization

Selected Counter(s)*

input-utilization

Formula* ①

1 avg(input-utilization_avg)

Object Filter ②

1 /network-device-mgr:network-devices/network-device[name='11.50.150.33']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.50.150.33:shelf-1:cardSlot-1:card:daughterCardSlot-1:daughterCard:port-1']

2 /network-device-mgr:network-devices/network-device[name='11.50.150.33']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.50.150.33:shelf-1:cardSlot-1:card:daughterCardSlot-1:daughterCard:port-2']

3 /network-device-mgr:network-devices/network-device[name='11.50.150.33']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.50.150.33:shelf-1:cardSlot-1:card:daughterCardSlot-1:daughterCard:port-3']

4 /network-device-mgr:network-devices/network-device[name='11.50.150.35']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.50.150.35:shelf-1:cardSlot-1:card:daughterCardSlot-1:daughterCard:port-10']

5 /network-device-mgr:network-devices/network-device[name='11.50.150.35']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.50.150.35:shelf-1:cardSlot-1:card:daughterCardSlot-1:daughterCard:port-11']

6 /network-device-mgr:network-devices/network-device[name='11.50.150.35']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.50.150.35:shelf-1:cardSlot-1:card:daughterCardSlot-1:daughterCard:port-12']

7

Resource List

Provider	Name	input-utilization	Resource
NFM-P	(11.50.150.33), Port 1/1/1	0	/network-device-mgr:network-devices/network-device[name='11.50.150.33']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.5...
NFM-P	(11.50.150.33), Port 1/1/2	0	/network-device-mgr:network-devices/network-device[name='11.50.150.33']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.5...
NFM-P	(11.50.150.33), Port 1/1/3	0	/network-device-mgr:network-devices/network-device[name='11.50.150.33']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.5...
NFM-P	(11.50.150.35), Port 1/1/10	0	/network-device-mgr:network-devices/network-device[name='11.50.150.35']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.5...
NFM-P	(11.50.150.35), Port 1/1/11	0	/network-device-mgr:network-devices/network-device[name='11.50.150.35']/root/nokia-nsp-source:fdn[id='fdn:realmsam:network:11.5...

Stop Verification

① Verify that the correct resources are found and that they are returning data

② Click Stop Verification

40503

8

Add thresholds if needed and click **Create**. See [7.4 “Creating a simple indicator” \(p. 124\)](#) for steps to create a threshold.

END OF STEPS

We’re done

The indicator appears in the list in **Data Collection and Analysis Management, Indicators**.

Nokia Network Services Platform									
Data Collection and Analysis Management NSP Indicators Indicators									
+ Indicator									
Admin State	Name	Description	Telemetry Type	Counters	Units	Thresholds	Actions	Multiple Output	Collectio (second
Enabled	Ring Ingress Utilization		telemetry./base/interf...	input-utilization	%	✓	✓		30

40476

8 Chart sample procedures

8.1 Creating baseline and anomaly charts

8.1.1 Purpose

This sample procedure shows how to use NSP to chart baselines and anomalies.

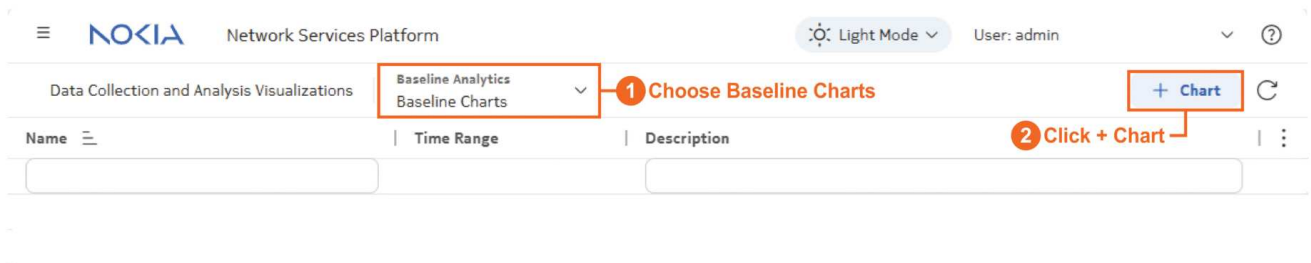
The following prerequisites must be completed:

- The required baselines have been created.
- The baselines have been in active learning status for enough time to detect anomalies.
Depending on the anomaly detector, this can be one or two seasons.

8.1.2 Steps

1

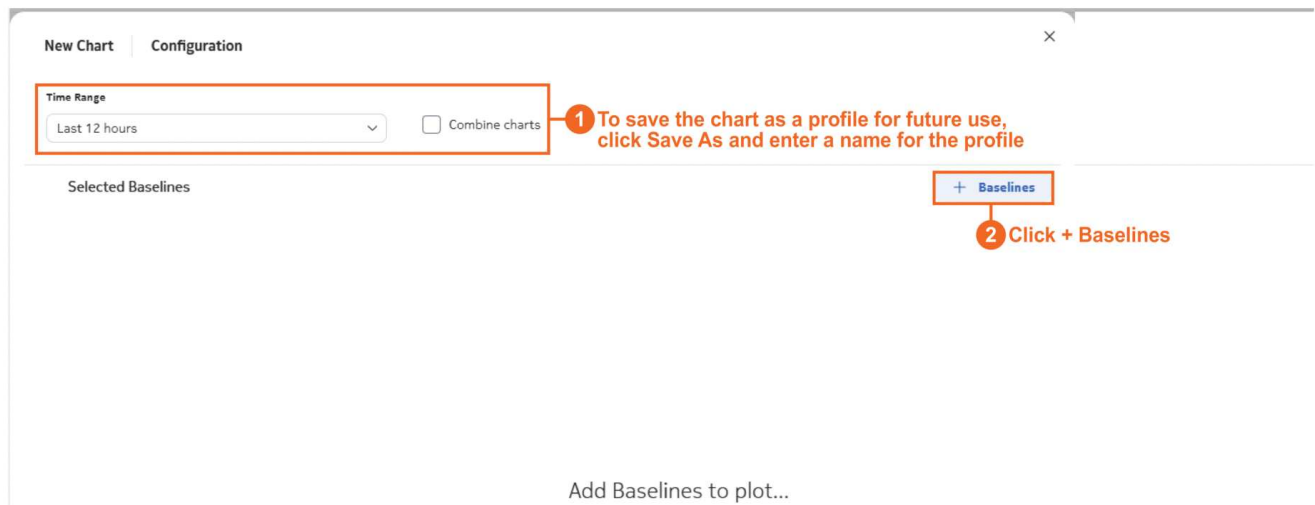
From **Data Collection and Analysis Visualizations, Baseline Charts**, we'll create a chart.



40494

2

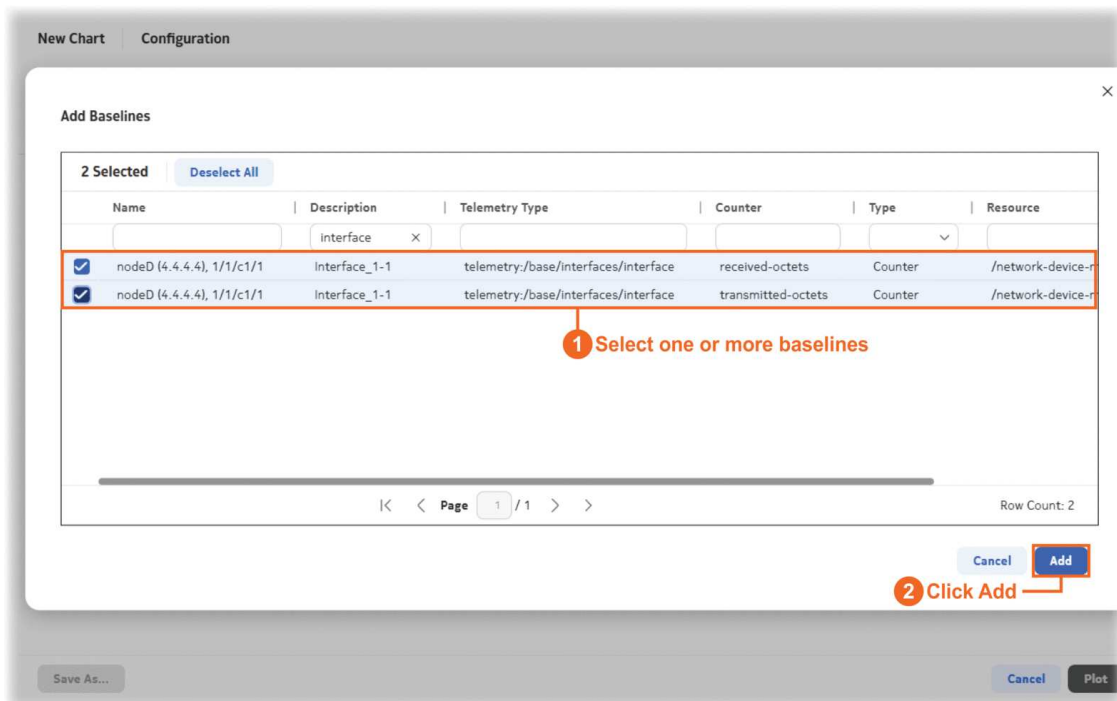
Configure the time range parameter. To display multiple baselines on the same chart, enable the **Combine Charts** check box.



40493

3

In the Add Baselines form, choose the baselines to chart and click **Add**.



40466

4

If you want to save the configuration for future use, click **Save As** and enter a name for the chart.

Click **Plot** to plot the chart.

The screenshot shows a 'New Chart' configuration window with a 'Configuration' tab. At the top, there's a 'Time Range' dropdown set to 'Last 12 hours' and a 'Combine charts' checkbox. Below this is a 'Selected Baselines' section with a table. The table has columns: Name, Description, Telemetry Type, Counter, Type, and Resource. It contains two rows of baseline data. A red box highlights the reorder handles (three dots) on the left of the second row. Below the table, an orange callout bubble with the number '1' says 'Click on the handles to reorder baselines in the list if needed'. At the bottom of the window, there are three buttons: 'Save As...', 'Cancel', and 'Plot'. An orange callout bubble with the number '2' points to the 'Save As...' button and says 'To save the chart as a profile for future use, click Save As and enter a name for the profile'. Another orange callout bubble with the number '3' points to the 'Plot' button and says 'Click Plot'.

Name	Description	Telemetry Type	Counter	Type	Resource
nodeD (4.4.4.4), 1/1/c1/1	Interface_1-1	telemetry:/base/interfaces/interface	received-octets	Counter	/network-device-mgr:netw
nodeD (4.4.4.4), 1/1/c1/1	Interface_1-1	telemetry:/base/interfaces/interface	transmitted-octets	Counter	/network-device-mgr:netw

40475

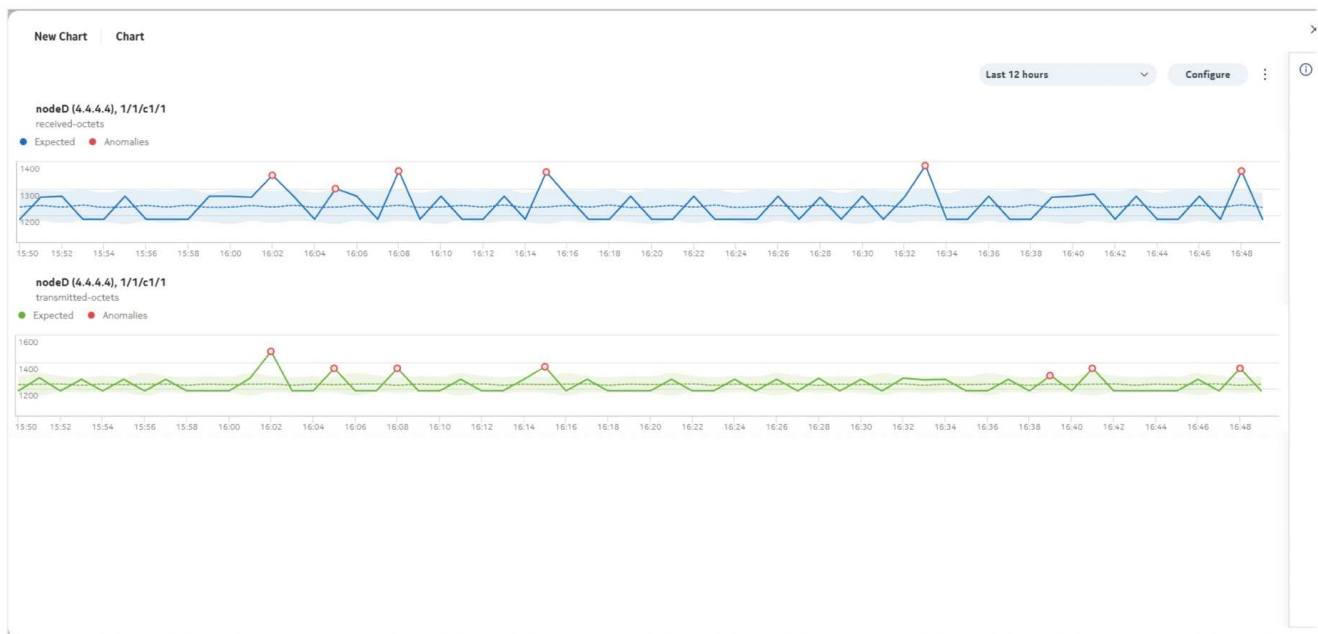
END OF STEPS

We're done

Visualizations displays a chart for each baseline.

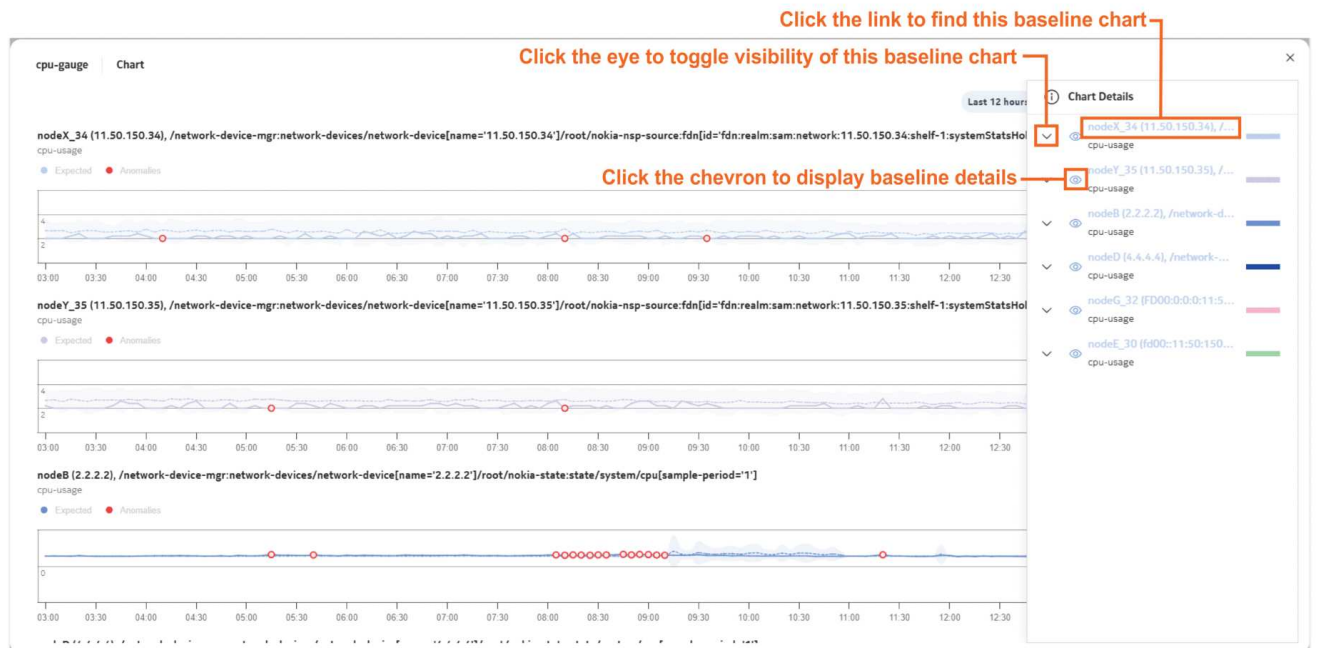
The dotted line shows the expected values. The shading shows the detector threshold range, that is, the range of values that is not considered anomalous.

The solid line shows the actual values. Anomalies are indicated with red dots.



40504

Click **Chart Details** ⓘ at the top right of a chart to display the Chart Details panel.



40473

8.2 Creating an indicator chart

8.2.1 Purpose

This sample procedure shows how to use NSP to chart an indicator.

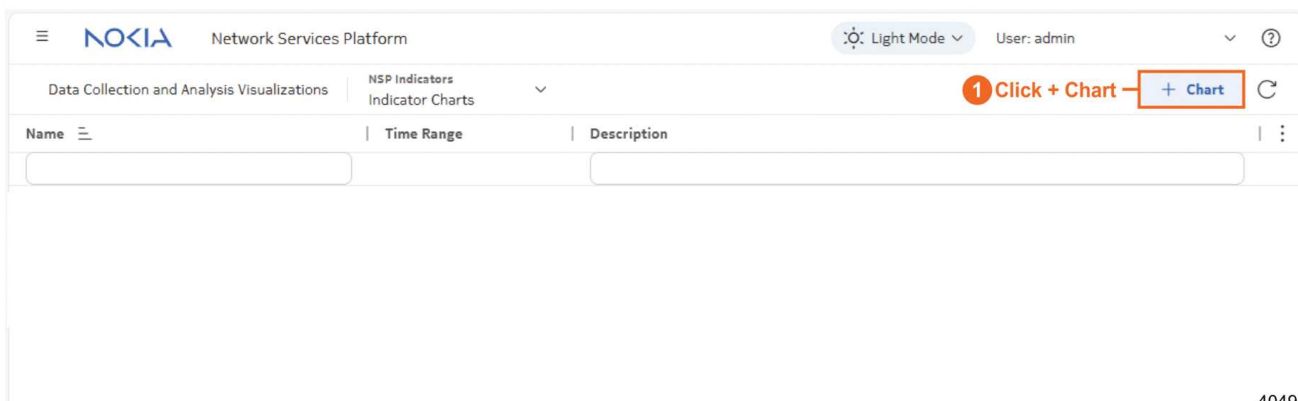
The following prerequisites have been completed:

- The indicator we need has been created.
- The indicator has been operating for at least one window duration.

8.2.2 Steps

1

From **Data Collection and Analysis Management, Indicator Charts**, we'll create a chart.



40495

2

Configure the time range parameter. To display multiple indicators on the same chart, enable the **Combine Charts** check box.

Click **+ Indicators**.

New Chart Configuration X

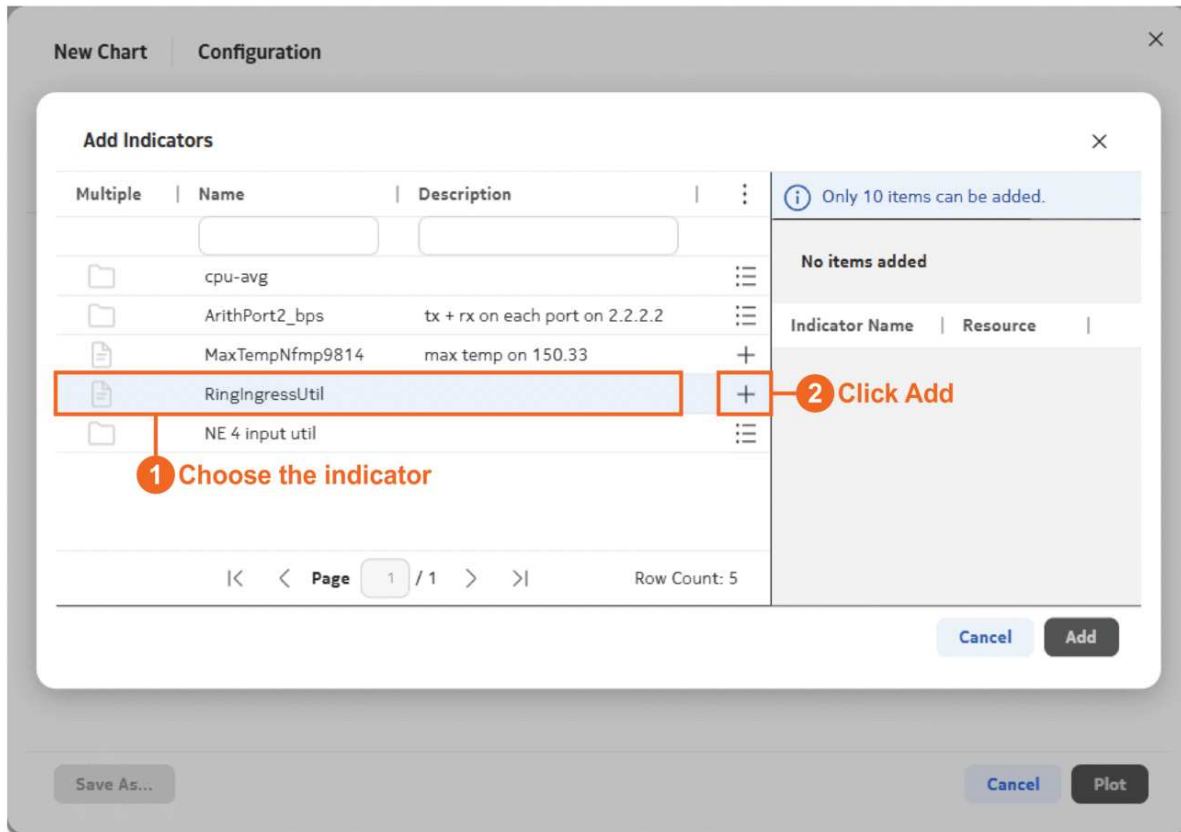
Time Range Last 12 hours Combine charts

Selected Indicators + Indicators

40470

3

In the Add Indicators form, choose the indicators to chart and click **Add**. The indicator we want to add has a file icon in the Multiple column, so the resources are aggregated and there is no need to select resources. We can just choose the Ring Ingress Util indicator and click **Add**.



40474

4

If you want to save the configuration for future use, click **Save As** and enter a name for the chart.

Click **Plot** to plot the chart.

New Chart

Configuration

Time Range

Last 12 hours

Combine charts

Selected Indicators

+ Indicators

Indicator Name	Description	Resource
RingIngressUtil		

2 To save the chart as a profile for future use, click Save As and enter a name for the profile

1 When all the indicators are added click Plot

Save As...

Cancel

Plot

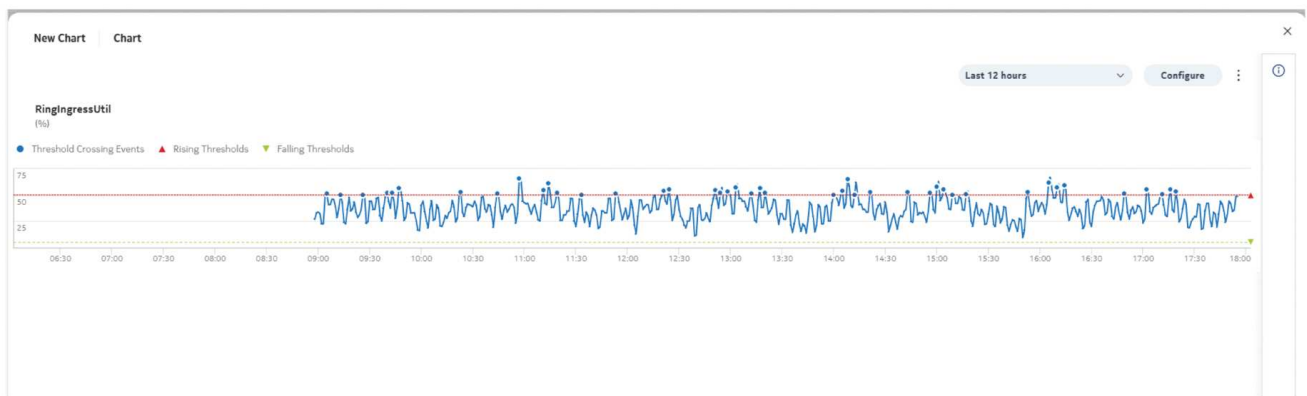
40492

END OF STEPS

We're done

Visualizations displays a chart for the indicator.

The dotted line shows the thresholds. Threshold crossing events are indicated with dots.



40491

9 OAM testing sample procedure

9.1 Discovering OAM tests from NFM-P to NSP

9.1.1 Purpose

This sample procedure shows how to use NSP to discover brownfield OAM tests that are running in NFM-P. For this scenario, we'll show CFM Loopback tests.

The following prerequisites must be completed:

- The required tests are configured in the NFM-P.
- Where applicable, the `Publish to Kafka` parameter was enabled in the NFM-P when the tests were configured.
- The tests are stopped in the NFM-P.

9.1.2 Steps

1

Open **Data Collection and Analysis Management, Tests**.

2

In the **Filter** column at the left of the view, choose the test type and click **Retrieve**. The list of tests is populated and filtered according to your input.

3

Choose a test and click  to open the **Test Details** panel to verify that the test parameters meet your needs.

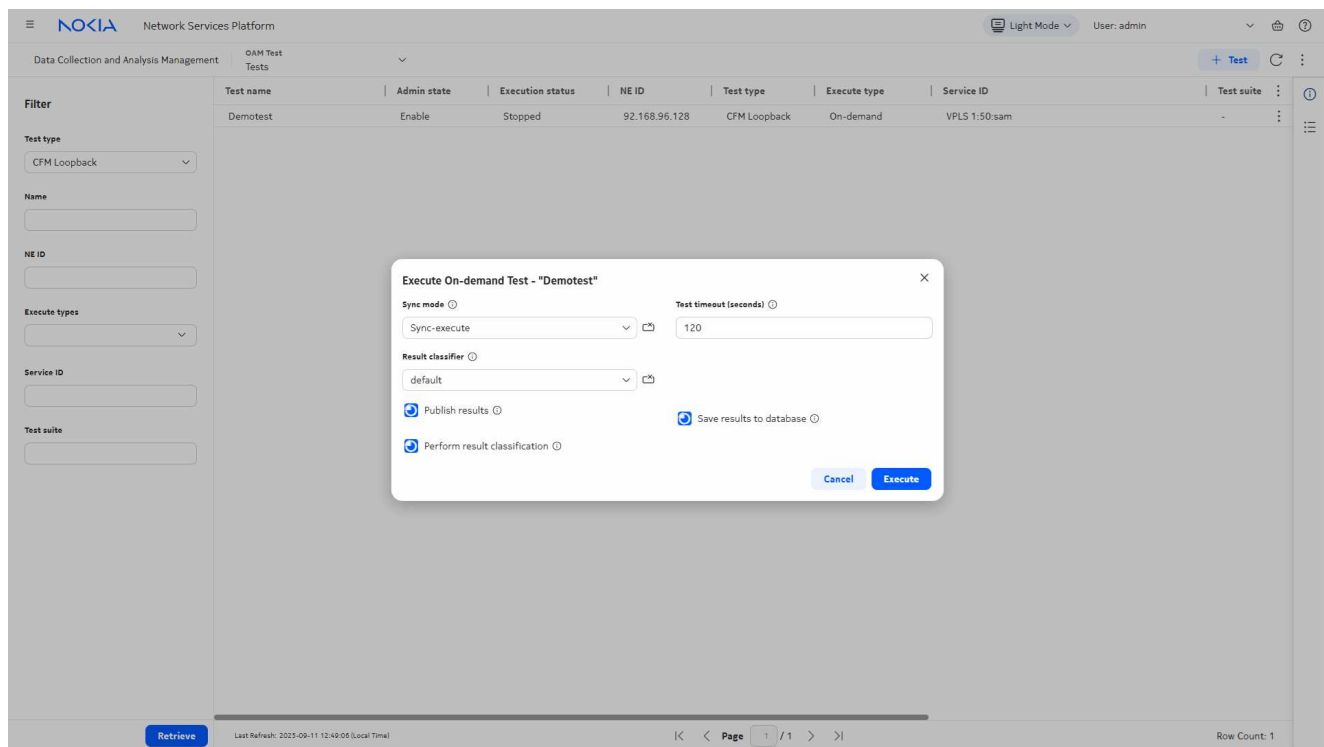
4

From your selected test row, click  (Table row actions), **Execute**. The Execute dialogue opens.

5

Configure the required parameters.

- The Sync mode parameter specifies whether or not execution requests generate notifications in the UI. The **Sync-execute** option (default) provides a notification if the execution fails.
- The Result Classifier parameter specifies the name of the result classifier used to determine test success or failure. Result classifiers are configurable using a REST API.
Check the **Perform result classification** check box to apply the classifications.
- The **Publish results** check box publishes results to Kafka.
- The **Save results to database** check box makes results available to the NSP UI. If this box is not checked, the UI does not display the test results.



Click **Execute**. The execution status changes to Running.

6

To stop a test that is running, choose the test in the list and click  (Table row actions), **Stop**.

END OF STEPS

We're done

The test is now running in the NSP.

To view results of a discovered test, the test must be stopped in NFM-P and executed from the NSP.

Select the test in the Tests view and click  (Table row actions), **View Results**.

The Test executions view opens, showing the executions and their results.

i Note: After a test has executed, there is a brief processing delay before results are available. For tests that have just finished running, Nokia recommends that you wait a minimum of 5 s before viewing the results.

Demotest (92.168.96.128) Test Results							
Last 1 day		telemetry:/base/oam-result/loopback-result (Default)					
Test execution ID	Result classification	Result status	Time captured	Probes sent	Probes received	Probes lost	Loss ratio
19	Passed	success	2025-09-11 12:44:02	1	1	0	0.00

To view the test result lifecycle, select *telemetry type-result/lifecycle* from the dropdown in the Test Results view.

Demotest (92.168.96.128) Test Results							
Last 1 day		telemetry:/base/oam-result/lifecycle					
Test execution ID	Result classification	Time captured	Event type	Service ID	Reason	Message	Test path
25	Started	2025-09-11 12:52:13	TEST_STARTED	VPLS 1:50.sam			/nsp-oam:tests/oam-test.t
23	Stopped	2025-09-11 12:51:45	TEST_STOPPED	VPLS 1:50.sam			/nsp-oam:tests/oam-test.t
23	Started	2025-09-11 12:51:06	TEST_STARTED	VPLS 1:50.sam			/nsp-oam:tests/oam-test.t
21	Stopped	2025-09-11 12:49:50	TEST_STOPPED	VPLS 1:50.sam			/nsp-oam:tests/oam-test.t
21	Error (timeout)	2025-09-11 12:49:50	TEST_RESULT	VPLS 1:50.sam			/nsp-oam:tests/oam-test.t
21	Started	2025-09-11 12:45:49	TEST_STARTED	VPLS 1:50.sam			/nsp-oam:tests/oam-test.t
19	Started	2025-09-11 12:40:05	TEST_STARTED	VPLS 1:50.sam			/nsp-oam:tests/oam-test.t
-	Test-created	2025-09-11 12:35:43	TEST_CREATED				/nsp-oam:tests/oam-test.t

